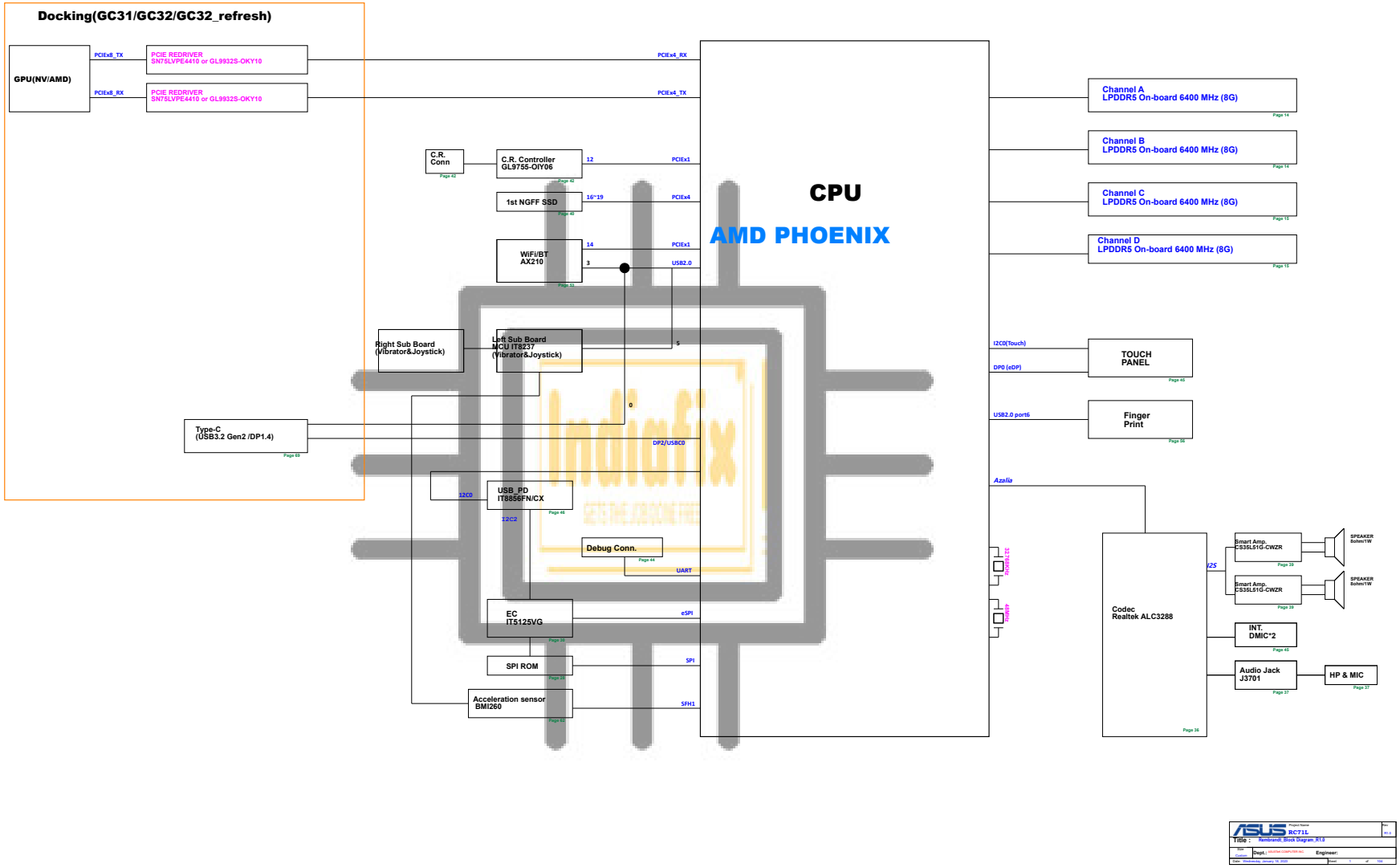
001_Block Diagram
002_System Setting
003.CPU_PCIE,SATA,WLAN,LAN
004.CPU_LPDDR5
005_CPU_GND
006.CPU_eDP/JTAG/SVI2/MISC
007.CPU_AZ/I2C/ACPI/GPIO
008.CPU_CLK/LPC/SD/EMMC/SPI
009.CPU_POWER
010.CPU_POWER_CAP
011_
012_
013_
014_LPDDR5_ON-BOARD_A
015_LPDDR5_ON-BOARD_B
016_
017_
018_
020_
021.CPU USB/CSI2
022_
023_
024_
025_
026_
027_
028_SMB_SPIROM
029_
030_EC_IT5125VG & SPI ROM
031_
032_RST_Reset Circuit_Policy
033_
034_
035_Mac&N_KEY_IT8299E-120A
036_AUD-ALC3288CG
037_
038_AUD_INT SPK
040_NGFF_SSD
041_
042_CR_GL9755
043_
044_DEBUG_LPC
045_CRT_eDP
046_PD_TPS65993BF
047_TYPE-C_KB8002_PI3DPX1207CZH
048_
049_
050_FAN_Thermal Sensor & Fan
051_
052_
053_WLAN_GIP2_AX210
056_LED & CON
057_DSG_Discharge
058_Power Good Detecter
059_OTH_EMI
060_DC & BAT IN
062_BSH_Sensor
063_SB Right
064_SB Left
065_screw hold
066_
067_
068_
069_Docking_Con
070_
071_
072_
073_
074_
075_
076_
077_

080_PW_APU (1)
081_PW_APU (2)
082_PW_0.75VS
083_PW_+0.75VSUS
084_PW_1P8VSUS/SYV738
085_PW_+VDD_MEM
086_PW_1.05V_1.8V_+0.9V/LPDDR5
087_PW_+3VSUS/+5VSUS
088_PW_LOAD SWITCH
089 PW_BB CHARGER
090_PW_PROTECTION
091_
092_
093_PW_+1.1VS_VDD_11/+0.5V
095_PW_V_LED_LCM/ +1V_ANX
096_PW_VSP_LCM_VSN_LCM
099_PW_FLOW CHART

100_Power On Timing--AC mode
101_Power On Timing--DC mode

RC71L AMD Block Diagram



ASUS	Project Name: RC71L		Rev: REV:X
Title : Rembrandt_Block Diagram_R1.0			
Size: Custom	Dept.: MB-PRJ&PM-1-DEPT-PRJ&PM-SEC	Engineer:	
Date: Wednesday, January 15, 2020	Sheet: 1	of: 102	

PCH_CPU GPIO

	Default	Use As	Signal Name	INT PU/PD	EXT PU/PD	Power
GPP_A0	SCI3#	Native	RC_I3#		PU 10K	+3VS
GPP_A1	LAD3	Native	LPC_AD3			
GPP_A2	LAD2	Native	LPC_AD2			
GPP_A3	LAD1	Native	LPC_AD1			
GPP_A4	LAD0	Native	LPC_AD0			
GPP_A5	LFRAME#	Native	LPC_FRAME#			
GPP_A6	SERIRQ	Native	SMI_SERIRQ		PU 10K	+3VS
GPP_A7	PIRQN#	Native	PCI_INTA#		PU 10K	+3VS
GPP_A8	CLKOUT0	Native	PM_CLKOUT0		PU 8.2K	+3VS
GPP_A9	CLKOUT_LPC0	Native	CLK_48CPCI_PCH_01			
GPP_A10	CLKOUT_LPC1	Native	CLK_DEBUG_01			
GPP_A11	PME#	GPI	PM_C7			
GPP_A12	SD3	GPO	SD_C3		PU 10K(E)	+3VDUO
GPP_A13	SUSWARN#	Native	WE_PushPullAuto			
GPP_A14	SUS_STAT#	Native	PCH_SUS_STAT#			
GPP_A15	SUS_ACK#	Native	PM_SUSACK#			
GPP_A16	Native/GPI	GPO	SD_C1			
GPP_A17	SD1	GPO	SD_C1			
GPP_A18	SD2	GPO	SD_C2			
GPP_A19	SD3	GPO	SD_C3			
GPP_A20	SD1	GPO	SD_C1			
GPP_A21	SD2	GPO	THIS_BRANCH? (Reserved)			
GPP_A22	SD3	GPO	PCH_RTC13 C3O_PWR_EN		PU 10K	+VCCSTG_EN
GPP_A23	SD3	GPO	PCH_RTC13 C3O_PWR_EN		PU 100K	

	Default	Use As	Signal Name	INT PU/PD	EXT PU/PD	Power
GPP_B0	SD2	GPO	SD_C2			
GPP_B1	SD3	GPO	TBT_I2C_INT_PCH		PU 10K	+3VDUO
GPP_B2	SD1	GPO	SD_C1			
GPP_B3	SD2	GPO	SD_C2			
GPP_B4	SD3	GPO	SD_C3			
GPP_B5	SD1	Native	CLK_REQ0#/CLKREQ0_VRO4#		PU 10K	+3VDUO
GPP_B6	SD2	Native	CLK_REQ1#		PU 10K(E)	+3VDUO
GPP_B7	SD3	Native	CLK_REQ2#/CLKREQ2_WLAN_ACK		PU 10K	+3VDUO
GPP_B8	SD1	Native	CLK_REQ3#		PU 10K(E)	+3VDUO
GPP_B9	SD2	Native	CLK_REQ4#/CLKREQ4_GLAN#		PU 10K	+3VDUO
GPP_B10	SD3	Native	CLK_REQ5#/CLKREQ5_TBT_#		PU 10K	+3VDUO
GPP_B11	SD0	GPO	SD_C0			
GPP_B12	SLP_S0#	Native	SLP_S0#			
GPP_B13	PLTRST#	Native	PLT_RST#			
GPP_B14	SPI0	Strap	PCH_SPI0IO_0			
GPP_B15	SPI0	Strap	PCH_SPI0IO_1	PU		
GPP_B16	SPI1	GPI	PCH_SPI0IO_2/TCH_LOAD_IO2		PU 10K	+3VS
GPP_B17	SPI1	GPO	PCH_SPI0IO_3/XDCH_WP0T		PU 10K	
GPP_B18	SPI0	Strap	PCH_SPI0_0	PU		
GPP_B19	SPI1	GPO	PCH_SPI0IO_0/BT_GM_PCH		PU 10K(E)	
GPP_B20	SPI1	GPO	PCH_SPI0IO_0/TOPCHOTOPT_PCH		PU 10K	+3VDUO
GPP_B21	SPI1	GPO	PCH_SPI0IO_0/SCHPWRM_PCH		PU 10K(E)	+3VS
GPP_B22	GPO	Strap	PCH_SPI0_2	PU		
GPP_B23	GPO	Native	SMOLLAERT#	PU	PU 150K	+3VDUO

	Default	Use As	Signal Name	INT PU/PD	EXT PU/PD	Power
GPP_C0	SMBCLK	Native	SMB_CLK		PU 2.2K	+3VDUO
GPP_C1	SMBDATA	Native	SMB_PDA		PU 2.2K	+3VDUO
GPP_C2	GPO	GPO	SPP_TO_BT (SAFE MODE_0	PU	PU 4.7K	+3VDUO
GPP_C3	SMLOCLK	Native	SMB2_CLK (Reserved)		PU 2.2K	+3VDUO
GPP_C4	SMLODATA	Native	SMB2_DAT (Reserved)		PU 2.2K	+3VDUO
GPP_C5	SD1	Strap	PCH_SPI0_1	PU		
GPP_C6	SMLICLK	Native	SMB1_CLK		PU 2.2K	+3VDUO
GPP_C7	SMLIDATA	Native	SMB1_DAT		PU 2.2K	+3VDUO
GPP_C8	SD1	GPO	SD_C1			
GPP_C9	SD1	GPO	SD_C1			
GPP_C10	SD2	GPO	SD_C2			
GPP_C11	SD2	GPO	SD_C2			
GPP_C12	SD2	GPO	SD_C2			
GPP_C13	SD2	GPO	SD_C2			
GPP_C14	SD2	GPI	CORE_DSL1			
GPP_C15	SD2	GPI	CORE_DSL2			
GPP_C16	SD1	GPO	SD_C1		PU 10K	+3VDUO
GPP_C17	SD1	Native	SD2_I2C_SDA_PCH		PU 2.2K	+3VDUO
GPP_C18	SD1	Native	SD2_I2C_SCL_PCH		PU 2.2K	+3VDUO
GPP_C19	SD1	Native	I2C1_SDA		PU 2.2K	+3VS
GPP_C20	SD1	GPI	PCH_SPI0IO1/DSPU_PWRDN		PU 10K	+3VS_EC
GPP_C21	SD1	GPO	PCH_SPI0IO1/SPU_RST#		PU 10K	
GPP_C22	SD1	GPO	SD_C1			
GPP_C23	SD1	GPI	CORE_DSTB		PU 10K	+3VS

	Default	Use As	Signal Name	INT PU/PD	EXT PU/PD	Power
GPP_D0	SD1	GPO	SD_C1			
GPP_D1	SD1	GPO	SD_C1			
GPP_D2	SD1	GPO	SD_C1			
GPP_D3	SD1	GPO	SD_C1			
GPP_D4	SD1	GPO	SD_C1			
GPP_D5	SD1	GPO	SD_C1			
GPP_D6	SD1	GPO	SD_C1			
GPP_D7	SD1	GPO	SD_C1			
GPP_D8	SD1	GPO	SD_C1			
GPP_D9	SD1	GPO	SD_C1			
GPP_D10	SD1	GPO	SD_C1			
GPP_D11	SD1	GPO	SD_C1			
GPP_D12	SD1	GPO	SD_C1			
GPP_D13	SD1	GPO	SD_C1			
GPP_D14	SD1	GPO	SD_C1			
GPP_D15	SD1	GPO	SD_C1			
GPP_D16	SD1	GPO	SD_C1			
GPP_D17	SD1	GPO	SD_C1			
GPP_D18	SD1	GPO	SD_C1			
GPP_D19	SD1	GPO	SD_C1			
GPP_D20	SD1	GPO	SD_C1			
GPP_D21	SD1	GPO	SD_C1			
GPP_D22	SD1	GPO	SD_C1			
GPP_D23	SD1	GPO	SD_C1			

	Default	Use As	Signal Name	INT PU/PD	EXT PU/PD	Power
GPP_E0	SD1	Native	SATAXPCIE_0#3		PU 10K(E)	+3VS
GPP_E1	SD1	Native	SATAXPCIE_0#4		PU 10K	+3VS
GPP_E2	SD1	GPO	SD_C1			
GPP_E3	SD1	GPO	SD_C1			
GPP_E4	SD1	GPO	SD_C1			
GPP_E5	SD1	GPO	SD_C1			
GPP_E6	SD1	GPO	SD_C1			
GPP_E7	SD1	GPO	SD_C1			
GPP_E8	SD1	GPO	SD_C1			
GPP_E9	SD1	GPO	SD_C1			
GPP_E10	SD1	GPO	SD_C1			
GPP_E11	SD1	GPO	SD_C1			
GPP_E12	SD1	GPO	SD_C1			
GPP_E13	SD1	GPO	SD_C1			
GPP_E14	SD1	GPO	SD_C1			
GPP_E15	SD1	Native	USB_OC#3		PU 10K	+3VDUO
GPP_E16	SD1	Native	USB_OC#4		PU 10K	+3VDUO
GPP_E17	SD1	Native	USB_OC#5		PU 10K	+3VDUO
GPP_E18	SD1	Native	USB_OC#7		PU 10K	+3VDUO
GPP_E19	SD1	GPO	SD_C1			
GPP_E20	SD1	GPO	SD_C1			
GPP_E21	SD1	GPO	SD_C1			
GPP_E22	SD1	GPO	SD_C1			
GPP_E23	SD1	GPO	PCH_GPP23		PU 10K	+3VS

	Default	Use As	Signal Name	INT PU/PD	EXT PU/PD	Power
GPP_F0	SD1	Native	SATAXPCIE_0#5		PU 10K(E)	+3VS
GPP_F1	SD1	Native	SATAXPCIE_0#6		PU 10K	+3VS
GPP_F2	SD1	GPO	SD_C1			
GPP_F3	SD1	GPO	SD_C1			
GPP_F4	SD1	GPO	SD_C1			
GPP_F5	SD1	GPO	SD_C1			
GPP_F6	SD1	GPO	SD_C1			
GPP_F7	SD1	GPO	SD_C1			
GPP_F8	SD1	GPO	SD_C1			
GPP_F9	SD1	GPO	SD_C1			
GPP_F10	SD1	GPO	SD_C1			
GPP_F11	SD1	GPO	SD_C1			
GPP_F12	SD1	GPO	SD_C1			
GPP_F13	SD1	GPO	SD_C1			
GPP_F14	SD1	GPO	SD_C1			
GPP_F15	SD1	Native	USB_OC#8		PU 10K	+3VDUO
GPP_F16	SD1	Native	USB_OC#9		PU 10K	+3VDUO
GPP_F17	SD1	Native	USB_OC#10		PU 10K	+3VDUO
GPP_F18	SD1	Native	USB_OC#7		PU 10K	+3VDUO
GPP_F19	SD1	GPO	SD_C1			
GPP_F20	SD1	GPO	SD_C1			
GPP_F21	SD1	GPO	SD_C1			
GPP_F22	SD1	GPO	SD_C1			
GPP_F23	SD1	GPO	PCH_GPP23		PU 10K	+3VS

	Default	Use As	Signal Name	INT PU/PD	EXT PU/PD	Power
GPP_G0	SD1	GPO	SD_C1			
GPP_G1	SD1	GPO	SD_C1			
GPP_G2	SD1	GPO	SD_C1			
GPP_G3	SD1	GPO	SD_C1			
GPP_G4	SD1	GPO	SD_C1			
GPP_G5	SD1	GPO	SD_C1			
GPP_G6	SD1	GPO	SD_C1			
GPP_G7	SD1	GPO	SD_C1			

	Default	Use As	Signal Name	INT PU/PD	EXT PU/PD	Power
GPP_H0	SD1	Native	CLK_REQ6#/CLKREQ6_NSPF14		PU 10K	+3VDUO
GPP_H1	SD1	Native	CLK_REQ7#/CLKREQ7_NSPF24		PU 10K	+3VDUO
GPP_H2	SD1	Native	CLK_REQ8#/CLKREQ8_NSPF34		PU 10K	+3VDUO
GPP_H3	SD1	Native	CLK_REQ9#		PU 10K(E)	+3VDUO
GPP_H4	SD1	Native	CLK_REQ10#		PU 10K(E)	+3VDUO
GPP_H5	SD1	Native	CLK_REQ11#		PU 10K(E)	+3VDUO
GPP_H6	SD1	Native	CLK_REQ12#		PU 10K(E)	+3VDUO
GPP_H7	SD1	Native	CLK_REQ13#		PU 10K(E)	+3VDUO
GPP_H8	SD1	GPO	SD_C1			
GPP_H9	SD1	GPO	SD_C1			
GPP_H10	SD1	GPO	SD_C1			
GPP_H11	SD1	GPO	SD_C1			
GPP_H12	SD1	GPO	SD_C1			
GPP_H13	SD1	GPO	SD_C1			
GPP_H14	SD1	GPO	SD_C1			
GPP_H15	SD1	GPO	SD_C1			
GPP_H16	SD1	GPO	SD_C1			
GPP_H17	SD1	GPO	SD_C1			
GPP_H18	SD1	GPO	SD_C1			
GPP_H19	SD1	GPO	SD_C1			
GPP_H20	SD1	GPO	SD_C1			
GPP_H21	SD1	GPO	SD_C1			
GPP_H22	SD1	GPO	SD_C1			
GPP_H23	SD1	GPO	SD_C1			

	Default	Use As	Signal Name	INT PU/PD	EXT PU/PD	Power
GPP_I0	SD1	GPO	SD_C1			
GPP_I1	SD1	GPO	SD_C1			
GPP_I2	SD1	GPO	SD_C1			
GPP_I3	SD1	GPO	SD_C1			
GPP_I4	SD1	GPO	SD_C1			
GPP_I5	SD1	GPO	SD_C1			
GPP_I6	SD1	GPO	SD_C1			
GPP_I7	SD1	GPO	SD_C1			
GPP_I8	SD1	GPO	SD_C1			
GPP_I9	SD1	GPO	SD_C1			
GPP_I10	SD1	GPO	SD_C1			
GPP_I11	SD1	GPO	SD_C1			
GPP_I12	SD1	GPO	SD_C1			
GPP_I13	SD1	GPO	SD_C1			
GPP_I14	SD1	GPO	SD_C1			
GPP_I15	SD1	GPO	SD_C1			
GPP_I16	SD1	GPO	SD_C1			
GPP_I17	SD1	GPO	SD_C1			
GPP_I18	SD1	GPO	SD_C1			
GPP_I19	SD1	GPO	SD_C1			
GPP_I20	SD1	GPO	SD_C1			
GPP_I21	SD1	GPO	SD_C1			
GPP_I22	SD1	GPO	SD_C1			
GPP_I23	SD1	GPO	SD_C1			

	Default	Use As	Signal Name	INT PU/PD	EXT PU/PD	Power
GPP_J0	SD1	GPO	SD_C1			
GPP_J1	SD1	GPO	SD_C1			
GPP_J2	SD1	GPO	SD_C1			
GPP_J3	SD1	GPO	SD_C1			
GPP_J4	SD1	GPO	SD_C1			
GPP_J5	SD1	GPO	SD_C1			
GPP_J6	SD1	GPO	SD_C1			
GPP_J7	SD1	GPO	SD_C1			
GPP_J8	SD1	GPO	SD_C1			
GPP_J9	SD1	GPO	SD_C1			
GPP_J10	SD1	GPO	SD_C1			
GPP_J11	SD1	GPO	SD_C1			

	Default	Use As	Signal Name	INT PU/PD	EXT PU/PD	Power
GPP_K0	SD1	GPO	SPIO_0_FORCE_PWR		PU 100K	
GPP_K1	SD1	GPI	PCH_TS0		PU 10K	
GPP_K2	SD1	GPI	PCH_TS1		PU 10K	
GPP_K3	SD1	GPO	SPP_K3/WLAN_BT_LED		PU 100K	
GPP_K4	SD1	GPO	SPP_K4/WLAN_LED		PU 100K	
GPP_K5	SD1	GPO	CPU_SOPWR-05-03			
GPP_K6	SD1	GPO	CPU_SOP(E)?			
GPP_K7	SD1	GPO	SPP_K7/TBT_RPU#		PU 10K	+3VS
GPP_K8	SD1	GPO	SPP_K8/CONFI_PWR_EN#		PU 10K(E)	+3VS
GPP_K9	SD1	GPO	WLAN_DIS#		PU 10K	
GPP_K10	Reserved	GPO	SPP_K10/CONFI_WP1#		PU 10K	+3VS
GPP_K11	Reserved	GPO	SPP_K11/TBT_RPU#		PU 10K	+3VS
GPP_K12	SD1	GPO	SD_C1			
GPP_K13	SD1	GPO	SD_C1			
GPP_K14	SD1	GPO	SD_C1			
GPP_K15	SD1	GPO	SD_C1			
GPP_K16	SD1	GPO	SD_C1			
GPP_K17	SD1	GPO	SD_C1			
GPP_K18	SD1	GPO	SD_C1			
GPP_K19	SD1	GPO	SD_C1			
GPP_K20	Reserved	GPO	SD_C1			
GPP_K21	Reserved	GPO	SD_C1			
GPP_K22	SD1	GPO	SD_C1			
GPP_K23	SD1	GPO	SD_C1			

	Default	Use As	Signal Name	INT PU/PD	EXT PU/PD	Power
GPD0	BATLOW#	Native	PM_BATLOW#		PU 10K	+3VA_SUS
GPD1	ACPRESENT	Native	PM_AC_PRESENT_01		PU 10K	+3VA_SUS
GPD2	LAN_WAKE#	Native	LAN_WAKE#(E.C)		PU 10K	+3VA_SUS
GPD3	PWRBTN#	Native	PWRBTN#		PU 10K(E)	+3VSUS
GPD4	SLP_S3#	Native	SLP_S3#		PU 100K	
GPD5	SLP_S4#	Native	SLP_S4#		PU 100K	
GPD6	SLP_A#	Native	SLP_A# (E.C)			
GPD7	GPO	GPO	PCH_GPD7		PU 10K	+3VA_SUS
GPD8	SUSCLK	Native	SUS_CLK#			
GPD9	SLP_WLAN#	Native	SLP_WLAN# (E.C)			
GPD10	SLP_S5#	Native	SLP_S5# (E.C)			
GPD11	LANPHYPC	GPO	LAN_PHYPC (E.C)		PU 10K(E)	+3VA_SUS

IT8225VG-128/BX GPIO

	Default	Use As	Signal Name	EXT PU/PD	Power
GPA0	0	GPO	PEN_LED	PU 1K	
GPA1	OD	GPO	CAPS_LED#		
GPA2	OD	GPO	CAPS_PULL_LED#		
GPA3	A15	PWM	PA3/_PWM		
GPA4	0	GPO	EC_GPA4(E.C)		
GPA5	A15	PWM	PA5/_PWM		
GPA6	A15	PWM	EC_GPA6(E.C)		
GPA7	0	GPO	EC_GPA7(E.C)		

	Default	Use As	Signal Name	EXT PU/PD	Power
GPB0	A15	PWM	PB0_ON_OCK	PU 100K	+3VA
GPB1	I	GPI	LID_DB#	PU 100K	+3VA
GPB2	OD	GPO	EC_GPB2(E.C)		
GPB3	A15	GPI	EC_GPB3/PWR_SW#	PU 100K	+3VA
GPB4	I	GPO	EC_GA_EC		
GPB5	0	GPO	EC_GPB5/TPPwr_EN#3	PU 10K	
GPB6	OD	GPO	EC_GPB	PU 10K	+3VS

	Default	Use As	Signal Name	EXT PU/PD	Power
GPC0	0	GPO	EC_GPC0		
GPC1	A15	PWM/ad	EC_GPC1/SDA	PU 4.7K	+3VA_EC
GPC2	A15	PWM/ad	EC_GPC2/SCL	PU 4.7K	+3VA_EC
GPC3	0	GPO	PM_PWRBTN#	PU 10K(E)	+3VSUS
GPC4	I	GPI	EC_GPC4/PM_SUS#3	PU 100K	
GPC5	I	GPI	PM_SUS4#	PU 100K	
GPC6	A15	GPI	BAT1_IN_OCK	PU 100K	
GPC7	0	GPO	EC_GPC7/WAKE_PWR_EN	PU 10K(E)	

	Default	Use As	Signal Name	EXT PU/PD	Power
GPD0	0	GPO	EC_GPD0/SUSB_EC#	PU 8.2K	
GPD1	0	GPO	EC_GPD1/SUSC_EC#	PU 8.7K	
GPD2	A15	GPI	EC_PW_ON		
GPD3	0	GPO	VGSUS_ON	PU 1K & PD 100K	+3VA_EC
GPD4	0	GPO	VSUS ON	PU 1K & PD 100K	+3VA_EC
GPD5	I	GPO	VCASUS_ON	PU 100K	+3VA_EC
GPD6	OD	GPO	WLAN_PWR_EN	PU 100K	
GPD7	OD	GPO	TABL_CPU#	PU 10K	+3VA_SUS

	Default	Use As	Signal Name	EXT PU/PD	Power
GPE0	0	GPO	CAP LED#		
GPE1	0	GPO	LOCK LED#		
GPE2	A15	SMBus	P_SMB1_CLK	PU 4.7K	+3VA_EC
GPE3	A15	SMBus	P_SMB1_DAT	PU 4.7K	+3VA_EC
GPE4	A15	PS2	TP_CLK	PU 4.7K	+3VS
GPE5	A15	PS2	TP_DAT	PU 4.7K	+3VS
GPE6	A15	PSC2	EC_PSC2	PU 10K(E)	
GPE7	0	GPO	PCH_SPI_CS		

	Default	Use As	Signal Name	EXT PU/PD	Power
GPF0	0	GPO	TSPU_LOAD#		
GPF1	0	GPO	PCH_SUSACK#		
GPF2	3	GPI	EC_GPI2	PU 100K	+3VA_EC
GPF3	0	GPO	EC_GPI3(E.C)		

	Default	Use As	Signal Name	EXT PU/PD	Power
GPG0	A15	PCI	PM_CLKRUN#	PU 8.2K	+3VS
GPG1	A15	SMBus	P_SMB3_CLK/TBT_I2C_SCL_EC	PU 4.7K	+3VA_EC
GPG2	A15	SMBus	P_SMB3_SDA/TBT_I2C_SDA_EC	PU 4.7K	+3VA_EC
GPG3	0	GPO	PM_SLP_S3#	PU 10K	
GPG4	0	GPO	SUSWARN#	PU 10K	
GPG5	0	GPO	PM_PWROK	PU 10K	
GPG6	0	GPO	PM_SYSRST#	PU 10K	
GPG7	0	GPO	LCD_BACKOFF#		

	Default	Use As	Signal Name	EXT PU/PD	Power
GPH0	I	GPI	PM_SLP_S0#4	PU 100K(E)	
GPH1	I	GPI	SYSPOK_PWROK	PU 100K	+3VSUS
GPH2	I	GPI	ALL_SYSTEM_PWRGD	PU 100K	+3VS
GPH3	I	GPI	VGPWR_PWRGD	PU 100K	+3VS
GPH4	I	GPI	3VA_SUS_PWRGD_EC	PU 100K	+3VA_SUS
GPH5	I	GPI	PCIEWAKE#	PU 10K(E)	+3VSUS
GPH6	A15	PWM	EC_GPH6_PCHRST		
GPH7	A15	AD	VR_VRAT_PCHEN	PU 100K(E)	+3VALW

	Default	Use As	Signal Name	EXT PU/PD	Power
GPI0	0	GPO	EC_GPI0		
GPI1	0	GPI	SCPUPWRG_PCH	PU 10K(E)	+3VS
GPI2	A15	GPI	EC_GPI2/3VSA_SIO_PWR_EC		
GPI3	0	GPO	EC_GPI3/SUS_ACI_2_GPP8		
GPI4	A15	GPO	EC_GPI4/3VSA10_PCHEN		
GPI5	0	GPI	EC_GPI5/PM_DOCK_PACK_DET#3	PU 100K(E)	+3VS
GPI6	A15	GPI	QUEST_CSTB	PU 10K	+3VA_SUS
GPI7	0	GPO	CPU_CPASS0_OUT		

	Default	Use As	Signal Name	EXT PU/PD	Power
GPJ0	0	GPO	LPC_AD0 #/LPC_AD0		
GPJ1	0	GPO	LPC_AD1 #/LPC_AD1		
GPJ2	LAD2/KSO2	GPO	LPC_AD2 #/LPC_AD2		
GPJ3	LAD3/KSO3	GPO	LPC_AD3 #/LPC_AD3		
GPJ4	LPCCLK/KSCK	GPO	CLK_33CPCI_PCH		
GPJ5	LFRAME#/KSI8	GPO	LPC_FRAME#		
GPJ6	SERIRQ/KSO09	GPO	LPC_SERIRQ	PU 10K	+3VS

G703GI Source:

SM_BUS ADDRESS :

PCH Master	
SM-Bus Device	**SM-Bus Address**
SO-DIMM 0(0)	A0h
SO-DIMM 0(1)	A4h
SO-DIMM 1(0)	A2h
SO-DIMM 1(1)	A6h

SC Master (WMI1)	
SM-Bus Device	**SM-Bus Address**
CPU Temperature Sensor	98h, 9Ch
CPU Thermal Sensor	90h
TI_TMP451A0YFF2	98h, 9Ah
ASM1542	60h
GPU (Thermistor sense)	74h
PWR(Thermistor sense)	76h, 7Ch, 7Ah, 78h, 7Eh

Device Identification

SPD Name/Vendor		
Ver	0000022200	ALC1150
Dvd		

SPD Revision		
Ver	0000-0000000	2PLA0004079E
Dvd		

USB3.0/PCIE/SATA Setting

1	USB3(0)	
2	USB3(1)	
3	USB3(2)	USB Port1
4	USB3(3)	
5	USB3(5)	USB Port3
6	USB3(6)	USB Port4
7	USB3(7)	Card Reader
8	PCIE(0)	
9	PCIE(1)	
10	PCIE(2)	
11	PCIE(3)	
12	PCIE(4)	ThunderBolt x4
13	PCIE(7)	
14	PCIE(8)	
15	PCIE(9)	
16	PCIE(10)	PCIE3.0 x 4 SSD
17	PCIE(11)	
18	PCIE(12)	
19	PCIE(13)(SATA(0))	SATA HDD
20	PCIE(14)(SATA(1))	
21	PCIE(15)	ODD
22	PCIE(16)	WLAN_11AC
23	PCIE(17)	
24	PCIE(18)	PCIE2.0 x 4 SSD
25	PCIE(19)	
26	PCIE(20)	
27	PCIE(21)	
28	PCIE(22)	PCIE3.0 x 4 SSD
29	PCIE(23)	
30	PCIE(24)	

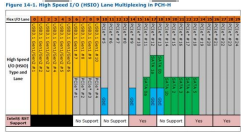
USB2.0 Setting

1	USB2(0)	
2	USB2(1)	
3	USB2(2)	USB Port1
4	USB2(3)	
5	USB2(4)	USB Port2
6	USB2(5)	USB Port3
7	USB2(6)	USB Port4
8	USB2(7)	Card Reader
9	USB2(8)	BACK WLAN
10	USB2(9)	Camera
11	USB2(10)	K-Key
12	USB2(11)	
13	USB2(12)	Tobii IR4
14	USB2(13)	TPC
15	USB2(14)	BT

Confidential

571391_CFL_H_PDG_Rev0p71, page 230

Figure 14-1. High Speed I/O (HSIO) Lane Multiplexing in PCH-H



571182_CNL_PCH_H_EDS_Vol_1_Rev_0_7, page24

Table 1-7. PCH HSIO Detail (SKU 8-11sf 11)

Pins I/O Lane	SKU		
	HH370	QM370	CM246
0	USB3.1 Gen1/Gen2	USB3.1 Gen1/Gen2	USB3.1 Gen1/Gen2
1	USB3.1 Gen1/Gen2	USB3.1 Gen1/Gen2	USB3.1 Gen1/Gen2
2	USB3.1 Gen1/Gen2	USB3.1 Gen1/Gen2	USB3.1 Gen1/Gen2
3	USB3.1 Gen1/Gen2	USB3.1 Gen1/Gen2	USB3.1 Gen1/Gen2
4	USB3.1 Gen1	USB3.1 Gen1/Gen2	USB3.1 Gen1/Gen2
5	USB3.1 Gen1	USB3.1 Gen1/Gen2	USB3.1 Gen1/Gen2
6	USB3.1 Gen1	USB3.1 Gen1	USB3.1 Gen1, PCIe*
7	USB3.1 Gen1	USB3.1 Gen1	USB3.1 Gen1, PCIe*
8	N/A	USB3.1 Gen1	USB3.1 Gen1, PCIe*
9	N/A	USB3.1 Gen1	USB3.1 Gen1, PCIe*
10	GbE	PCIe*, GbE	PCIe*, GbE
11	N/A	PCIe*	PCIe*
12	N/A	PCIe*	PCIe*
13	N/A	PCIe*	PCIe*
14	PCIe*, GbE	PCIe*, GbE	PCIe*, GbE
15	PCIe*	PCIe*	PCIe*
16	PCIe*, SATA 0A	PCIe*, SATA 0A	PCIe*, SATA 0A
17	PCIe*, GbE, SATA 1A	PCIe*, GbE, SATA 1A	PCIe*, GbE, SATA 1A
18	PCIe*, GbE, SATA 0B	PCIe*, GbE, SATA 0B	PCIe*, GbE, SATA 0B
19	PCIe*, SATA 1B	PCIe*, SATA 1B	PCIe*, SATA 1B
20	PCIe*	PCIe*	PCIe*, SATA 2
21	PCIe*	PCIe*	PCIe*, SATA 3
22	PCIe*, SATA 4	PCIe*, SATA 4	PCIe*, SATA 4
23	PCIe*, SATA 5	PCIe*, SATA 5	PCIe*, SATA 5
24	PCIe*	PCIe*	PCIe*, SATA 6
25	PCIe*	PCIe*	PCIe*, SATA 7
26	PCIe*	PCIe*	PCIe*
27	PCIe*	PCIe*	PCIe*
28	PCIe*	PCIe*	PCIe*
29	PCIe*	PCIe*	PCIe*

ASUS	Project: G703GI		Rev:
Title:	CFL-H System Setting Rl.3E		
Size:	Document Number:	Engineer:	MB SB 9022
Date: Monday, May 14, 2018		Sheet 1 of 1	

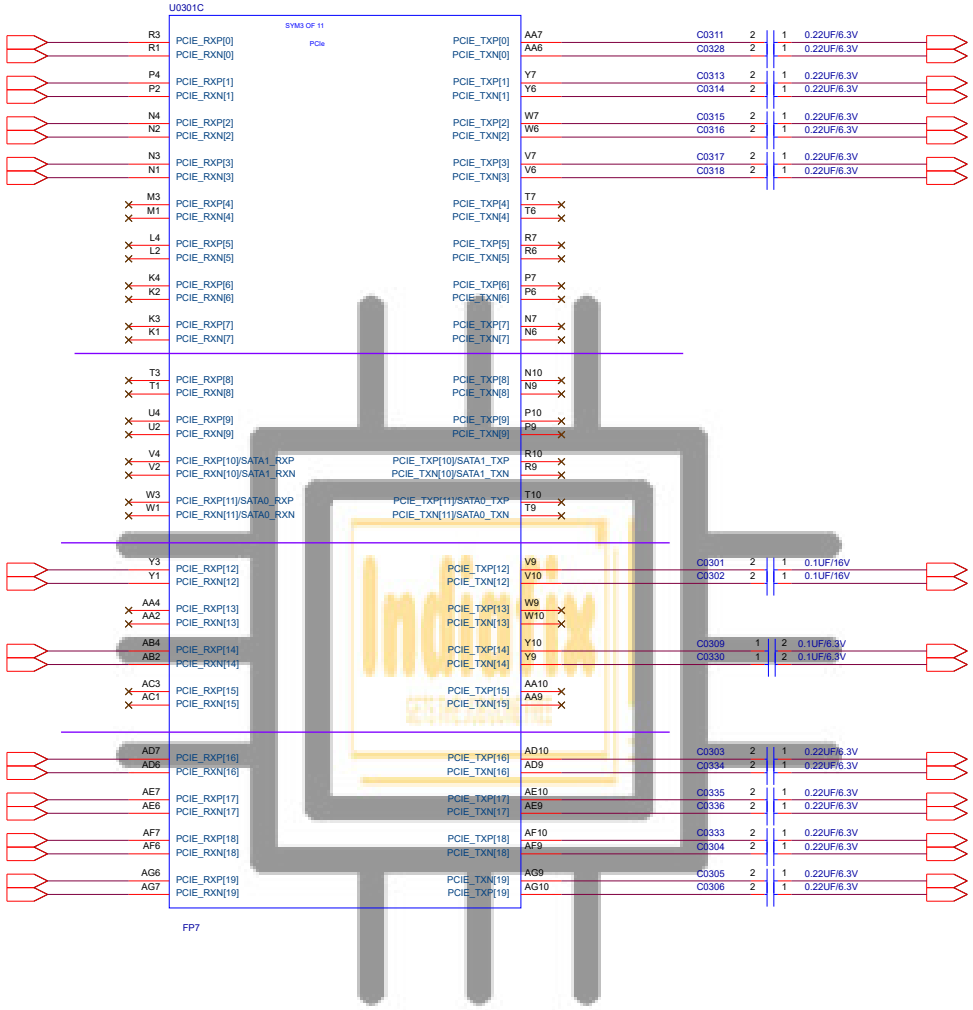
PCIE/WLAN/LAN/SSD

RX Side

TX Side

PCIENB RX [0~7]

PCIEG TX [0~7]



PCIe x1 CR1 lane 4

PCIe x1 lane 5

PCIe x1 5G lane 6

PCIe x1 WLAN lane 7

PCIe x4 SSD lane 0 / lane 1 / lane 2 / lane 3

CPU TX to GPU RX (Onboard device)
CPU RX to GPU TX
CPU TX to M.2 TX (CON.)
CPU RX to M.2 RX
CPU TX to WLAN TX (CON.)
CPU RX to WLAN RX
CPU TX to LAN RX (Onboard device)
CPU RX to LAN TX

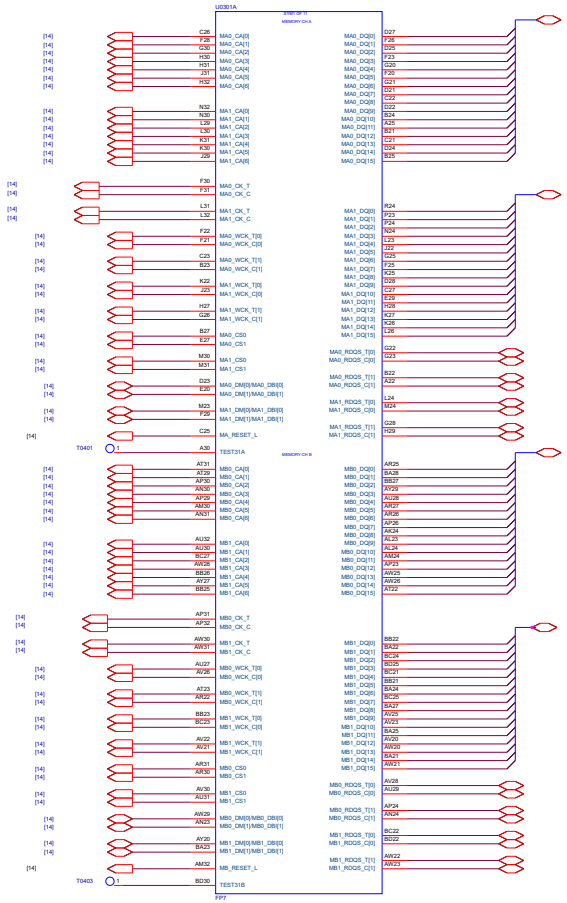
20191029, cap. selection related to PCIE speed
Gen3: with 220nF
Gen2: with 100nF

Project Name: GV301		Rev: R1.0
Title : AMD_CPU_PCIE		
Size: B	Dept.: ASUSTeK COMPUTER INC.	Engineer: NR EE RD3
Date: Wednesday, January 18, 2023	Sheet 3 of 104	

Memory Channel A/B

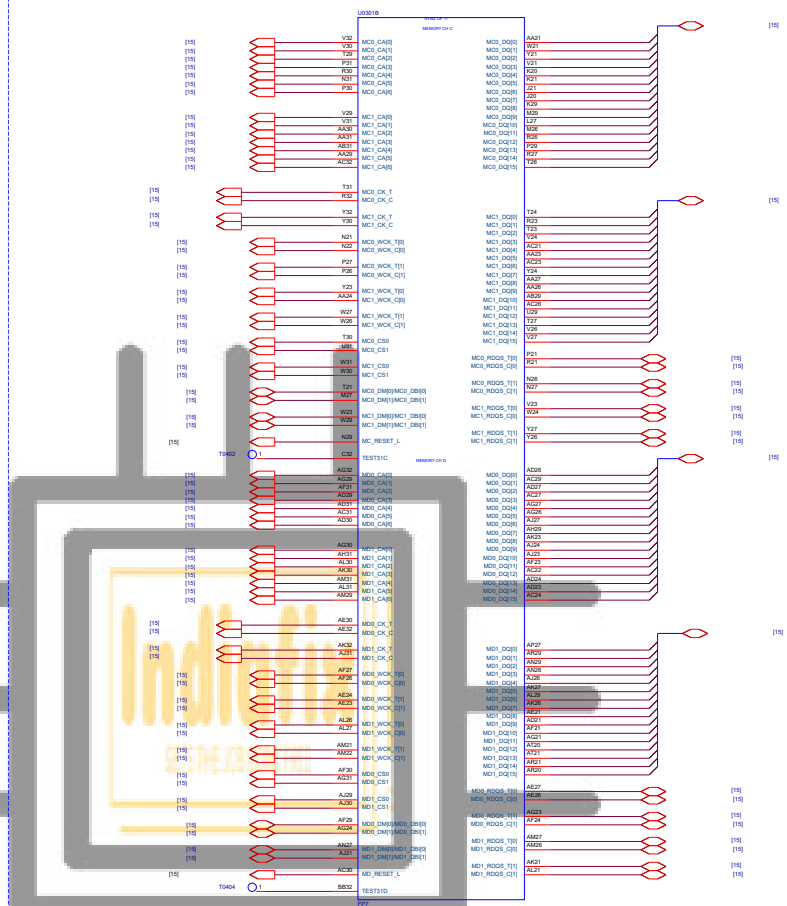
20210305 LPDDR5 DRAM down/pin swap follow MAYAN CRB

On Board LPDDR5



Memory Channel C/D

On Board LPDDR5

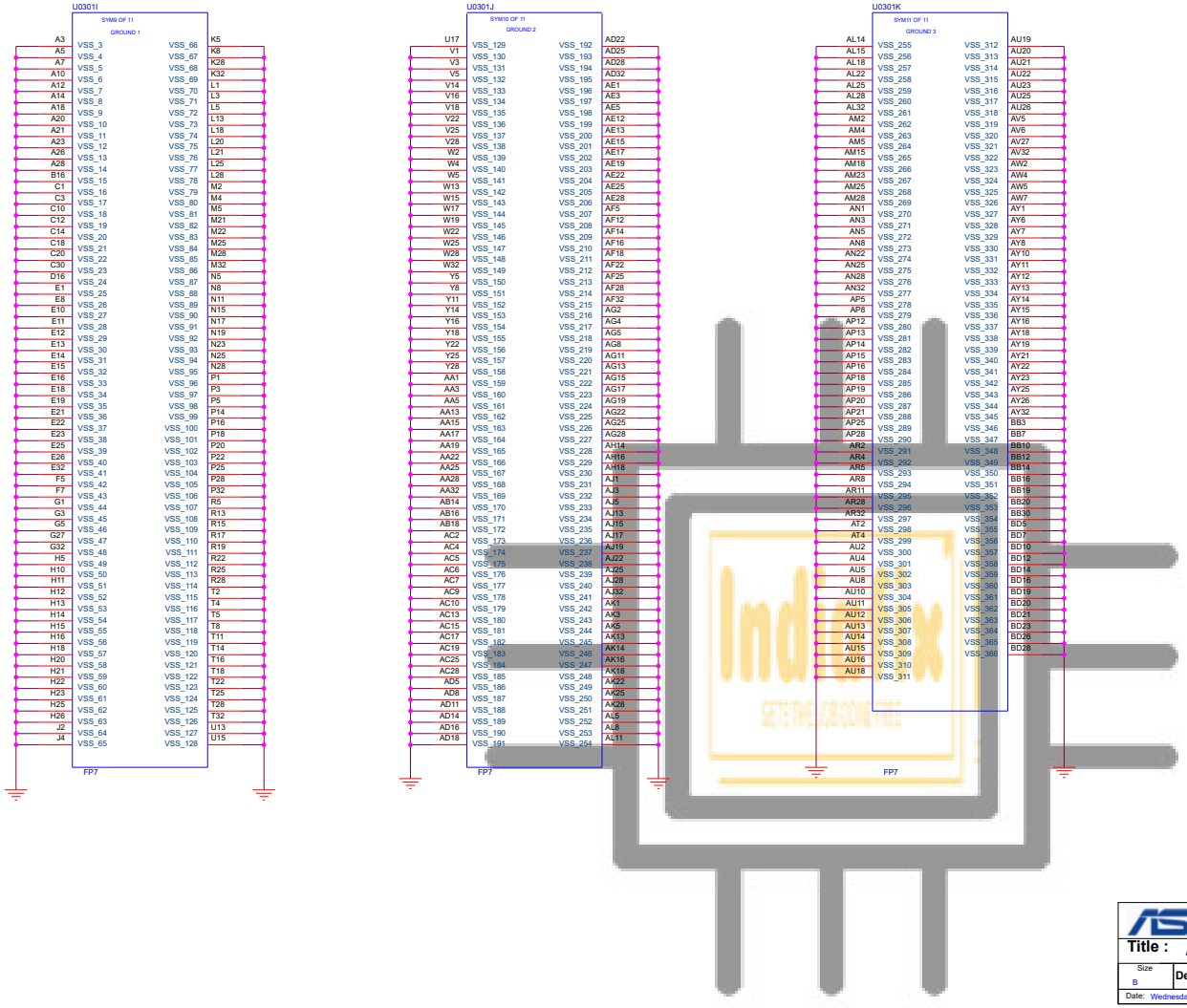


Main Board

ASUS	Project Name: GV301		Rev: R1.0
Title : AMD_CPU_DDR4			
Size: C	Dept.: ASUSTeK COMPUTER INC.	Engineer: NR EE RD3	
Date: Wednesday, January 18, 2023	Sheet 4 of 104		

Main Board

CPU_GND



Project Name: FX505DY			Rev: R1.0
Title : AMD_CPU_GND			
Size: B	Dept.: ASUSTeK COMPUTER INC.	Engineer: NR EE RD3	
Date: Wednesday, January 18, 2023	Sheet 5 of 104		

Display

Display Port TX/RX
O-IOVP-D_VDDP (S0_0.75)_FP6 (FDS)

Display Panel
O-IO18-S (S0_1.8)_FP6 (FDS)

Display Port HPD
I-IO33-S (S0_3.3)_FP6 (FDS)

eDP

LCD

DP2 GC CONN TYPE-C port A

DP3 TYPE-C port B

20200205 ashton remove R0602

PU FOR INTERNAL
PD FOR CUSTOMER

Test pin follow AMD BP Probing header

20191119 ashton remove 0ohm R0632 R0633 R0634

20200205 ashton removeC0602, C0604, C0605, C0606

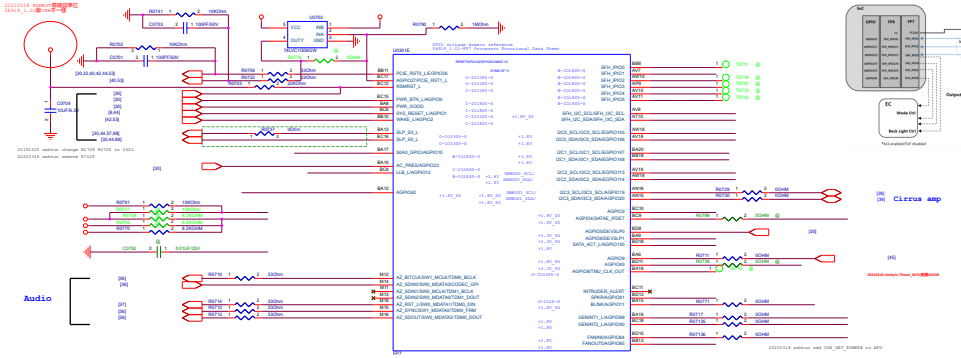
與POWER討論此PIN改由SW動作拉降頻(cost down)
故此PIN不接
P.30 &P.78 DGPU_LIMIT
@20181015E

PD RST

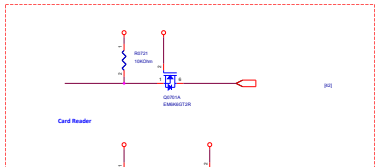
PD INT

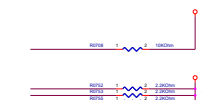
Indiafix

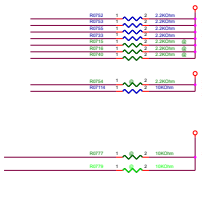
ASUS	Project Name: FX505DY	Rev: R1.0
Title :		
Size: C	Dept.: NB-SZ-RD3	Engineer:
Date: Wednesday, January 18, 2023	Sheet 6 of 104	


Audio
Cirrus amp

OD, Low active


Card Reader





CPU CLK

@20190624A

CLKREQ6_GPU# --> CLKREQ0_GPU#

CLKREQ0_SSD1# --> CLKREQ4_SSD1#
CLKREQ3_SSD2# --> CLKREQ5_SSD2#

CLKREQ1_WLAN# --> CLKREQ6_WLAN#

- GPU
- 5G
- CR2
- SSD2
- CR
- WLAN

PCIE CLK P/N
後端記得預留0 ohm

20200318 ashton add EGPU_PWROK

DGPU CLKReq#

20191119 ashton remove SL0805 SL0802

design guide pull high

WLAN_SUS_CLK Update

follow CRB reserve XTAL 156MHZ

eSPI

SPI

PCB ID1
EGPIO105

PCB ID0
EGPIO104

20210309 Chiehyin ESPI_ALERT# PU 10k 1.8VSUS follow DG CRB
20210309 Chiehyin ESPI_CS# PU 10k 1.8VSUS follow CRB
20210309 Chiehyin ESPI_RESET# PU NOT SURE

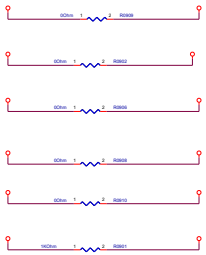
Wistron GETACHANGE

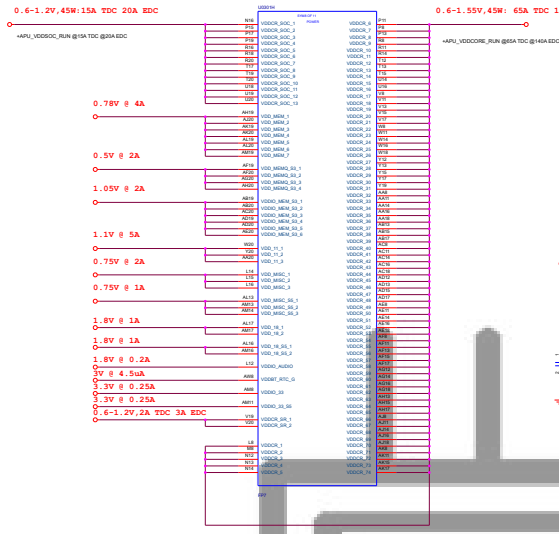
<Variant Name>

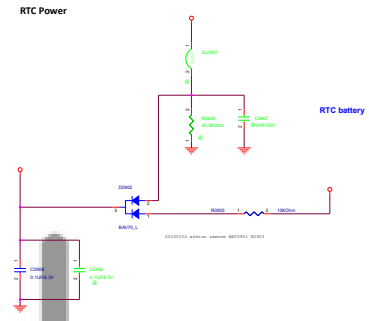
Project Name: FX505DY
Rev: R1.0
Title :
Size: Custom
Dept.: NB-SZ-RD3
Engineer:
Date: Wednesday, January 18, 2023
Sheet 8 of 104

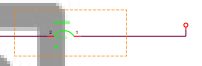
Main Board

CPU_Power









11G232210526360 (1UF/10V) change to 11G232210515320 (1UF/6.3V)
C0902
@20190705C

Indiafix
GET THE CRACKING FREE

ASUS	Project Name: FX505DY		Rev: R0.6
Title :	CFL-H_CPU_PWR(1)_R1.3B		
Size: D	Dept.: ASUSTeK COMPUTER INC	Engineer:	NR EE RD3
Date: Wednesday, January 18, 2023		Sheet: 1 of 104	

CPU_CAP

BO (Bottom side outside SOC)
BU (on bottom side under SOC)

+APU_VDD_MEM_RUN
7pcs 10UF /1pcs 1UF follow FP7 CRB
@1.0a

+VDD_ADD_ALW
2pcs 10UF /1pcs 1UF follow FP7 CRB
@1.0a

+APU_VDD_MEMQ_SUS
3pcs 10UF /3pcs 1UF follow FP7 CRB
@1.0a

+APU_VDD_11_RUN
2pcs 10UF /2pcs 1UF follow FP7 CRB
@1.0a

+3.3V_RUN
2pcs 10UF /1pcs 1UF/1pcs 180PF follow FP7 CRB
@1.0a

+3.3V_ALW
2pcs 10UF /1pcs 1UF/1pcs 180PF follow FP7 CRB
@1.0a

+1.8V_RUN
3pcs 10UF /1pcs 1UF/1pcs 180PF follow FP7 CRB
@1.0a

+1.8V_ALW
3pcs 10UF /1pcs 1UF/1pcs 180PF follow FP7 CRB
@1.0a

+VDD_MISC_RUN
6pcs 10UF /6pcs 1UF/1pcs 180PF follow FP7 CRB
@1.0a

+VDD_MISC_ALW
3pcs 10UF /2pcs 1UF follow FP7 CRB
@1.0a

+VDDBT_RTC
1pcs 0.22uF / 1pcs 1uF

+APU_VDDCR_SR_RUN
3pcs 10UF /2pcs 1UF follow FP7 CRB
@1.0a

BO (Bottom side outside SOC)
BU (on bottom side under SOC)

+APU_VDDCORE_RUN
40pcs 10UF /1pcs 180PF follow FP7 CRB
@1.0a

All BU(on bottom side under SOC)

20191120 ashton change MLCC 220F 11G233222625150 to 11203-0071F000

2022/4/28 Kent modify SI suggest

+APU_VDDSOC_RUN
26pcs 10UF /1pcs 1UF/1pcs 180PF follow FP7 CRB
@1.0a

All BU(on bottom side under SOC)

2022/5/11 Kent modify SI suggest

+APU_VDDIO_SUS
8pcs 10UF /2pcs 1UF follow FP7 CRB
@1.0a

All BU(on bottom side under SOC)

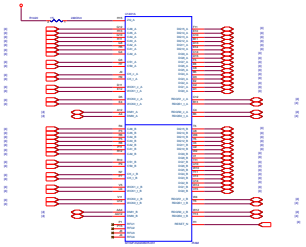
If the VSS plane is cut to create a VDDIO_MEM_S3 plane, ceramic capacitors with NP0 or C0G dielectric are connected across the VDDIO_MEM_S3 and VSS plane split.

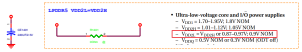
Indiafix

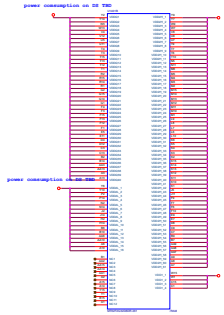
ASUS	Project Name: FX505DY	Rev: R3.0
Title : CFL-H_CPU_PWR(1)_R1.3B		
Size: D	Dept.: ASUSTeK COMPUTER INC	Engineer: NR EE RD3
Date: Wednesday, January 18, 2023	Sheet: 10 of 104	

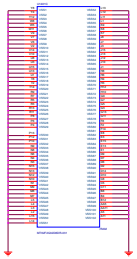
LPDDR5 Channel A

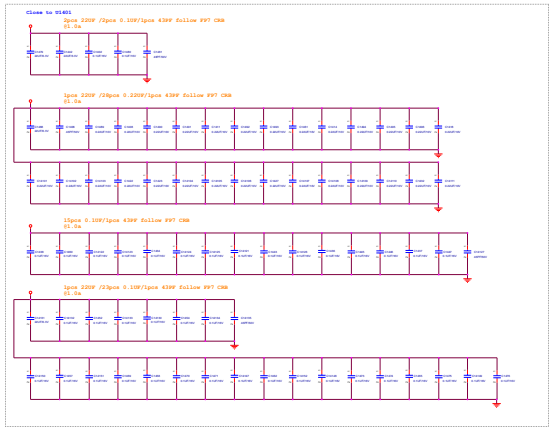
DRAM0





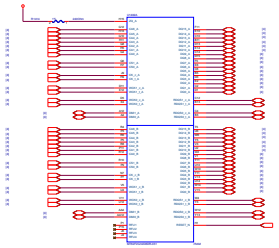


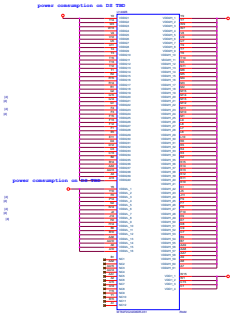


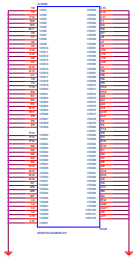


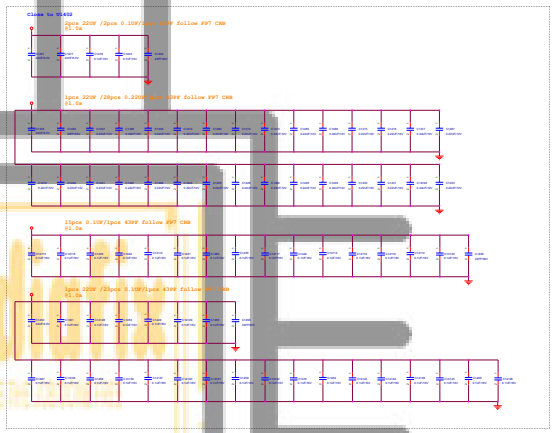
LPDDR5 Channel B

DRAM1





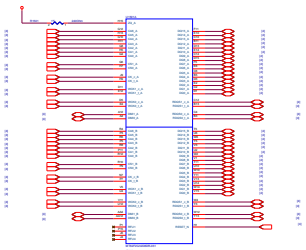






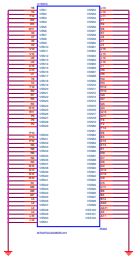
LPDDR5 Channel C

DRAM2

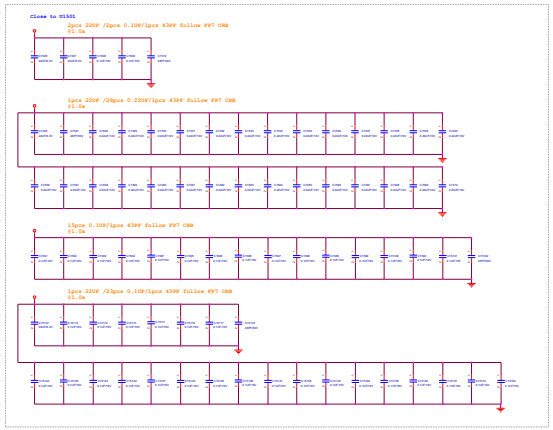


power consumption on DS TBD



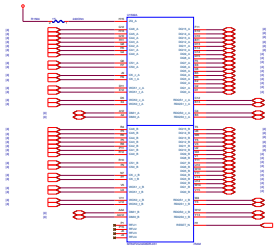


Close to U1501

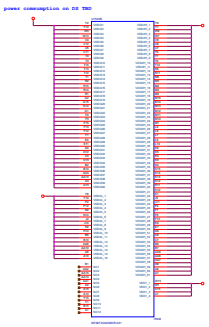


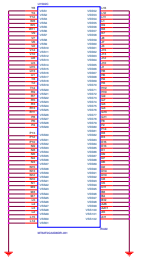
LPDDR5 Channel D

DRAM3

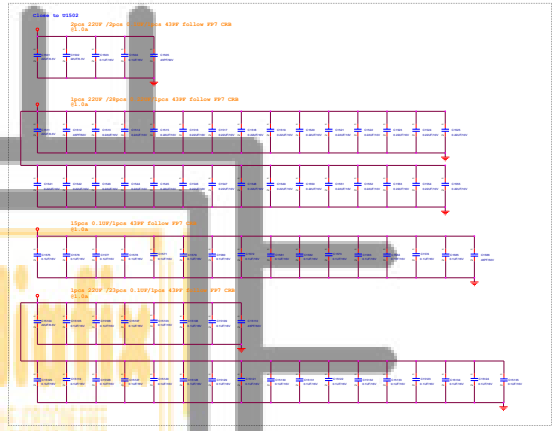


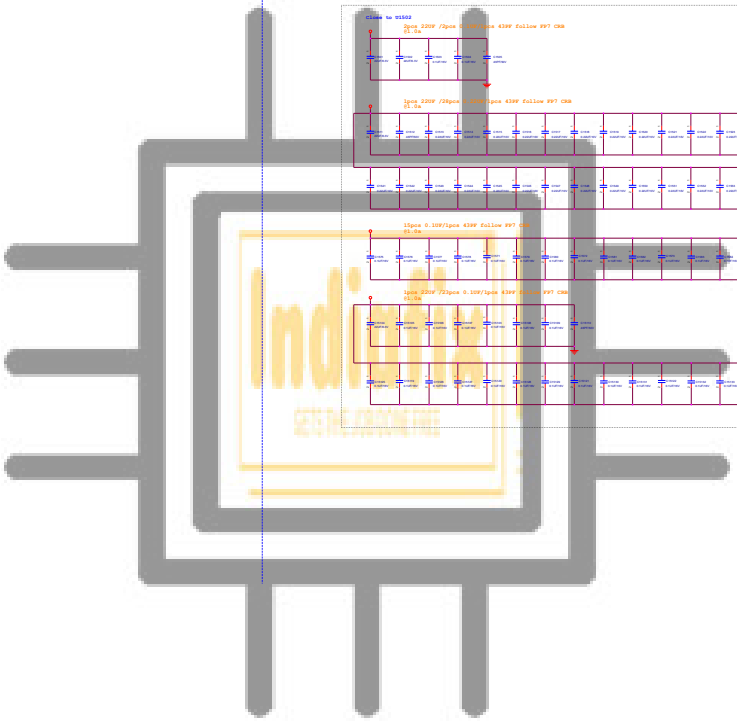
power consumption on DS TBD





Close to U1502





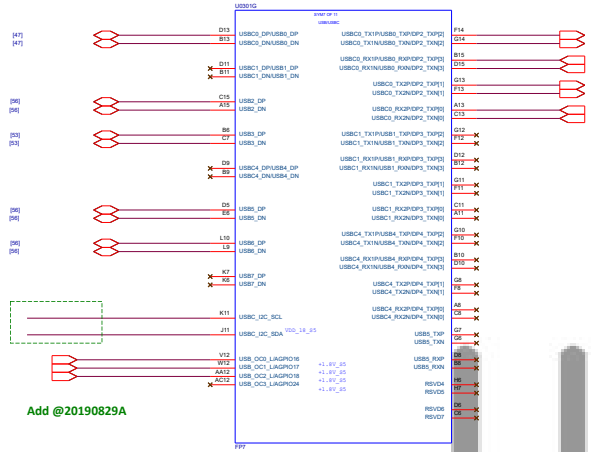


CPU_USB

20210305 CHIENYIN change USB2.0 port , USBC0 for USBC port A, USBC1 for USBC portB(USBC0/1 support USB4.0), USB5 for typeA

USB3.0 TX/RX
I/O-IOVPS5-D_VDDP (S5_0.75)_FP6 (FDS)

USB Port0	Type-C Port0
USB Port1	Type-C Port1
USB Port2	
USB Port3	BT
USB Port5	Type-A Port5
USB Port6	FP



USB3.2 Gen2/USB4.0 USB-C Poart A

USB3.2 Gen2/USB4.0 USB-C Poart B

USB3.2 Gen2 Type-A

AMD Design check

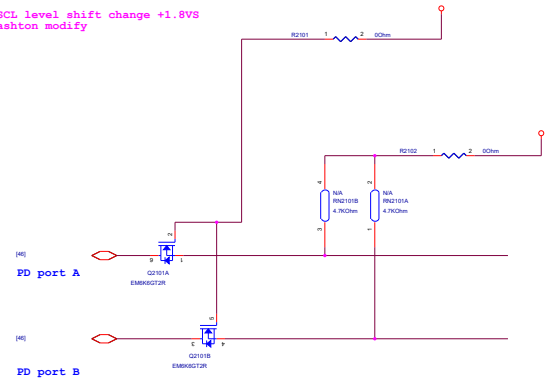
AGPIO13. If unused, enable internal pull down by software.



Add @20190829A

USB Port 0,1,4,5
USB-A USB3.2 Gen1(5Gbps): 304.8mm (12000mil)
USB-C USB3.2 Gen1(5Gbps): 177.8mm (7000mil)
USB-A & USB-C USB3.2 Gen2(10Gbps): 152.4mm (6000mil)

USBC_SDA/SCL level shift change +1.8VS
20200901 ashton modify

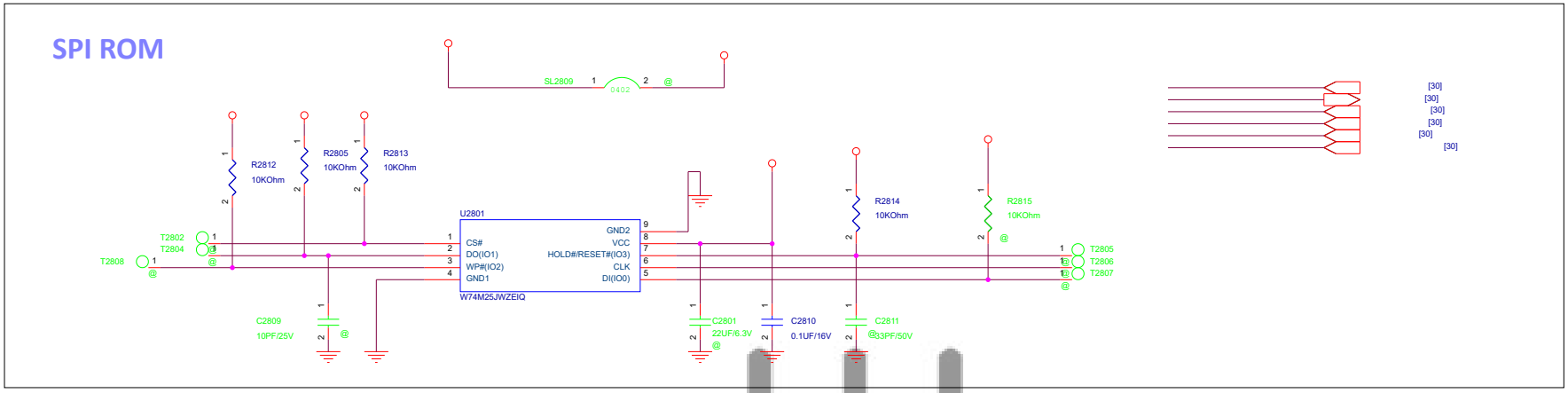


reserve I2C3 Isolater follow CRB 20211025

<Variant Name>

ASUS	Project Name: FX505DY	Rev: R1.0
Title :		
Size: C	Dept.: NB-SZ-RD3	Engineer:
Date: Wednesday, January 18, 2023	Sheet 21 of 104	

SPI ROM

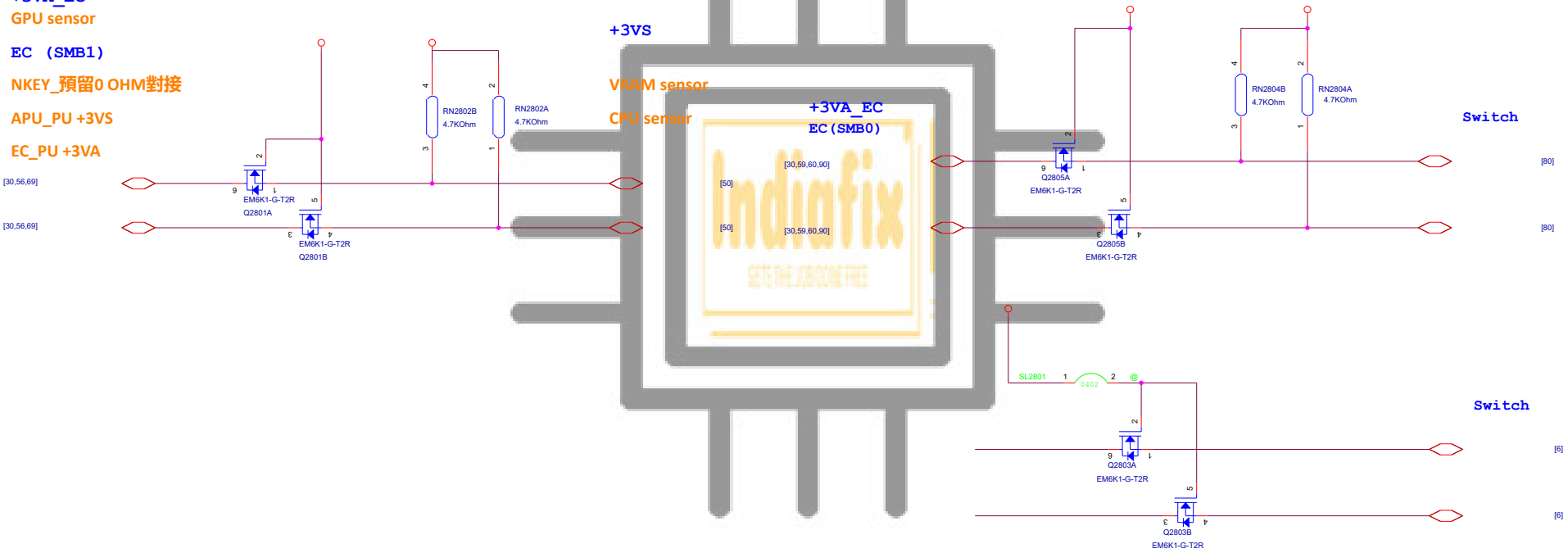


+3VA_EC
GPU sensor

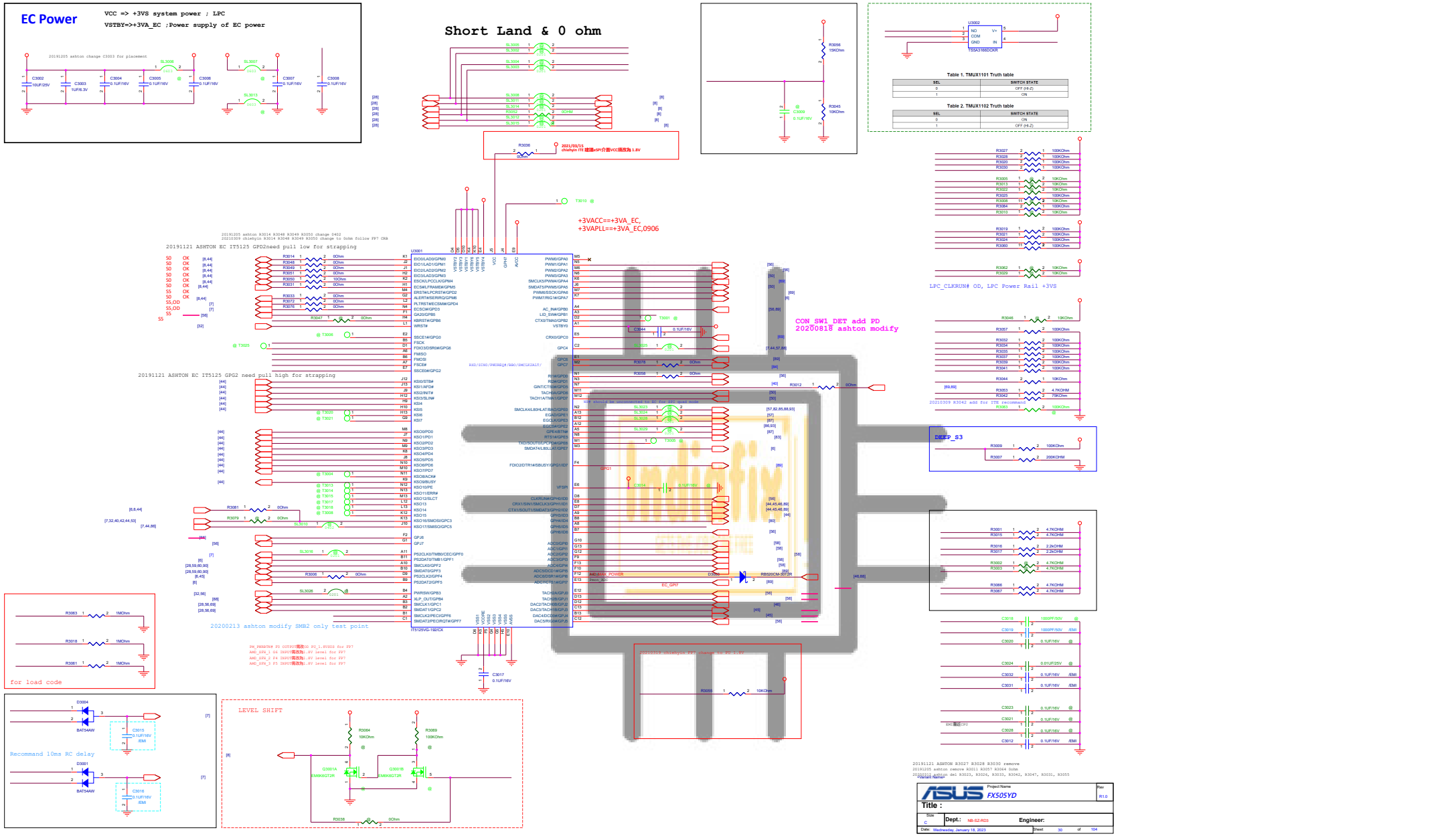
EC （SMB1）
NKEY_預留0 OHM對接
APU_PU +3VS
EC_PU +3VA

+3VS
VRAM sensor
CPU sensor

+3VA_EC
EC（SMB0）



ASUS	Project Name: GX531GX	Rev: 1.0
Title : PCH-SPI ROM,OTH		
Size: B	Dept.: ASUSTeK COMPUTER	Engineer: NR EE RD3
Date: Wednesday, January 18, 2023	Sheet 28 of 104	

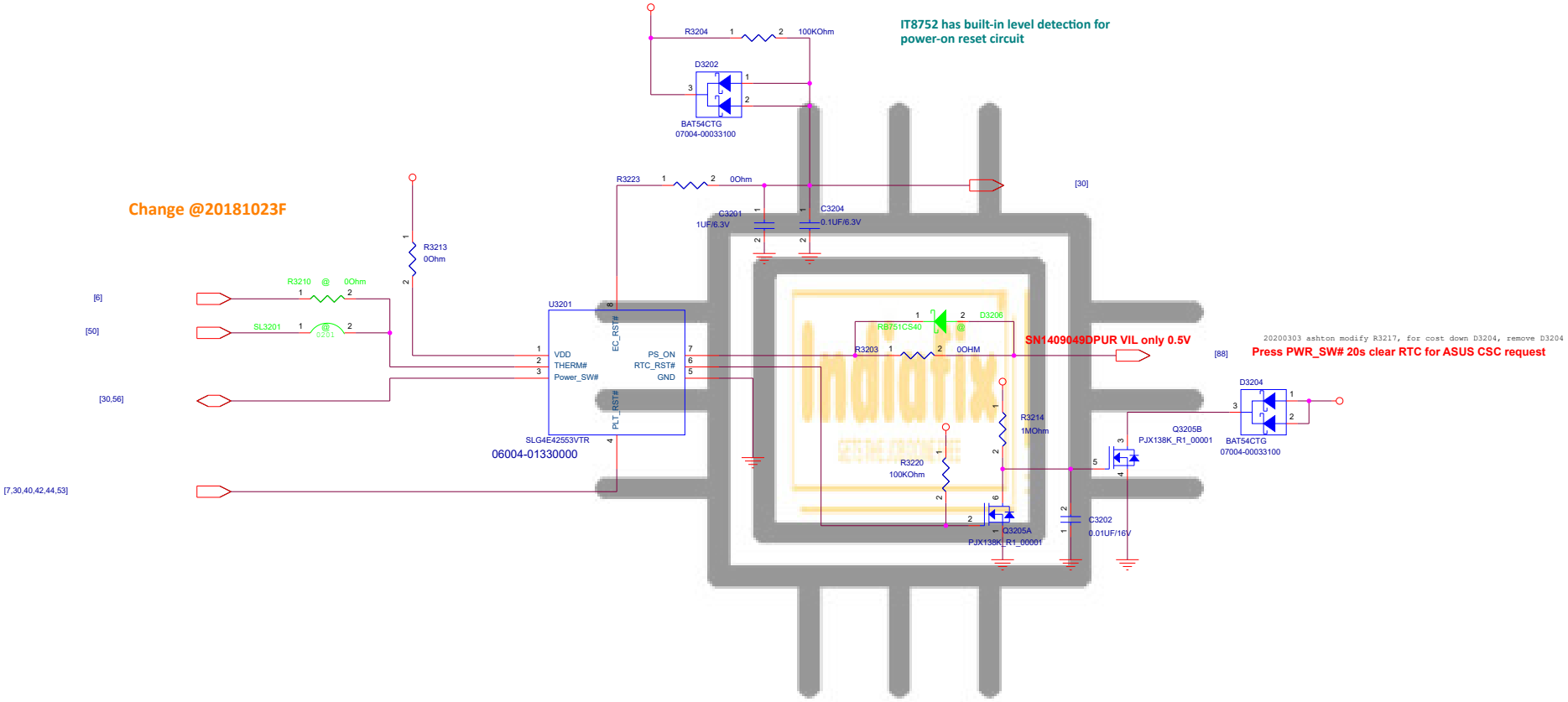

EC Power
VCC => +3VS system power ; LPC
VSTBY=>+3VA_EC ;Power supply of EC power
Short Land & 0 ohm
+3VACC==+3VA_EC,
+3VAPLL==+3VA_EC,0906
20191121 ASHTON EC IT5125 GPD2need pull low for strapping
20191121 ASHTON EC IT5125 GPG2 need pull high for strapping
CON_SW1_DET add PD
20200818 ashton modify
LPC_CLKRUN# OD, LPC Power Rail +3VS
DEEP_S3
20200213 ashton modify SMB2 only test point
for load code
Recommand 10ms RC delay
LEVEL SHIFT
Table 1. TMUX1101 Truth table
Table 2. TMUX1102 Truth table
20191121 ASHTON R3027 R3028 R3030 remove
20191205 ashton remove R3011 R3057 R3044 5ohm
20200212 ashton del R3023, R3026, R3033, R3042, R3047, R3031, R3055
ASUS
FX505YD
Title :
Engineer:
Wednesday, January 18, 2023

Modern standby project should use Silego solution for EC/RTC reset (Microsoft hardware requirements)

6.6.2 Power button behavior

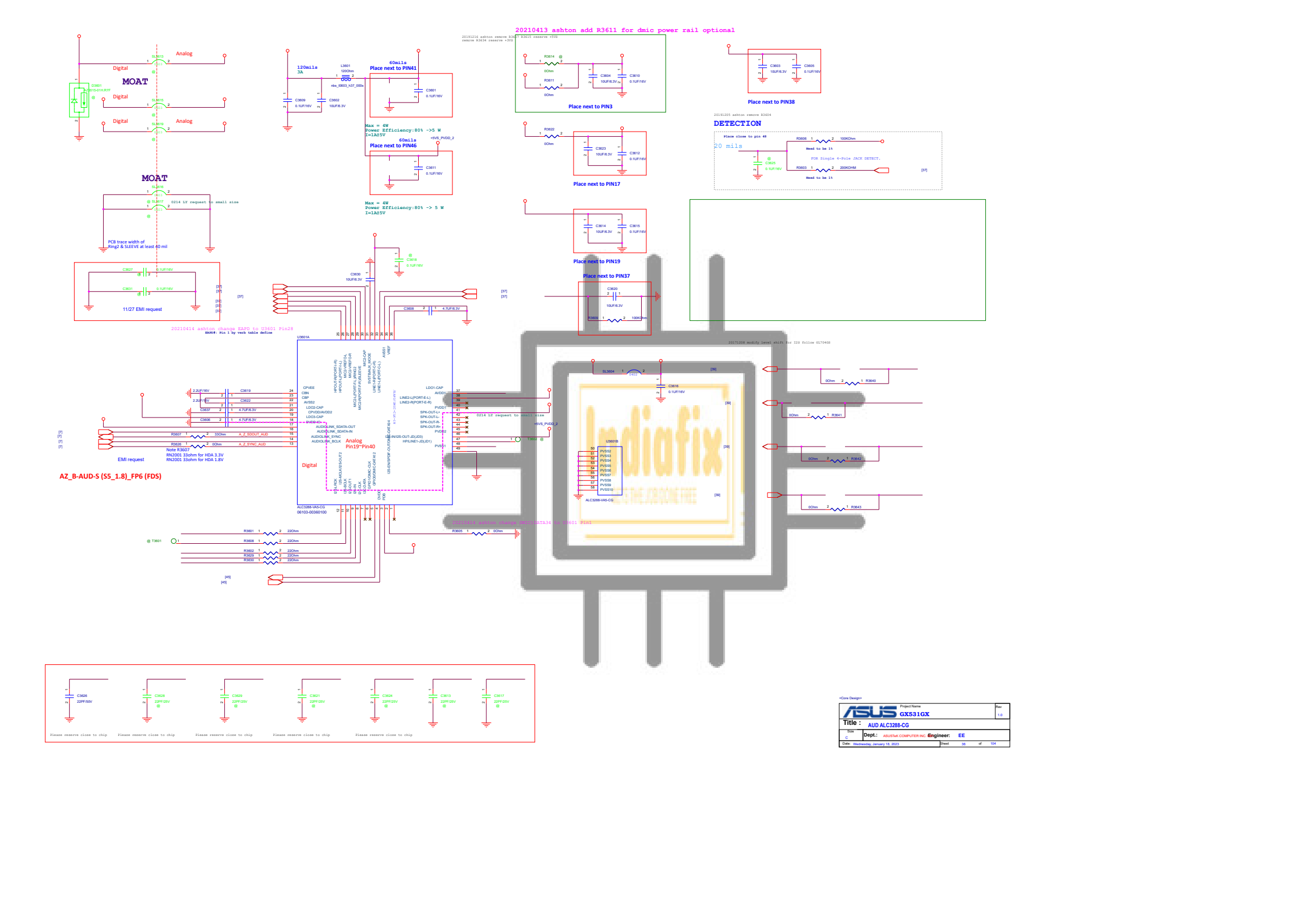
https://docs.microsoft.com/en-us/windows-hardware/design/minimum/minimum-hardware-requirements-overview#section-60---shared-minimum-hardware-requirements-for-components

UX362FA R1.3 board will verify this circuit 7/E



<Variant Name>

ASUSTeK COMPUTER INC. NB4	Title :	RST_Reset Circuit
	Engineer:	NR EE RD3
Size: B	Project Name: FX505DY	Rev: R1.0
Date: Wednesday, January 18, 2023	Sheet 32 of 104	



audiofix

<core_design>
Project Name: GX531GX
Rev: 1.0
Title : AUD ALC3288-CG
Size: C
Dept.: ASUSTeK COMPUTER INC.
Engineer: EE
Date: Wednesday, January 18, 2023
Sheet 36 of 104
</core_design>

*** POWER

** [P.36]

** [P.30]

** [P.36]

*** SINGAL

** Line to Codec [36]

[36]
[36]

** Headset from Codec [36]

[36]
[36]
[36]
[36]
[36]

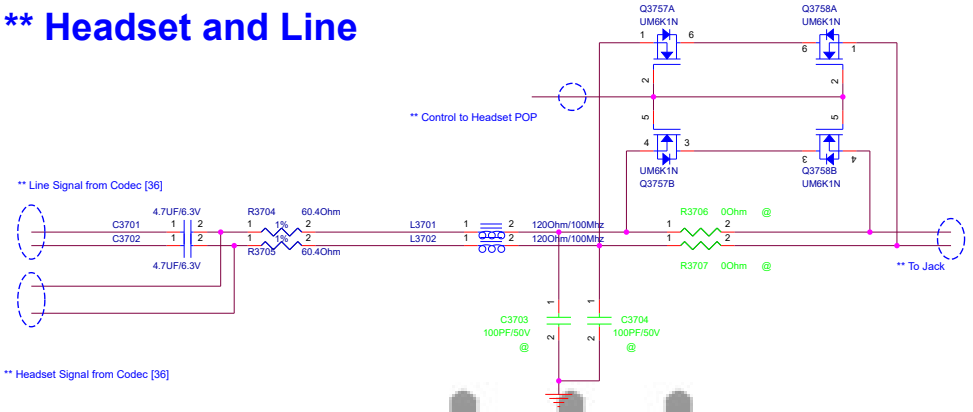
** Control Pin from Codec [36]

[36]
[36]

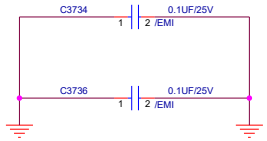
** Control Pin from APU [7]

[7]

** Headset and Line

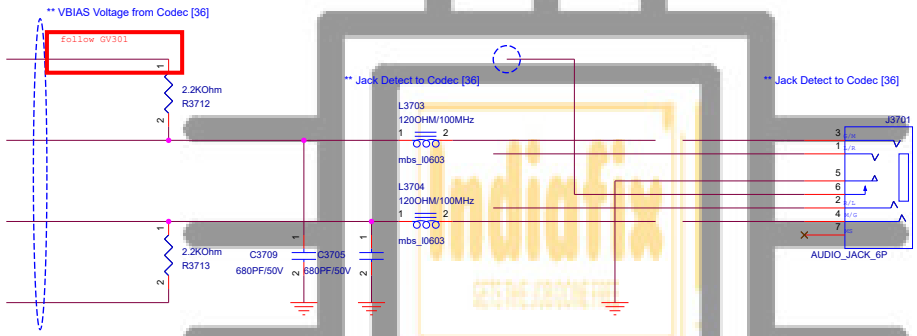


** A_GND / GND

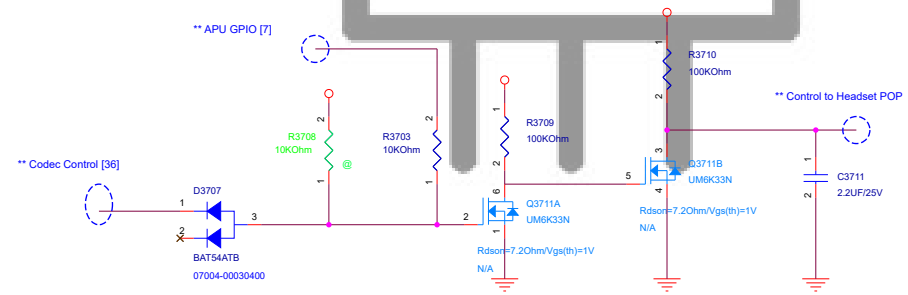


** Jack and MIC

Add more than 100pf cap for Headset detect



** TVS

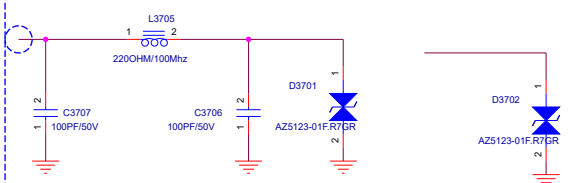


20210413 ashton change Q3711 Vgs=1V for DVDD 1.8V

** TVS

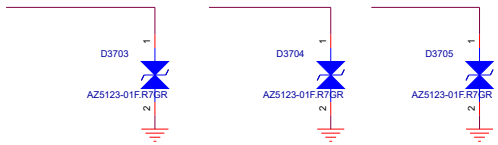
HP & MIC Connector

HP ESD Protect

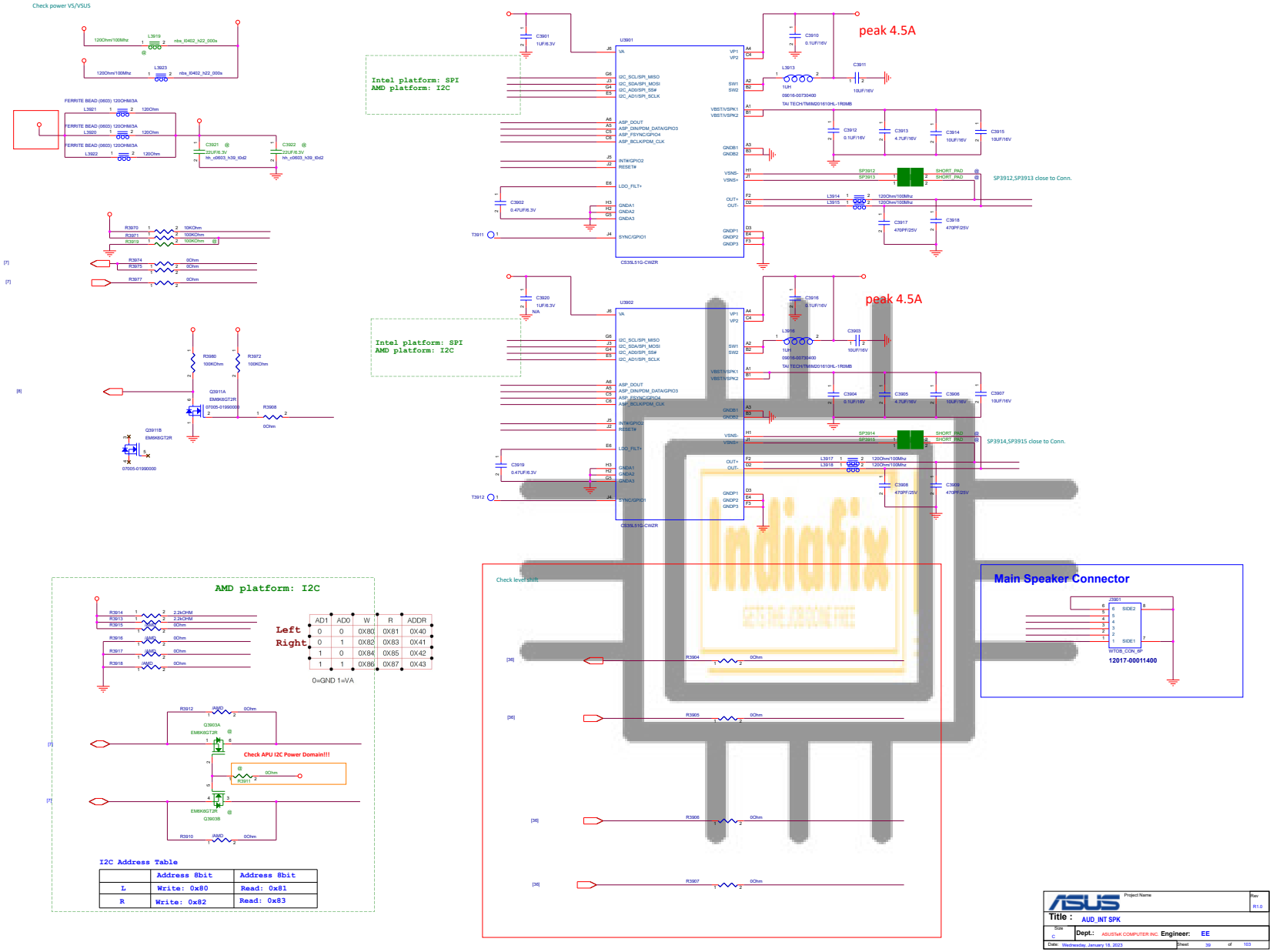


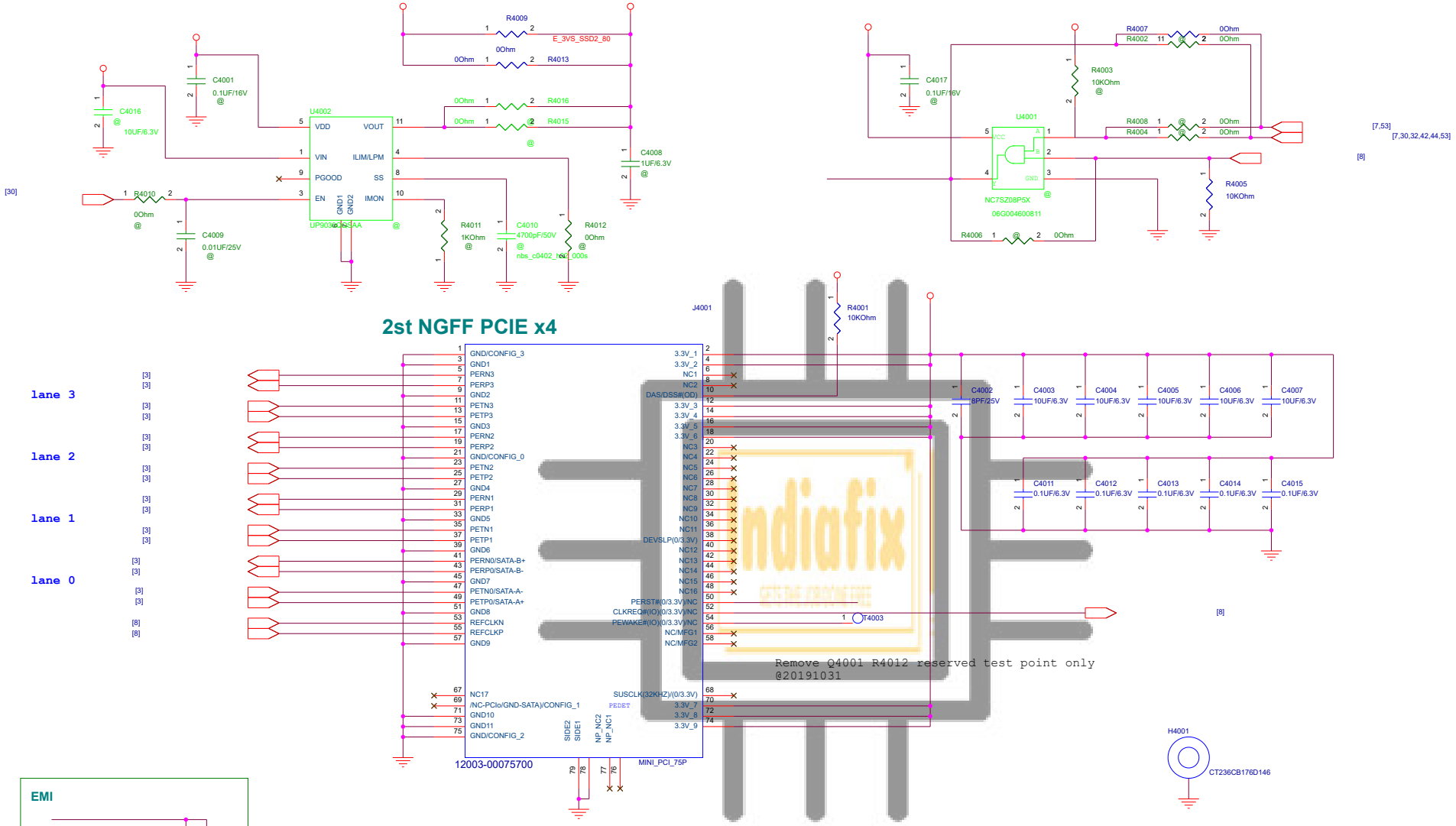
Design IP 0419 選用 07024-01152200 EOL
故使用 07024-01152200 ESD

EXTERNAL MICROPHONE



ASUS	Project Name	Rev R1.2
Title : AUD_EXT Jack		
Size B	Dept.: ASUSTeK COMPUTER INC.	Engineer: EE
Date: Wednesday, January 18, 2023	Sheet 37 of 104	


Check power VS/VSUS
Intel platform: SPI
AMD platform: I2C
peak 4.5A
SP3912,SP3913 close to Conn.
Intel platform: SPI
AMD platform: I2C
peak 4.5A
SP3914,SP3915 close to Conn.
Check level shift
Main Speaker Connector
12017-00011400
AMD platform: I2C
Left
Right
0=GND 1=VA
Check APU I2C Power Domain!!!
I2C Address Table
Address 8bit
Address 8bit
L
Write: 0x80
Read: 0x81
R
Write: 0x82
Read: 0x83
Indiafix
Title : AUD_INT SPK
Engineer: EE
Wednesday, January 18, 2023



Project Name: GX502GX		Rev: 1.0
Title : NGFF_SSD_PCIE_CON		
Size: B	Dept.: ASUSTeK COMPUTER	Engineer: NR EE RD3
Date: Wednesday, January 18, 2023	Sheet 40 of 104	

Power

Power_3.3V

UHS-II consuption
+3VS_CR : 238mA+0.8A=1.1A

Power_3.3V

Power_1.2V

Safety

EMI

close to connector

ESD

Micro SD Connector

Reset

GL9755

Support RTD3

M 06108-00260000 GL9755-06
S 06108-00250000 RTS5250S

SD WP if not in use
please connect to GND directly

Pull UP

SD HW option

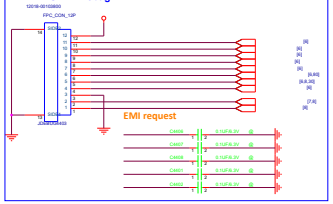
To GL9755 (GD603)

To GL9755 (GD603)

To GL9755 (GD603)

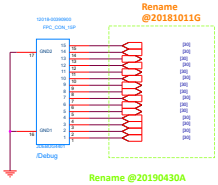
ASUS	Project Name: GU603Z		Rev: R1.0
Title : CR_GL9755			
Size: Custom	Dept.: ASUSTeK COMPUTER INC.	Engineer: EE	
Date: Wednesday, January 18, 2023	Sheet 42 of 103		

HDT + UART Debug



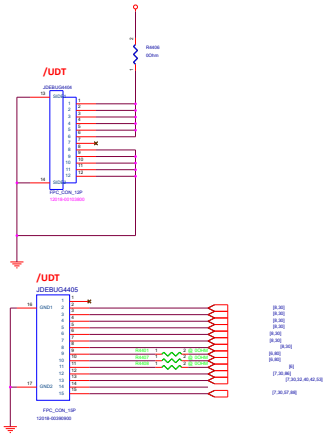
2017/11/10 **Flash BIOS**

Rename @20181011G

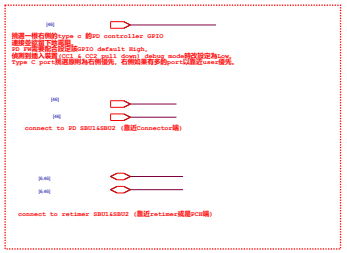


Rename @20190430A

UDT Debug Board



Interface



Debug switch

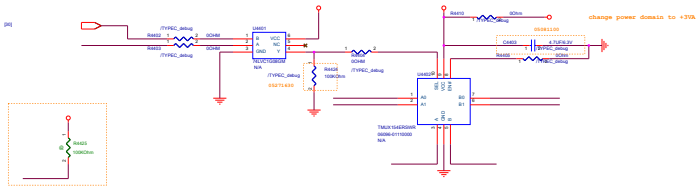
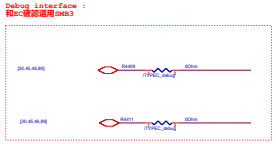


Table 1. Truth Table

SEL	EN	FUNCTION
X	H	Disconnect
L	L	A = A0 B= B0
H	L	A = A1 B = B1

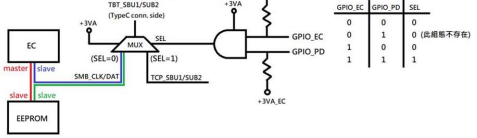
Debug interface :
和EC確認選用SMB3



TYPE-C connector和Hayden要件 : 目前BU12為固定位置
- 不Disable時 : 開始轉給預期用TOP/BOT腳所連接至
- 才能達到Label正確為正確方向

TOP腳位 : SBU1: DAT, SBU2: CLK
BOT腳位 : SBU1: CLK, SBU2: DAT

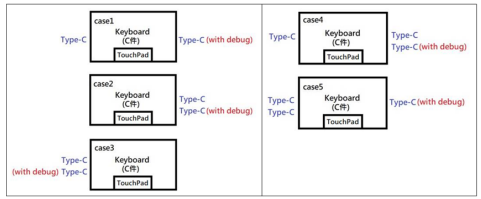




GPIO_EC	GPIO_PD	SEL
0	0	0
0	1	0 (此組態不存在)
1	0	0
1	1	1

Reserve for bypass MUX --> HDMI mode

R1.1 0505
use_GPIO0_RPDB to control debug mode (EC SMB3)
SMB2->VGA7 SMB1->CLK
L debug
H Normal



EEPROM提供兩種Package選用，依據各專案Layout空間自行挑選。

05002-00120400

05002-00120900
05002-00121300

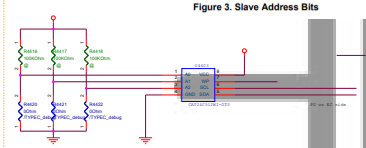
EC_EEPROM

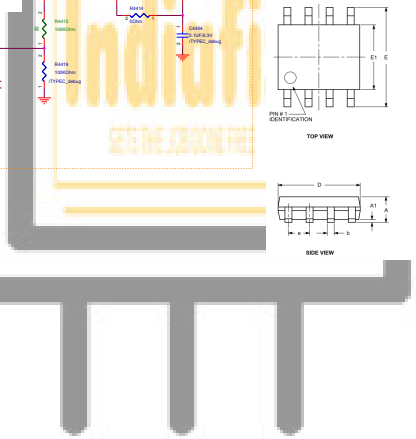
換料號和EC 確認address setting

1	0	1	0	A_2	A_1	A_0	R/W

DEVICE ADDRESS

Figure 3. Slave Address Bits





PACKAGE DIMENSIONS

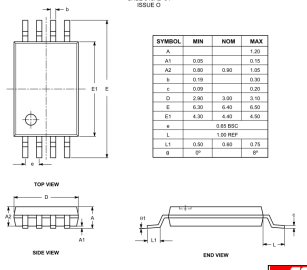
SOIC 8, 150 mils
CASE 751BD-01
ISSUE O

SYMBOL	MIN	NOM	MAX
A	1.35		1.75
A1	0.10		0.25
b	0.33		0.51
c	0.19		0.25
D	4.80		5.00
E	5.80		6.20
E1	3.80		4.00
e		1.27 BSC	
h	0.25		0.50
L	0.40		1.27
θ	0°		8°

PACKAGE DIMENSIONS

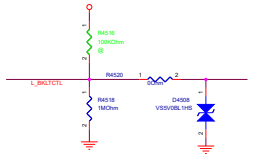
TSSOP8, 4.4x3
CASE 948AL-01
ISSUE O

SYMBOL	MIN	NOM	MAX
A			1.20
A1	0.05		0.15
A2	0.80	0.90	1.05
b	0.19		0.30
c	0.09		0.20
D	2.90	3.00	3.10
E	6.30	6.40	6.50
E1	4.30	4.40	4.50
e	0.65 BSC		
L	1.00 REF		
L1	0.50	0.60	0.75
θ	0°		8°

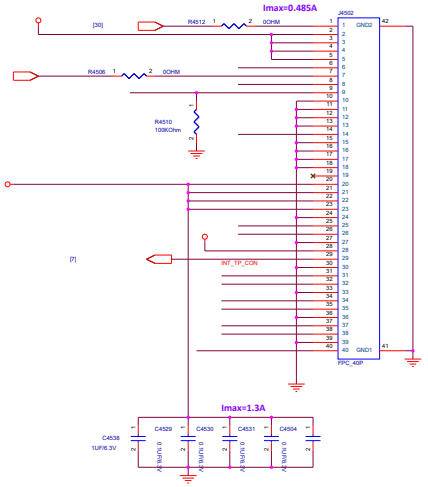


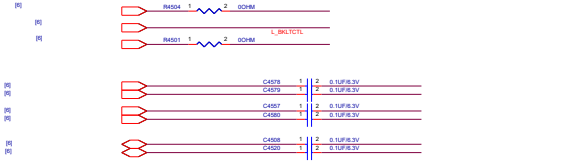
Title :	DEBUG_LPC
Engineer:	NR EE RD3
Size: Custom	Project Name: GX502GX
Date: Wednesday, January 16, 2019	Sheet 46 of 104

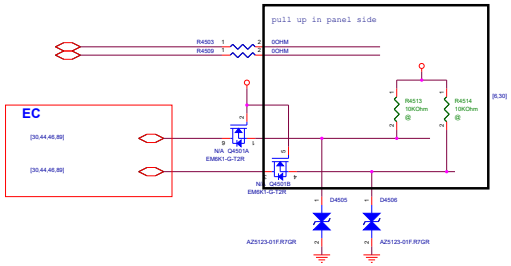
eDP_BL PWM



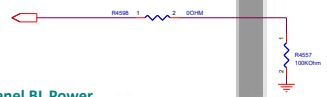
eDP Panel Conn.



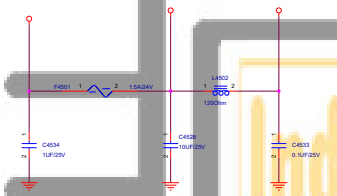




eDP_HPD (CPU)

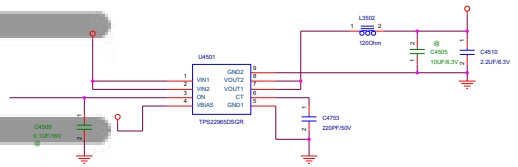


Panel BL Power

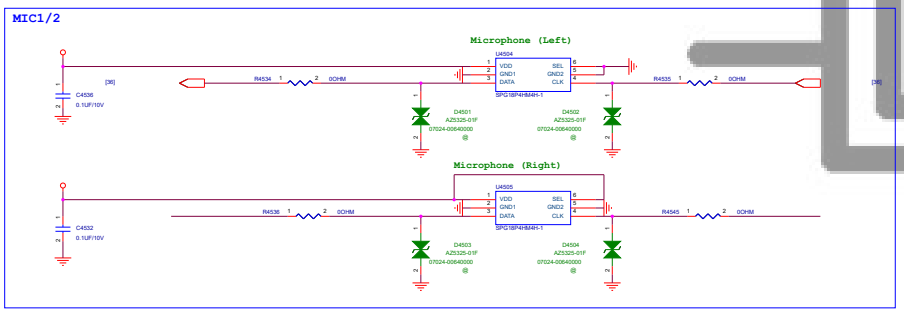


Display Panel
O-IO18-S (S0_1.8)_FP6 (FDS)

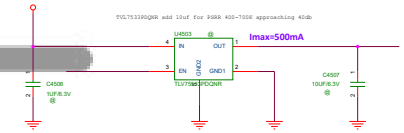
+3VS_LCD [For Panel]







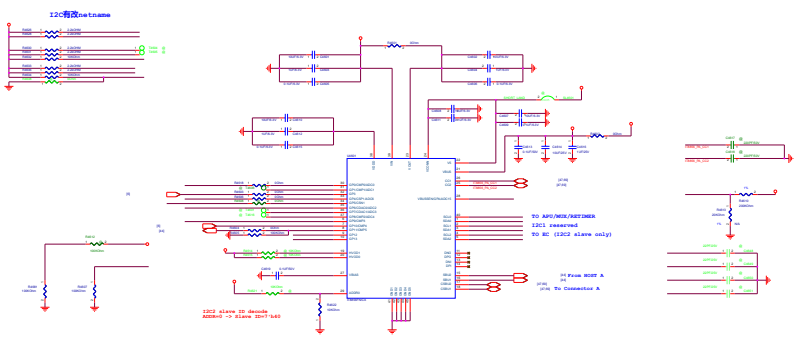
+3VSUS_TP [For Touch]

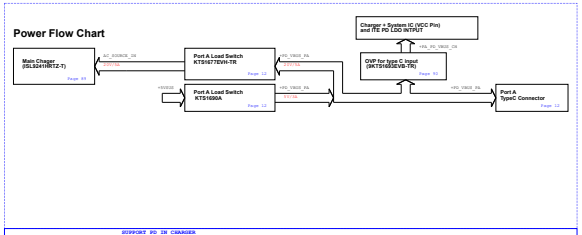


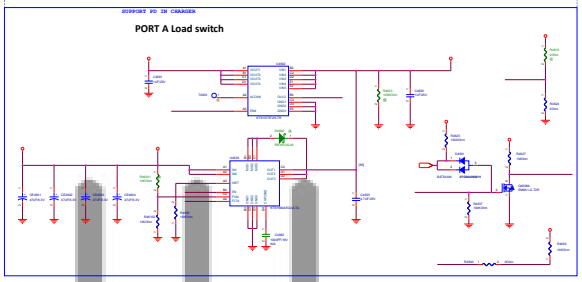


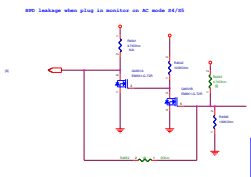
<Core Design>

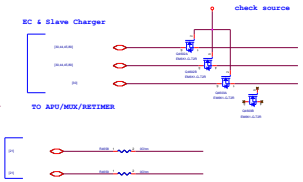
ASUSTeK COMPUTER	Title : CRT_eDP	
	Engineer: NR EE RD3	
Size: Custom	Project Name: GX502GX	Rev: 1.0
Date: Wednesday, January 18, 2023	Sheet 45 of 104	


I2C2 Pin netname
TO APU/MUX/RETIMER
I2C1 reserved
TO EC (I2C2 slave only)
I2C slave ID decode
ADDR=0 -> Slave ID=7'b40


Power Flow Chart
Main Charger (ISL9241IRTZ-T)
Port A Load Switch KTS1677EVH-TR
Port A Load Switch KTS1650A
Charger + System IC (VCC Pin) and ITE PD LDO INPUT
OVP for type C Input (SKTSH6SEVB-TR)
Port A Type-C Connector


SUPPORT PD IN CHARGER
PORT A Load switch


EMC leakage when plug in monitor on AC mode S4/S5


EC & Slave Charger
check source
TO APU/MUX/RETIMER

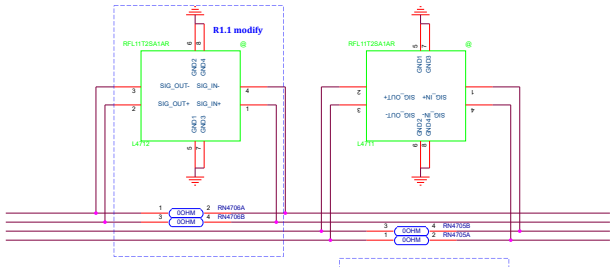

Options by DP/Retimer Host Function
PD SINK connect reserve for debug

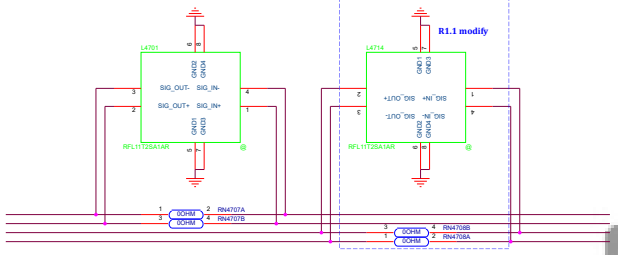

TITLE : PD_ITE83X

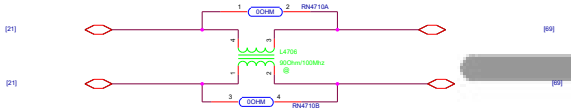
USB3.0 EMI

EMPASS footprint 调用
Check上线：nbs_filter_6p_001c
排阻上线：nbs_filter_6p_p4_001c

20191202 ashton L4711 / L4712 swap







NOTE 8.PIN ASSIGNMENT (FRONT VIEW)

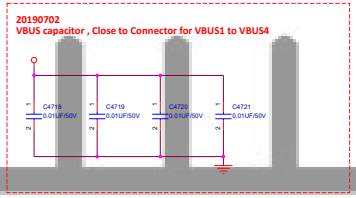
Pin No.	A1	A2	A3	A4	A5	A6	A7	A8	A9	A10	A11	A12
	GND	TX1+	TX1-	V_{BUS}	CC1	D+	D-	SBU1	V_{BUS}	RX2-	RX2+	GND
	GND	RX1+	RX1-	V_{BUS}	SBU2	D-	D+	CC2	V_{BUS}	TX2-	TX2+	GND
Pin No.	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1

NOTE 9.LASER WELD POINTS MAY BE DISCOLORED.

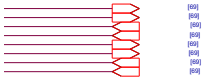
Irat=6A / 0603



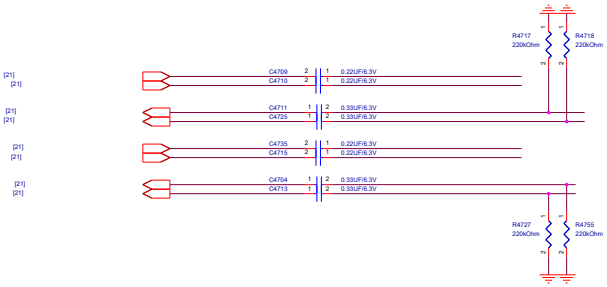
20190702
VBUS capacitor , Close to Connector for VBUS1 to VBUS4



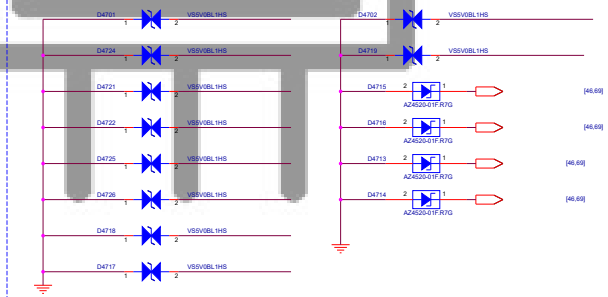
TYPE-C Connector



TYPE-C PORT A



USB3.2/USB2.0/SBU/CC
ESD-Protection





ASUS	Project Name: Type-C_ALT_USB3-10G + DP + PD + Slave_Charger_TI		Rev: 2.0
Title :	USB3.1 TYPE-C+TI 1046A		
Size: C	Dept.: ASUSTeK COMPUTER INC.	Engineer:	NR EE RD3
Date: Wednesday, January 18, 2023	Sheet 47 of 104		

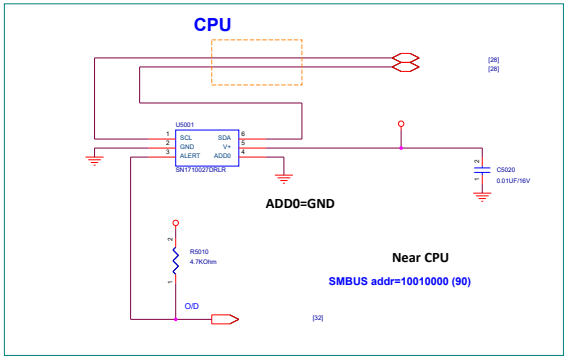
Thermal Sensor : SN170027

ALERT/SDA/SCL: Open-drain output; pullup resistor 5Kohm

Pin function Supply voltage.: 1.62 V to 3.6 V

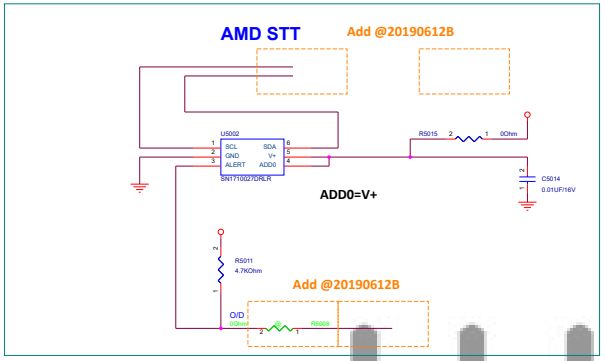
power rail : 3.3V

SMBUS1 to EC

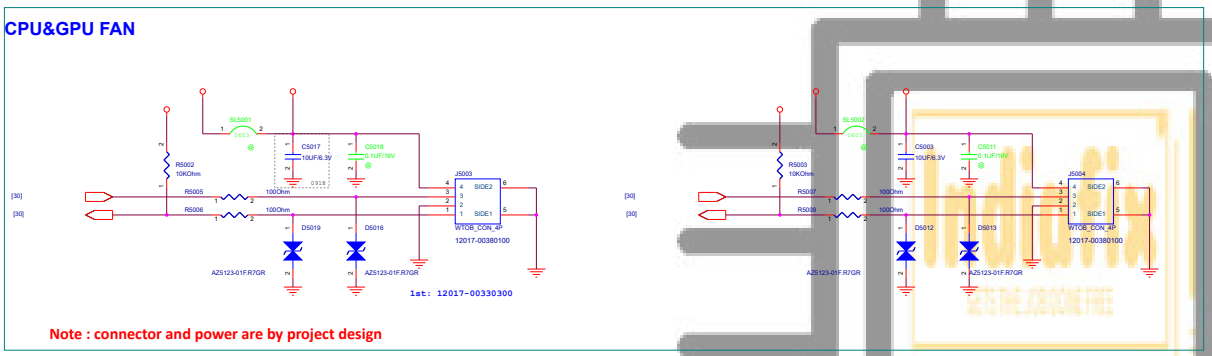


Near CPU

SMBUS addr=10010000 (90)



Near VRAM



Note : connector and power are by project design

ADD0: Address select. Connect to GND, SDA, SCL, or V+

DEVICE TWO-WIRE ADDRESS		ADD0 PIN CONNECTION	Output
1001000	90	Ground	CPU
1001001	91	V+	VRAM
1001010	92	SDA	GPU
1001011	93	SCL	

<Core Design>

ASUSTeK COMPUTER

Title : FAN_Thermal Sensor & Fan

Engineer: NR EE RD3

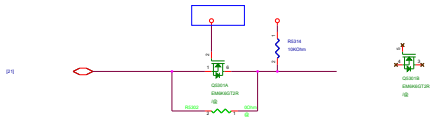
Size	Project Name	Rev
C	GX502GX	1.0

Date: Wednesday, January 18, 2023 Sheet 50 of 104

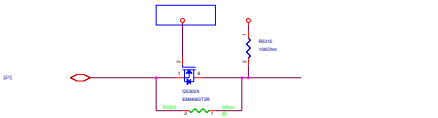
Control

WLAN card Vendor recommend: (Intel/Realtek)
1. Module internal PU
2. Placeholder for protect unexpected issue

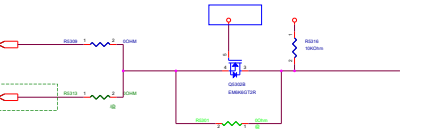
WLAN Ext_HW_Kill



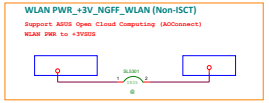
BT Ext_HW_Kill



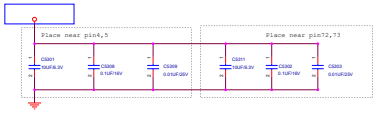
PCIE Wake



Power



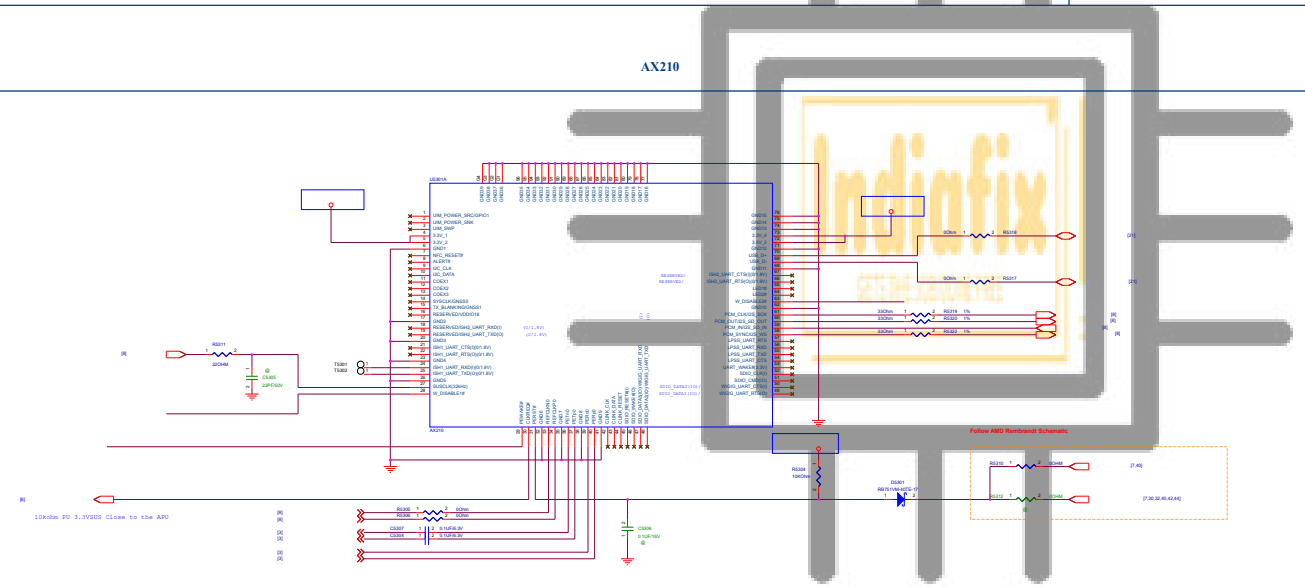
Bypass capactor



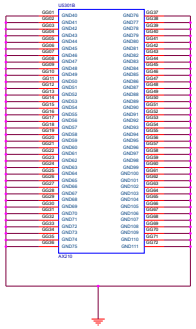
For EMI



AX210

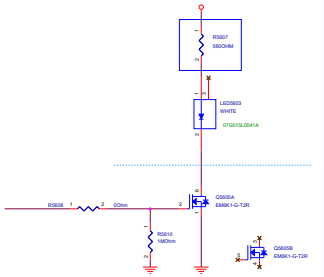


AX210_GND

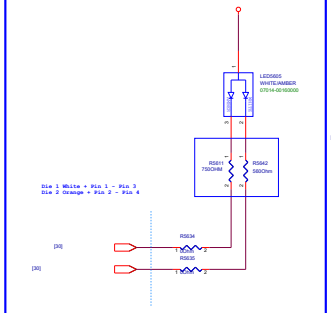


Project Name: IP2023			Rev: 1.0
Title: WLAN&BT_PCIE			
Size: D	Dept.: AsusTek COMPUTER INC.	Engineer: NR EE RD3	
Date: Wednesday, January 18, 2023	Sheet: 69	of: 104	

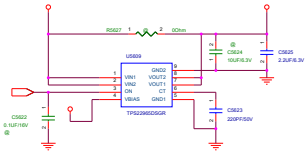
PWR LED (GU603)



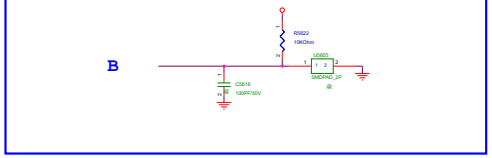
Charger LED (GU603)



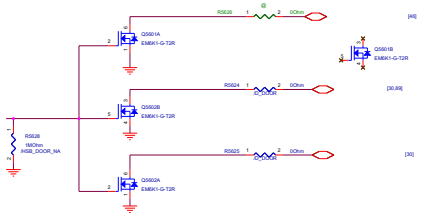
小板+5VSUS



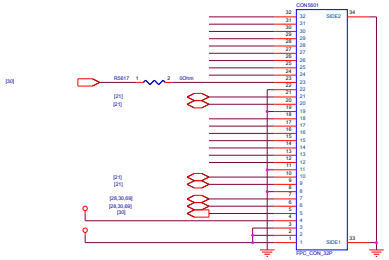
B鍵
(Maximunm rating 12V/50mA)
(Minimum rating 1V/10uA)



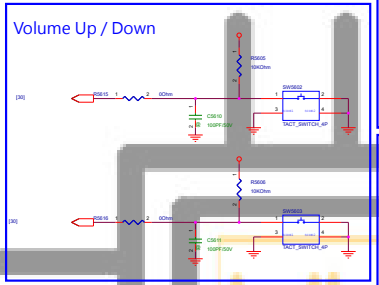
D Door(為了讓CON5601多一根GND)



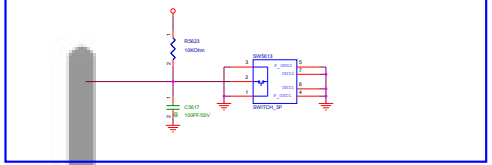
Connect to Left Board (page 71)



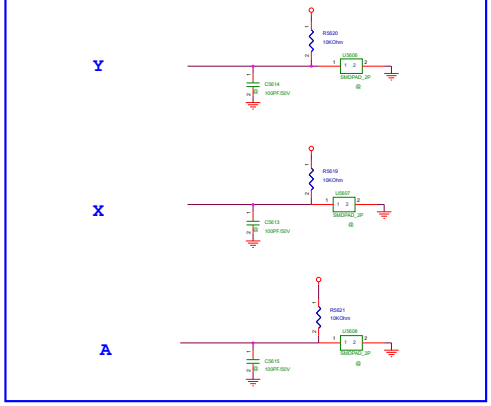
Volume Up / Down



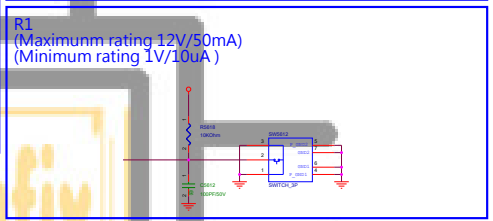
L1
(Maximunm rating 12V/50mA)
(Minimum rating 1V/10uA)



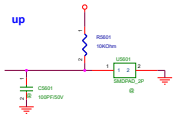
YXA鍵
(Maximunm rating 12V/50mA)
(Minimum rating 1V/10uA)



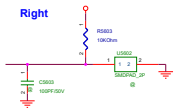
R1
(Maximunm rating 12V/50mA)
(Minimum rating 1V/10uA)



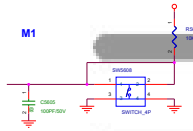
up



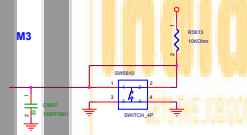
Right



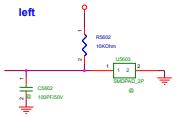
M1



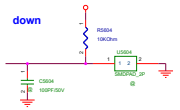
M3



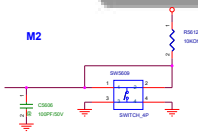
left



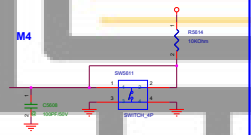
down



M2

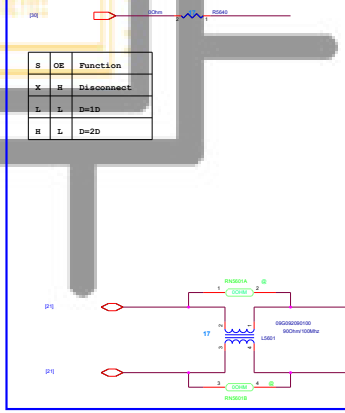


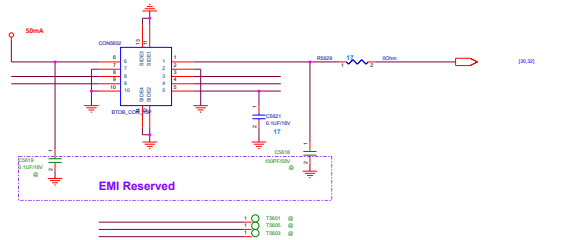
M4



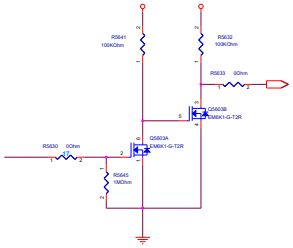
Power Key and FigerPrinter (GX703)

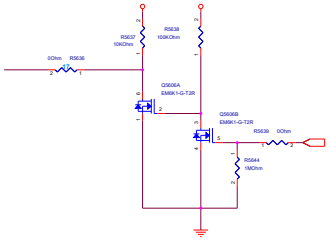
S	OE	Function
X	H	Disconnect
L	L	D=1D
H	L	D=2D





EMI Reserved

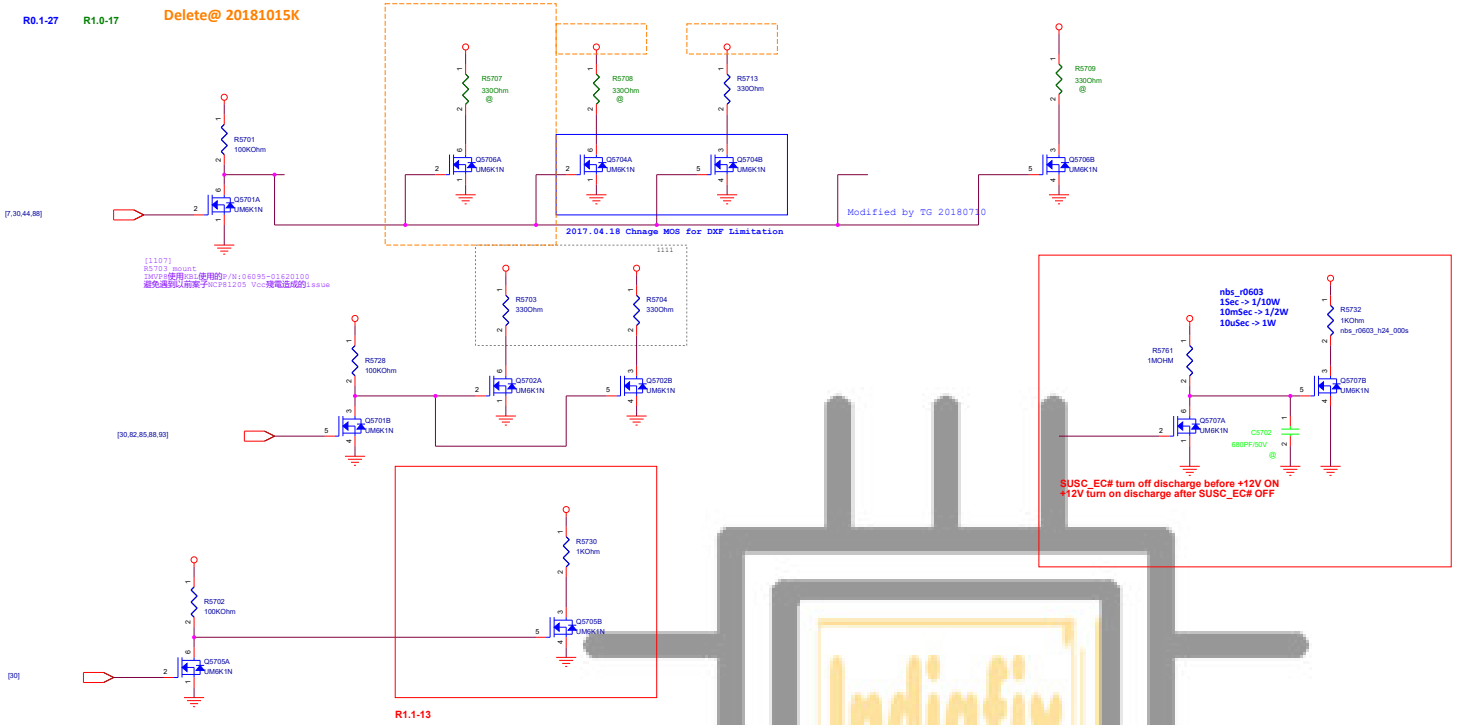




Title :	LED & LID	
Engineer:	NR EE RD3	
Size: Custom	Project Name: GMS31GX	Rev: 0.0
Date: Wednesday, January 19, 2022	Sheet 55 of 104	

R0.1-02 R0.1-27 R1.0-17

Delete@ 20181015K



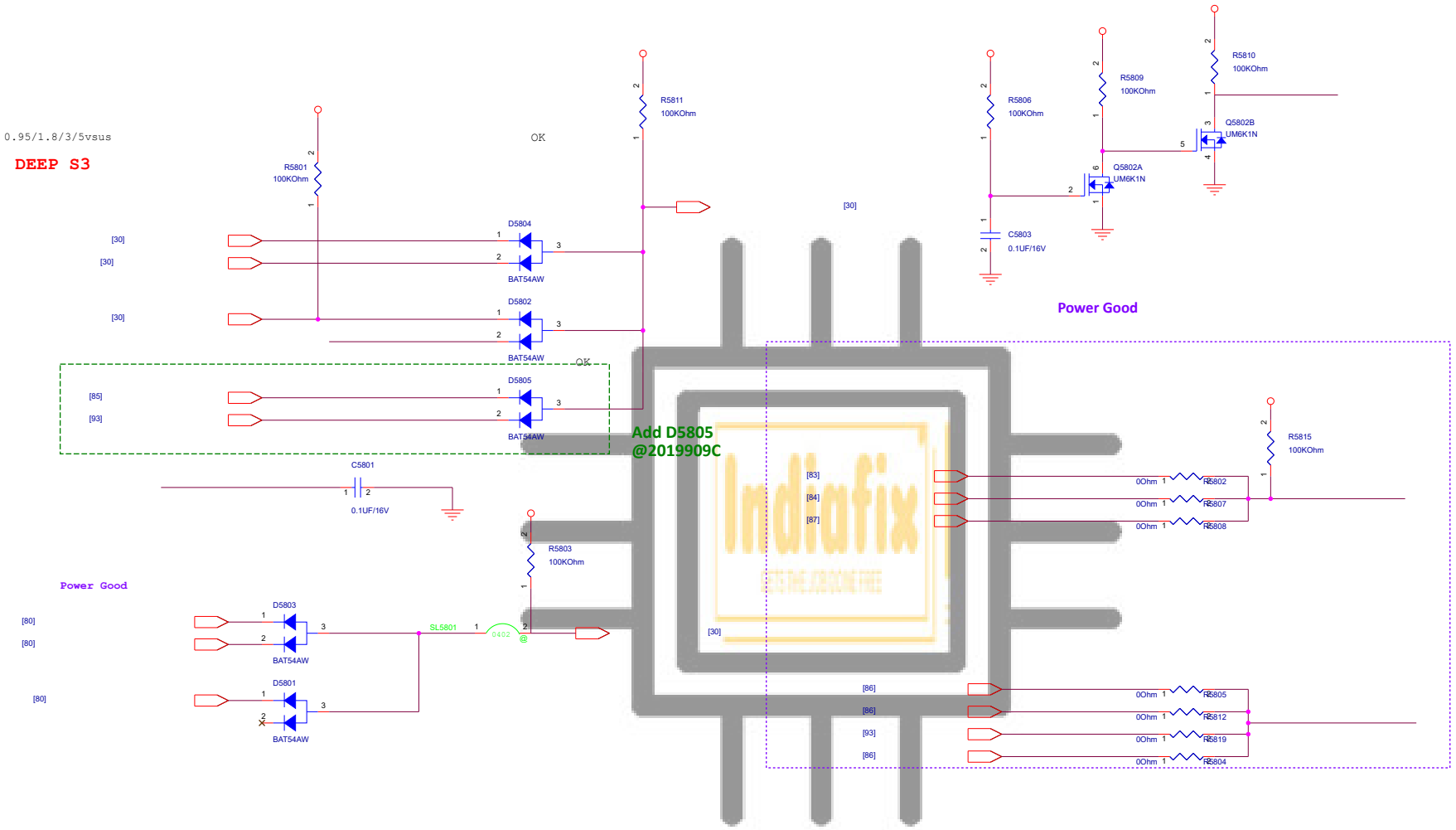
Change DSW to SUS
@20181009

<Core Design>

ASUSTeK COMPUTER	Title :	DSG_Discharge
	Engineer:	NR EE RD3
Size: Custom	Project Name: GX502GX	Rev: 1.0
Date: Wednesday, January 18, 2023	Sheet 57 of 104	

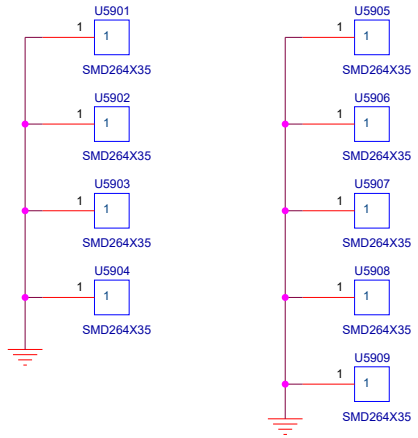
Indiafix

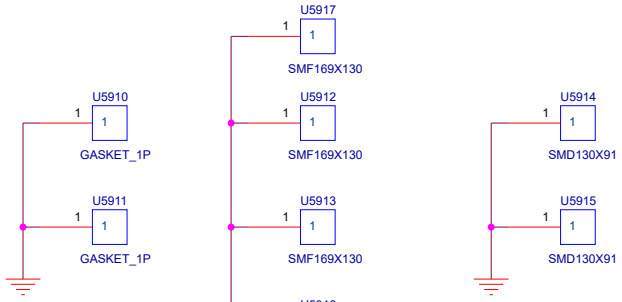
POWER GOOD DETECTER



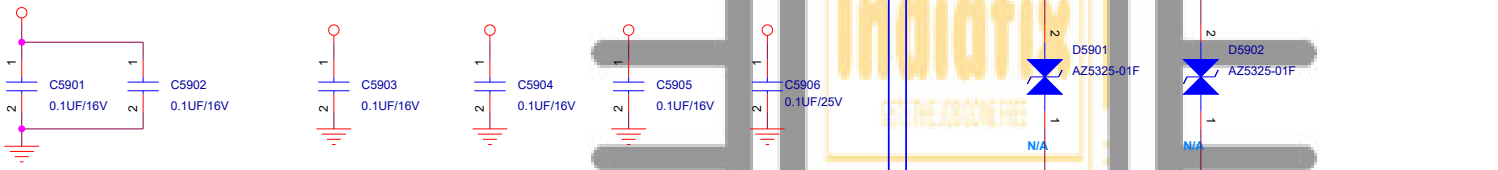
Project Name: FX505DY		Rev: R1.0
Title : POWER GOOD DETECTER		
Size: B	Dept.: ASUSTeK COMPUTER INC. Engineer: NR EE RD3	
Date: Wednesday, January 18, 2023	Sheet 58 of 104	

Clip





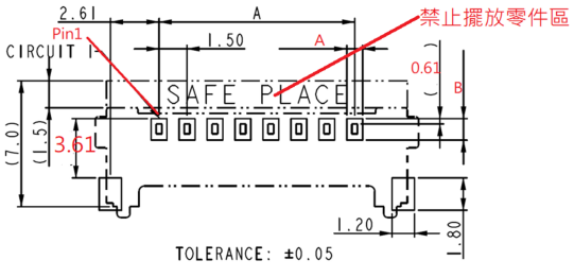
EMI



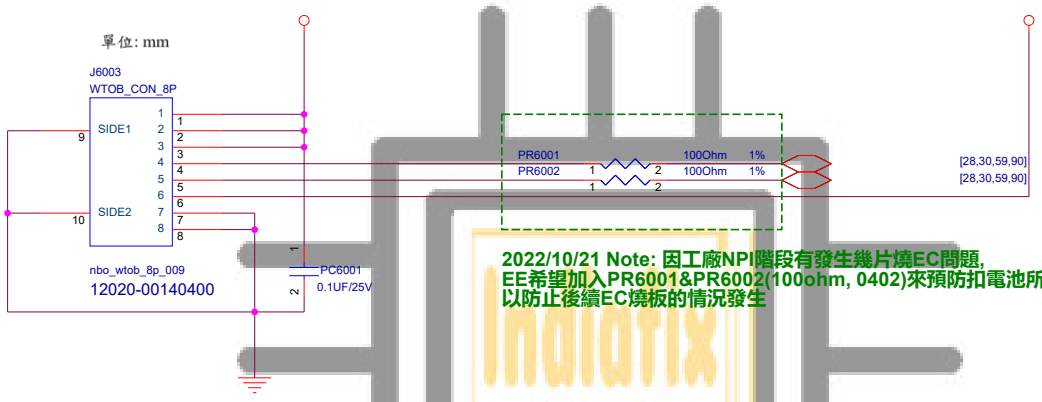
<Core Design>

ASUSTeK COMPUTER	Title : OTH_EMI	
	Engineer: NR EE RD3	
Size: A	Project Name: GX502GX	Rev: 1.0
Date: Wednesday, January 18, 2023	Sheet 59 of 104	

Battery Connector



BAT :
40Wh
4S1P
1C能耐=2.584A
滿充電壓=17.8V
充電能力=1.1C=2.843A
放電能力=1.5C=3.724A



2022/10/21 Note: 因工廠NPI階段有發生幾片燒EC問題,
EE希望加入PR6001&PR6002(100ohm, 0402)來預防扣電池所產生的突波,
以防止後續EC燒板的情況發生

Note:Battery Connector 正確性與BAT1_IN_OC#是否預留!

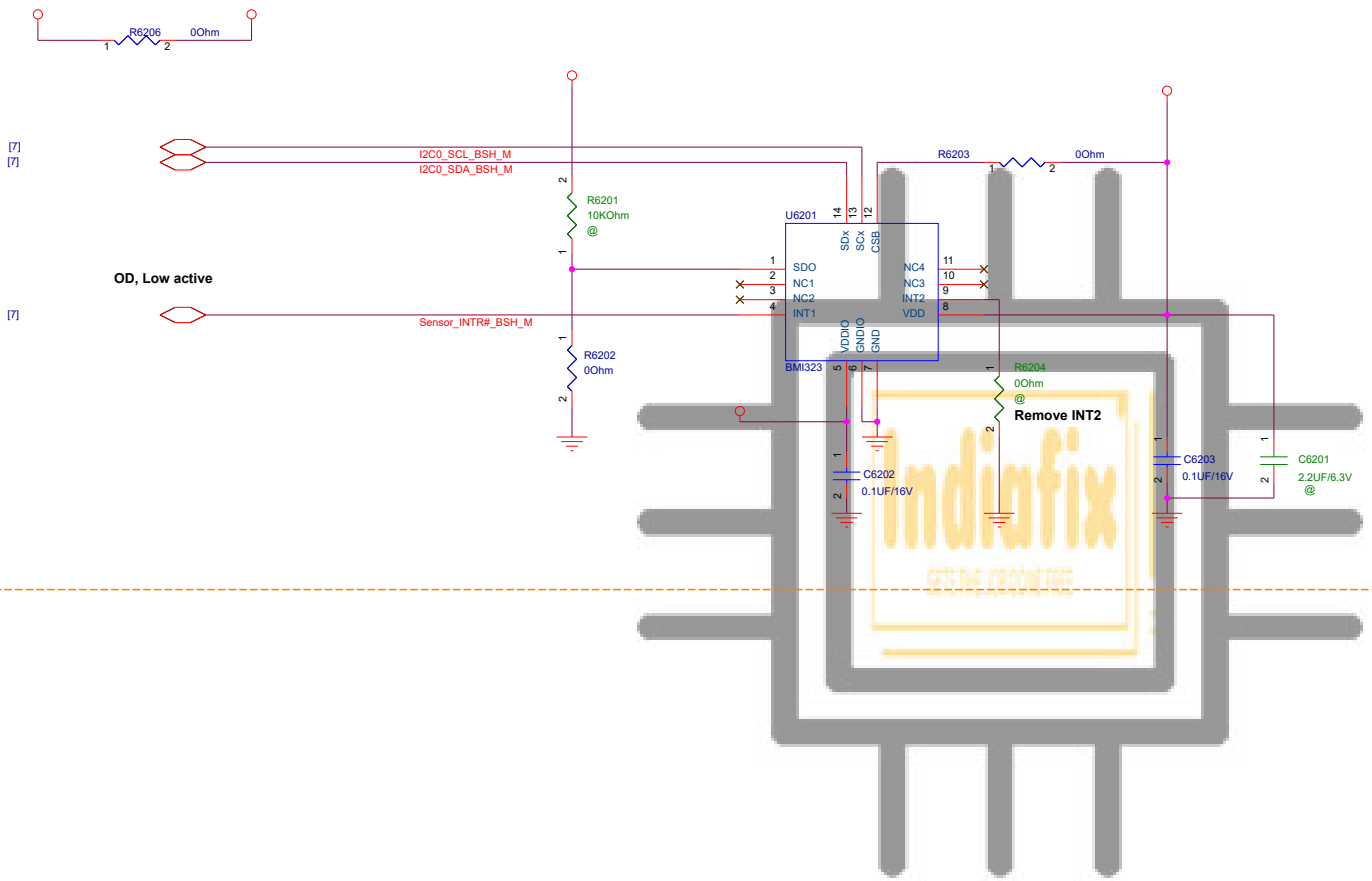
2022/6/22 Note: Follow需求修改Footprint為nbo_wtob_8p_008

2022/8/17 Note: Follow EE需求更換料號為12020-00140400, Footprint為nbo_wtob_8p_009

2022/8/31 Note: 12020-00140400 Footprint nbo_wtob_8p_009加入鐵片後推禁制區

ASUS	Project Name: GV301	Rev: R1.0
Title : DC & BAT IN		
Size: A4	Dept.: NB_Power team	Engineer: NR EE RD3
Date: Wednesday, January 18, 2023	Sheet 60 of 104	

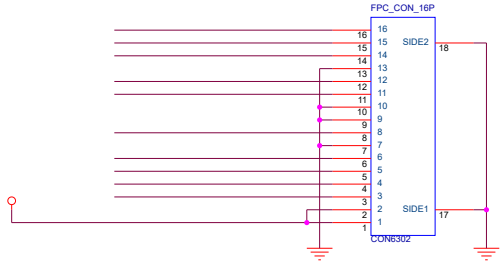
BMI323



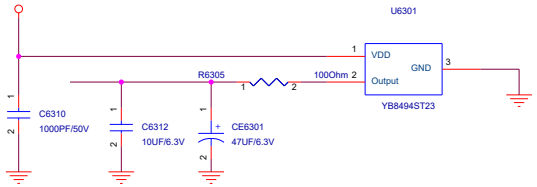
<Variant Name>

ASUSTeK COMPUTER	Title : BSH_Sensor	
	Engineer: NR EE RD3	
Size: Custom	Project Name: GL752VW	Rev: 1.0
Date: Wednesday, January 18, 2023	Sheet 62 of 104	

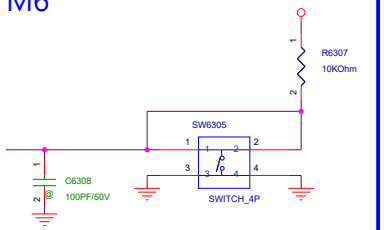
Connector LeftBoard(P71)



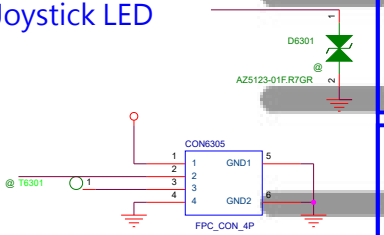
R2



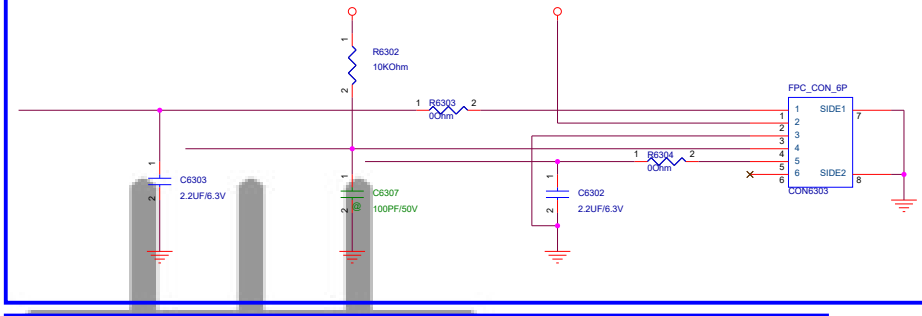
M6



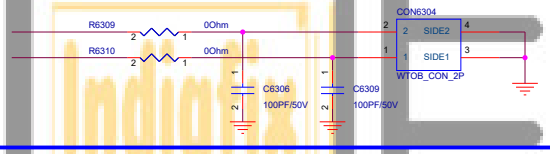
Joystick LED



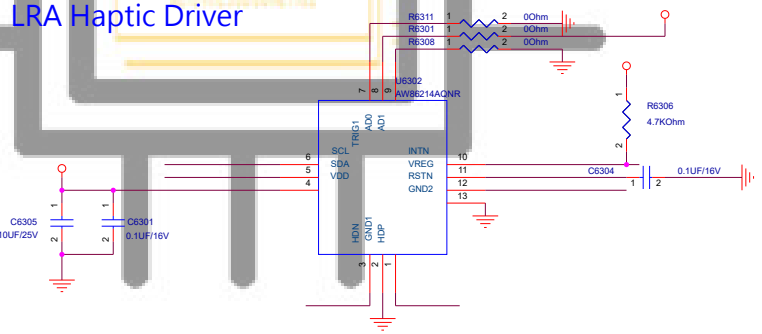
Right Joystick connector (Maximunm rating 5V/1mA) (Minimum rating 3.3V/60µA)



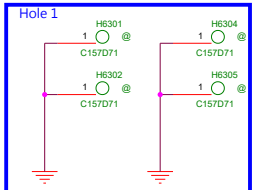
Right Vibrator connector (Maximunm rating 5V/0.55A)2022/8/15



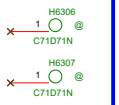
LRA Haptic Driver



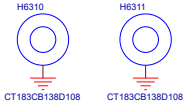
Hole 1



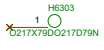
Hole 2



Nut 3

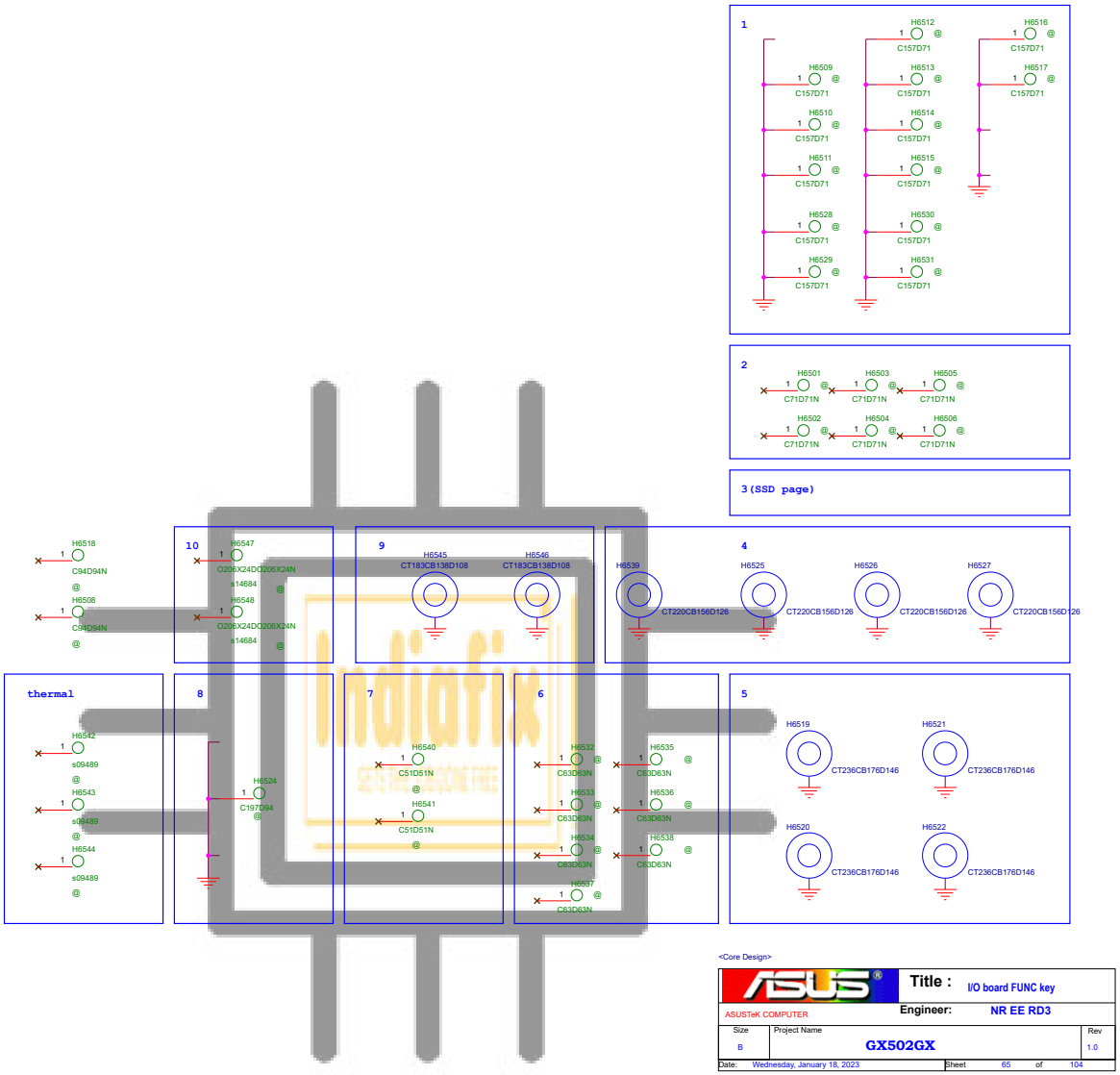


Hole 4



<Variant Name>

ASUSTeK COMPUTER	Title : NGFF SSD(MAXIM)	
	Engineer: NR EE RD3	
Size: Custom	Project Name: GL752VW	Rev: 1.0
Date: Wednesday, January 18, 2023	Sheet 63 of 104	



<Core Design>

ASUSTeK COMPUTER	Title : I/O board FUNC key	
	Engineer: NR EE RD3	
Size: B	Project Name: GX502GX	Rev: 1.0
Date: Wednesday, January 18, 2023	Sheet 65 of 104	

20200318 ashton add +3VS_PE_EN, add all PEG re-drivers and PEG all switch power cap

eGPU TX

to APU RX

Swap for layout 2022/02/18

APU TX

to eGPU RX

Swap for layout 2022/02/18

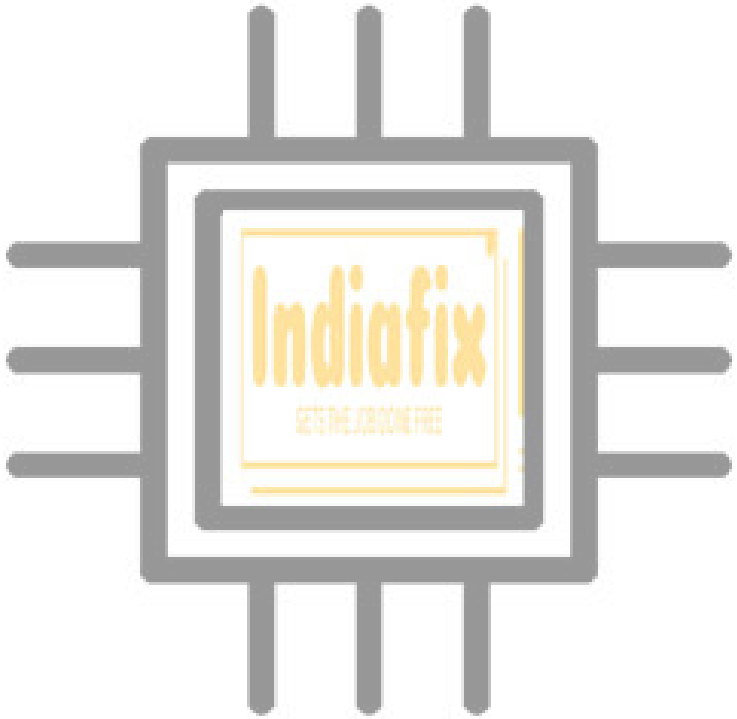
DOCKING_CON_80P

20191126 ashton add AGPU level shift
20191126 ashton remove EC_ACGPU_IRQ, only use P_AC_LOSS

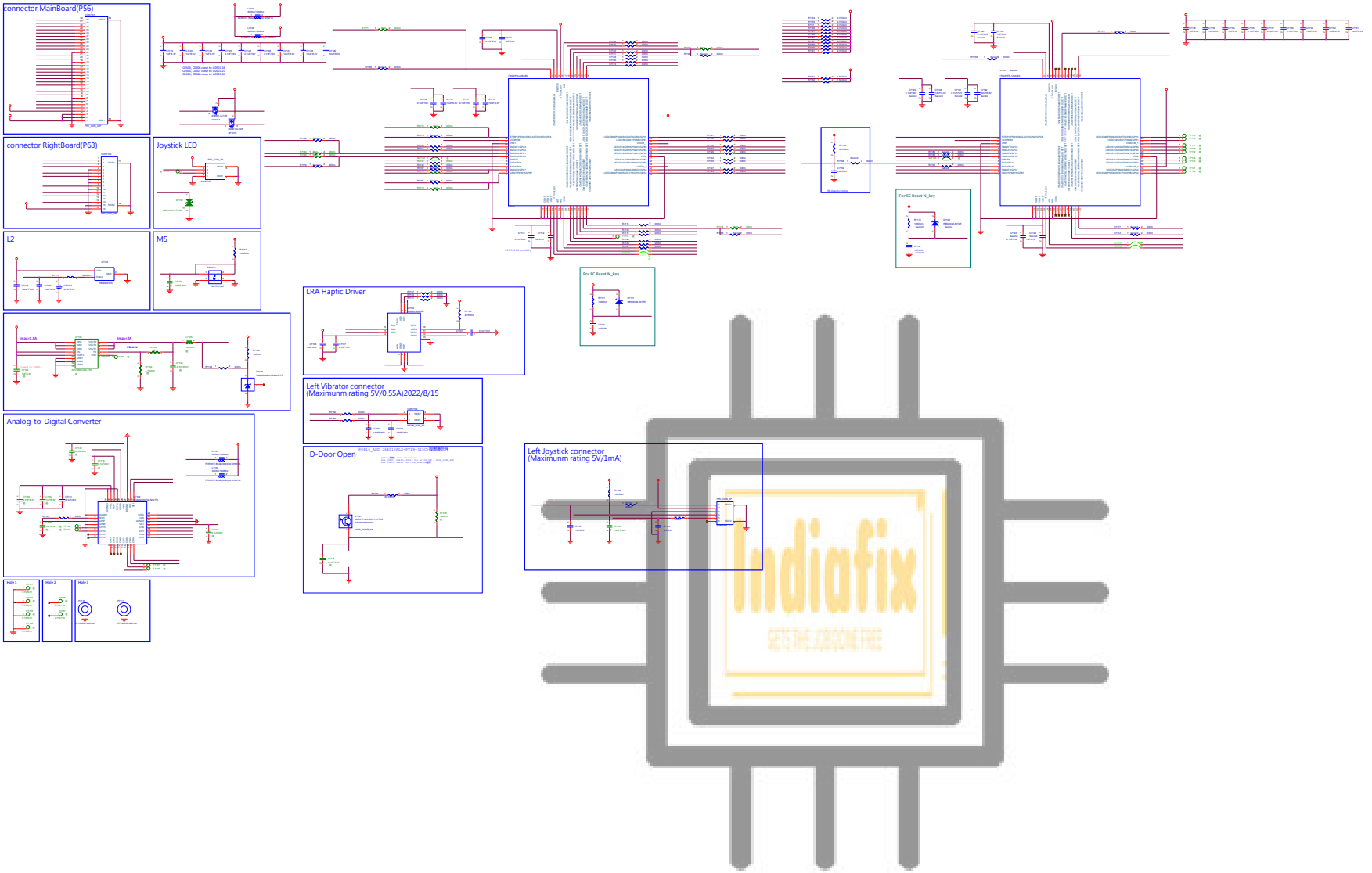
20200916 ashton modify

<Core Design>

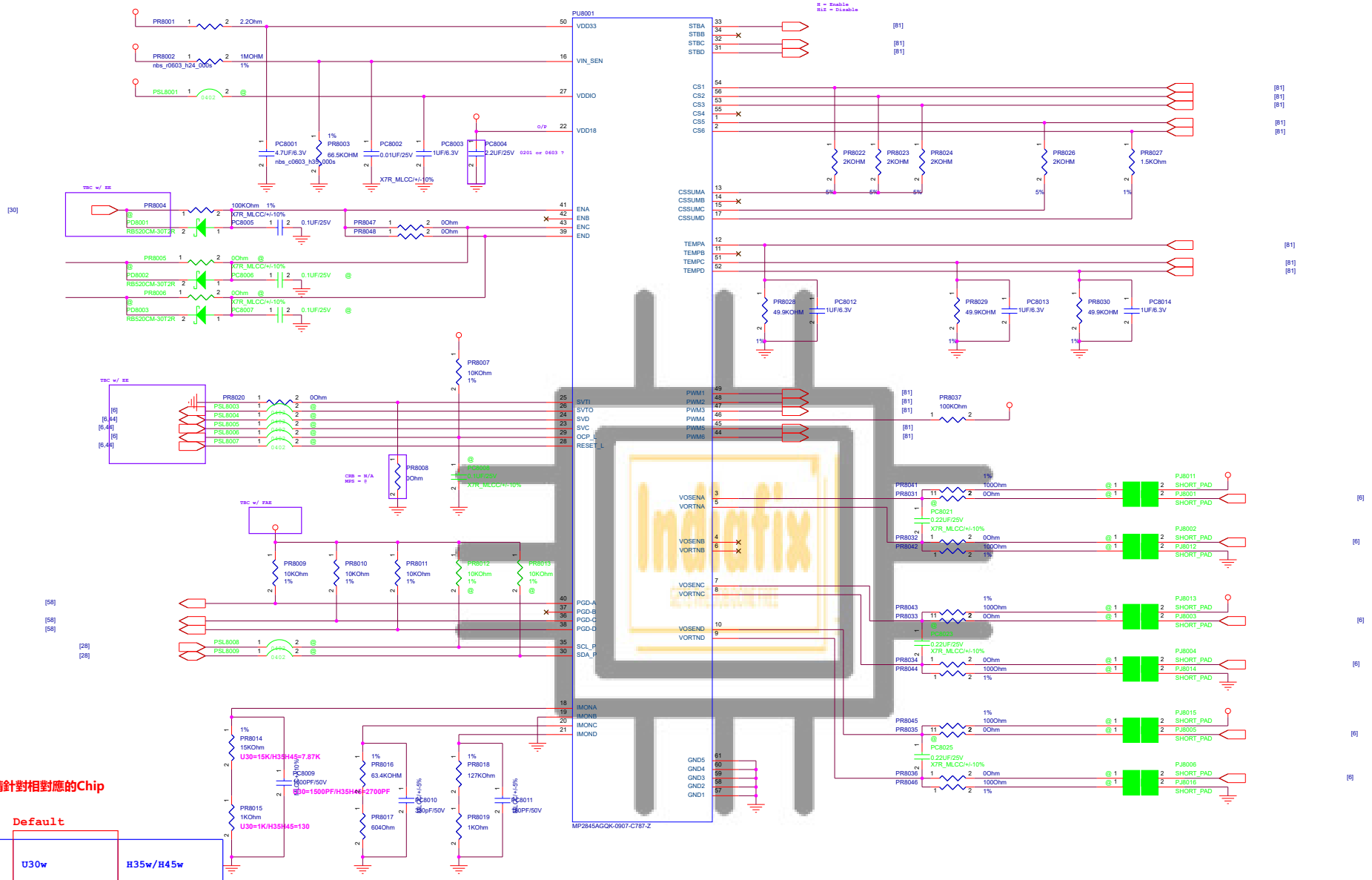
ASUSTeK COMPUTER	Title :	I/O_Main board Conn.	
	Engineer:	NR EE RD3	
Size: Custom	Project Name: GX502GX		Rev: 1.0
Date: Wednesday, January 18, 2023	Sheet	of	



ASUS	Project Name: G512LI	Rev: 4.0
Title :	GPU_PCIE IF	
Size: Custom	Dept.: ASUSTeK COMPUTER	Engineer: NR EE RD3
Date: Wednesday, January 18, 2023	Sheet 70 of 104	


connector MainBoard(P56)
connector RightBoard(P63)
Joystick LED
L2
M5
LRA Haptic Driver
Left Vibrator connector
(Maximunm rating 5V/0.55A)2022/8/15
Analog-to-Digital Converter
D-Door Open
Left Joystick connector
(Maximunm rating 5V/1mA)
For EC Reset N_key
For EC Reset N_key
Mode 1
Mode 2
Mode 3

indiafix



***注意,EE請針對相對應的Chip SPEC更換**

Default

PHX瓦數	U30w	H35w/H45w
PR8014	15K Ohm, N/A 10G2121502140l0	7.87K Ohm, N/A 10G2127871140z0
PR8015	1K Ohm, N/A 10G2121001140l0	130 Ohm, N/A 10G2121301040l0
PC8009	1500PF/50V, N/A 11G2321152143z0	2700PF/50V, N/A 11G2321272140z0

PHX = U30/H35/H45w
Peak = 105/130/140A
Rmon = 16 / 8 / 8K
Cmon = 1.5/2.7/2.7uF

+APU_VDDCORE_RUN

PHX = U30/H35/H45w
Peak = 23/26/26A
Conti. = 64/64/64K

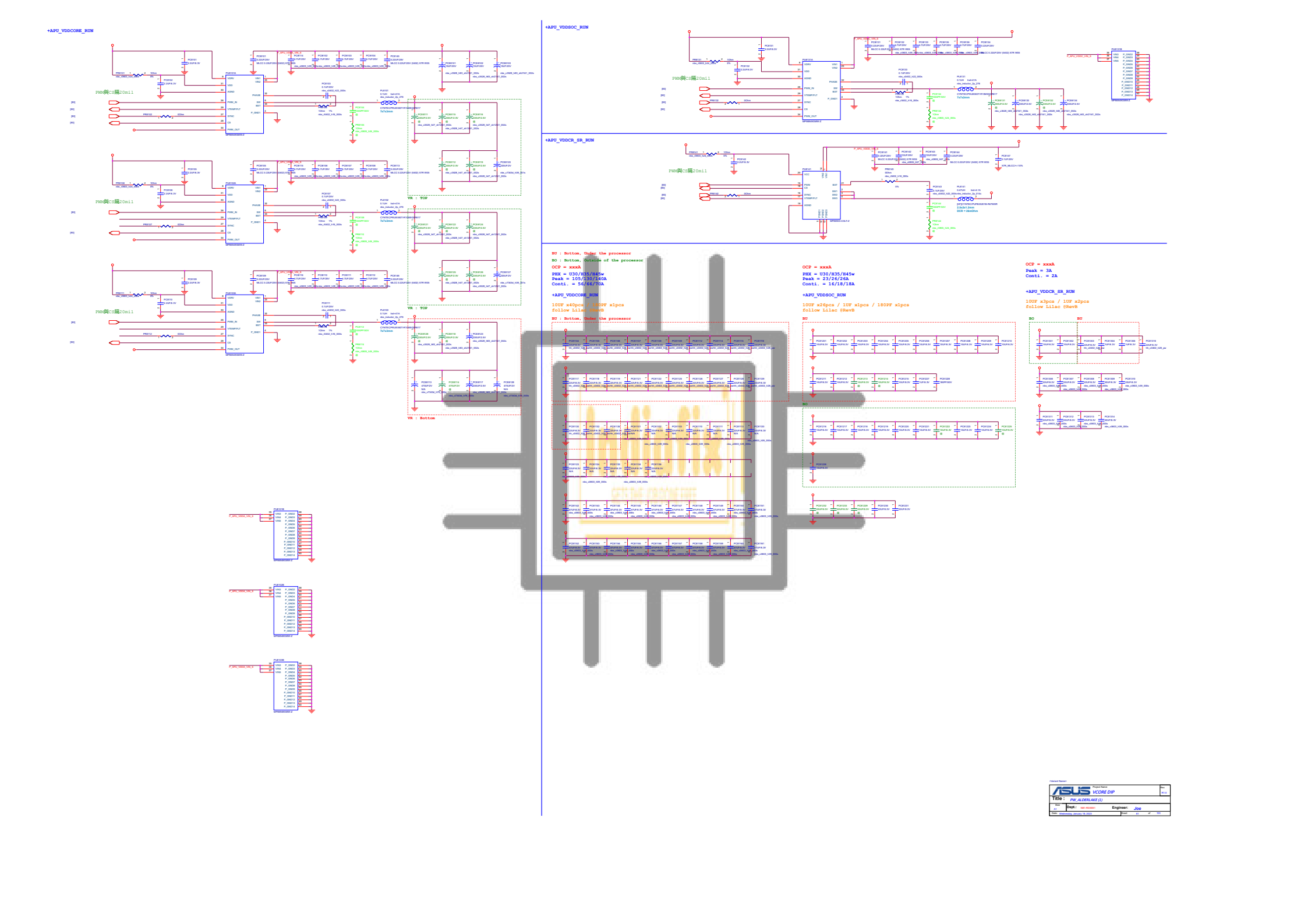
+APU_VDDSOC_RUN

Peak = 3A
Conti. = 2A

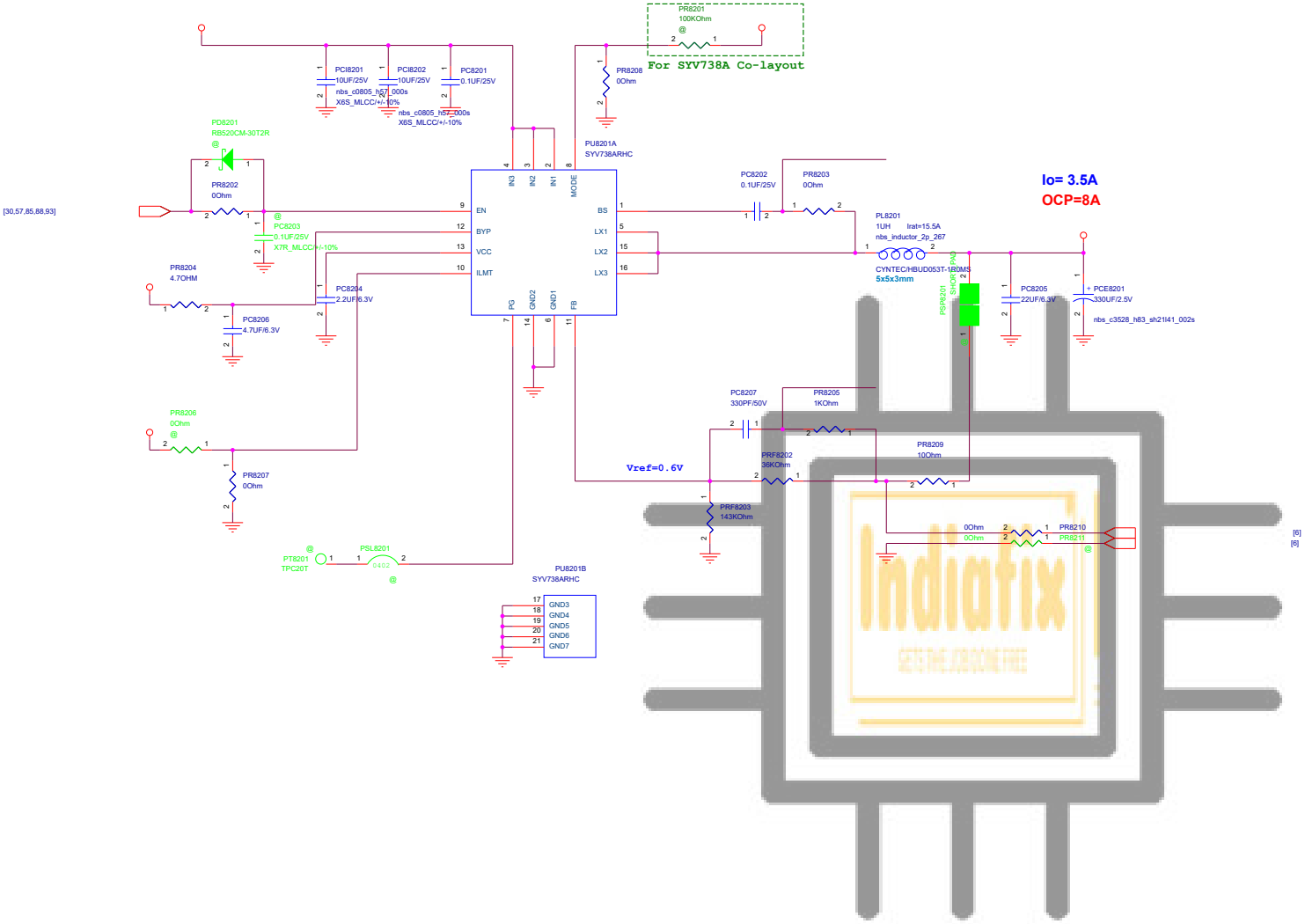
+APU_VDDCR_SR_RUN

<Variant Name>

ASUS	Project Name: VCORE DIP	Rev: R1.0
Title :	PW_ALDERLAKE (1)	
Size: A3	Dept.: NB1-RD3EE1	Engineer: Joe
Date: Wednesday, January 18, 2023	Sheet 80 of 103	


+APU_VDDCORE_RUN
+APU_VDDSOC_RUN
+APU_VDDCR_SR_RUN
VR : TOP
VR : Bottom
BU : Bottom, Under the processor
BO : Bottom, Outside of the processor
OCP = xxxA
PHX = U30/H35/H45w
Peak = 105/130/140A
Conti. = 56/66/70A
+APU_VDDCORE_RUN
10UF x40pcs / 10UF x1pcs
follow Lilac 8RevB
BU : Bottom, Under the processor
OCP = xxxA
PHX = U30/H35/H45w
Peak = 23/26/26A
Conti. = 16/18/18A
+APU_VDDSOC_RUN
10UF x26pcs / 10F x1pcs / 180FP x1pcs
follow Lilac 8RevB
OCP = xxxA
Peak = 3A
Conti. = 2A
+APU_VDDCR_SR_RUN
10UF x3pcs / 10F x2pcs
follow Lilac 8RevB
VCORE DIP
Title : PW_ALDERLAKE (1)
Engineer: Joe

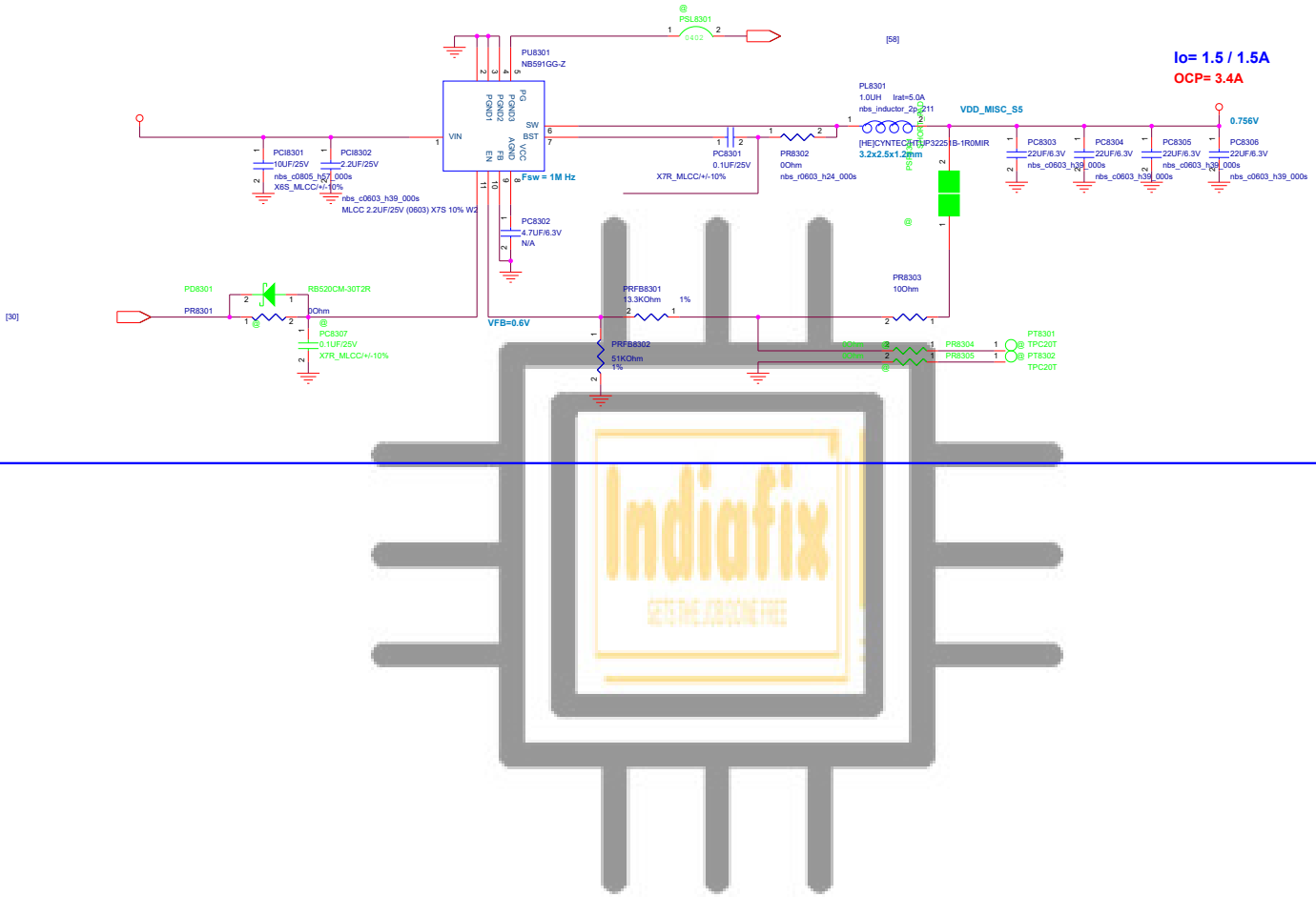
+0.75VS [For APU]



<Variant Name>

ASUS	Project Name: GA503QS	Rev: R1.0
Title : PW_+0.75VS		
Size: A3	Dept.: NB Power team	Engineer: CS Lin
Date: Wednesday, January 18, 2023	Sheet 82 of 104	

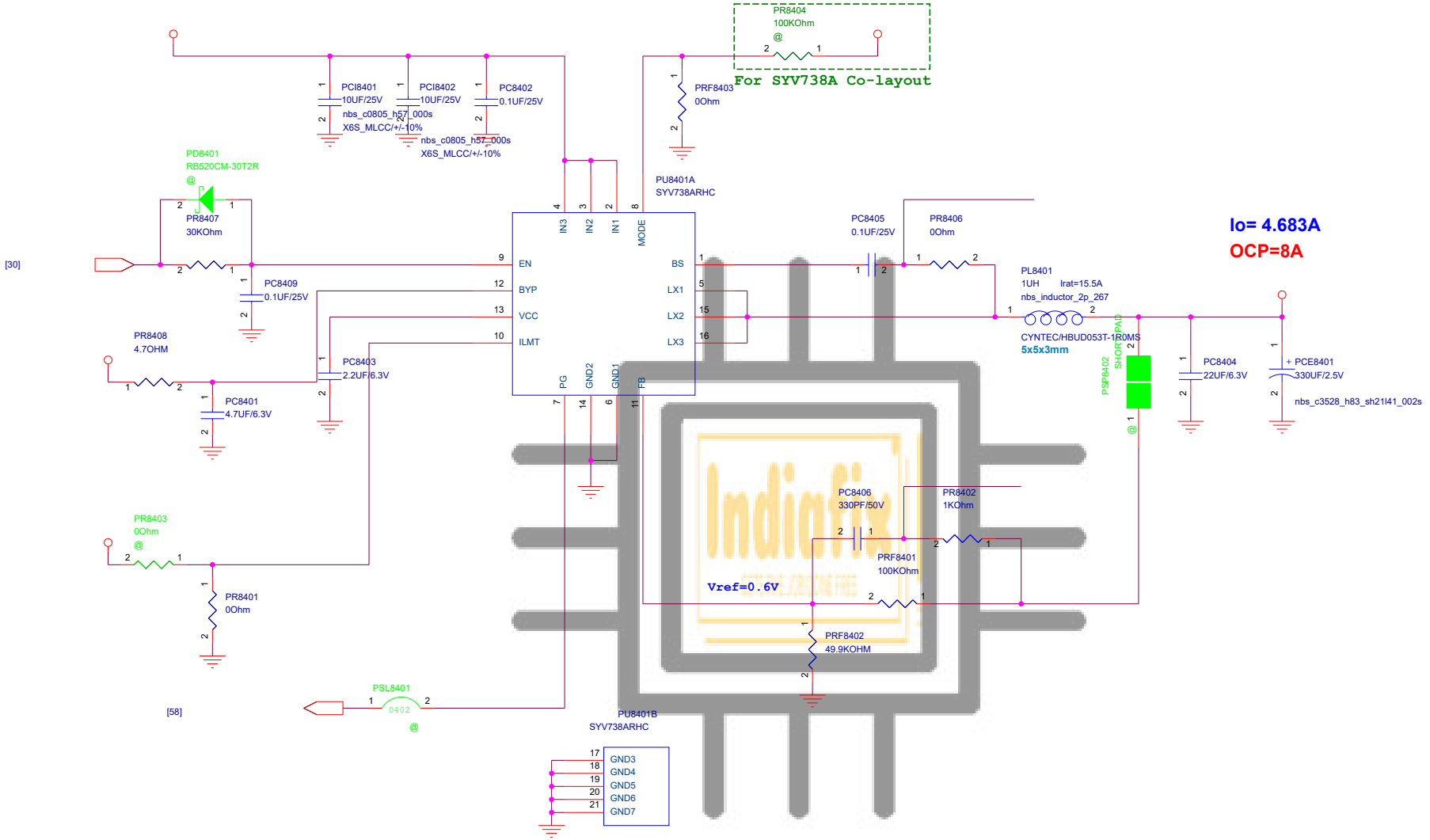
+0.75VSUS [For APU]



<Variant Name>

ASUS	Project Name: GA503QS	Rev: R1.0
Title : PW_FLOWCHART		
Size: A3	Dept.: NB Power team	Engineer: CS Lin
Date: Wednesday, January 18, 2023	Sheet 83 of 104	

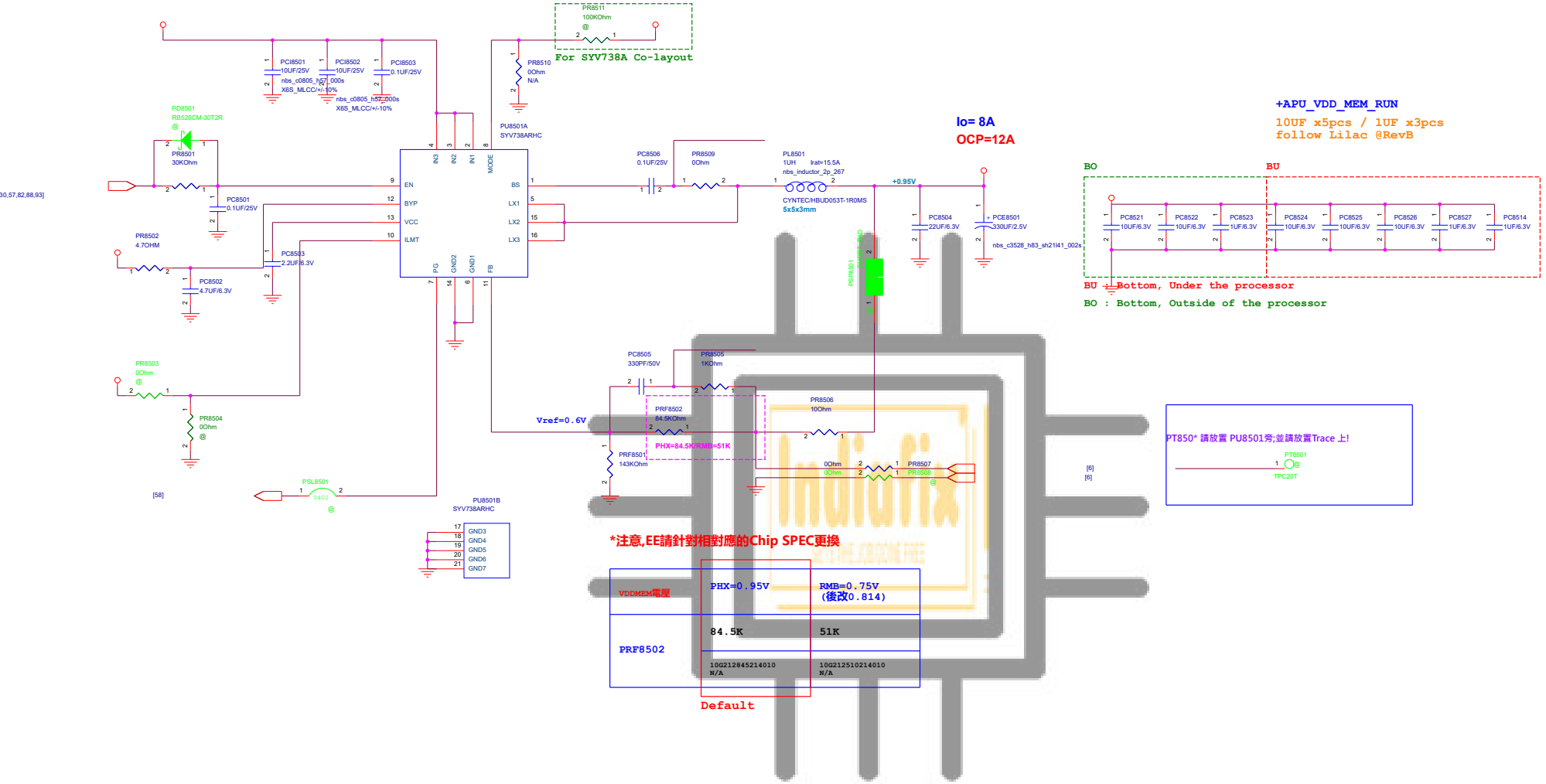
PW_1P8VSUS [For GPU]



<Core Design>

ASUS	Project Name: DESIGN_IP	Rev: R1.0
Title : PW_+1.8VSUS		
Size: A4	Dept.: NB Power team	Engineer: Power
Date: Wednesday, January 18, 2023	Sheet 84 of 102	

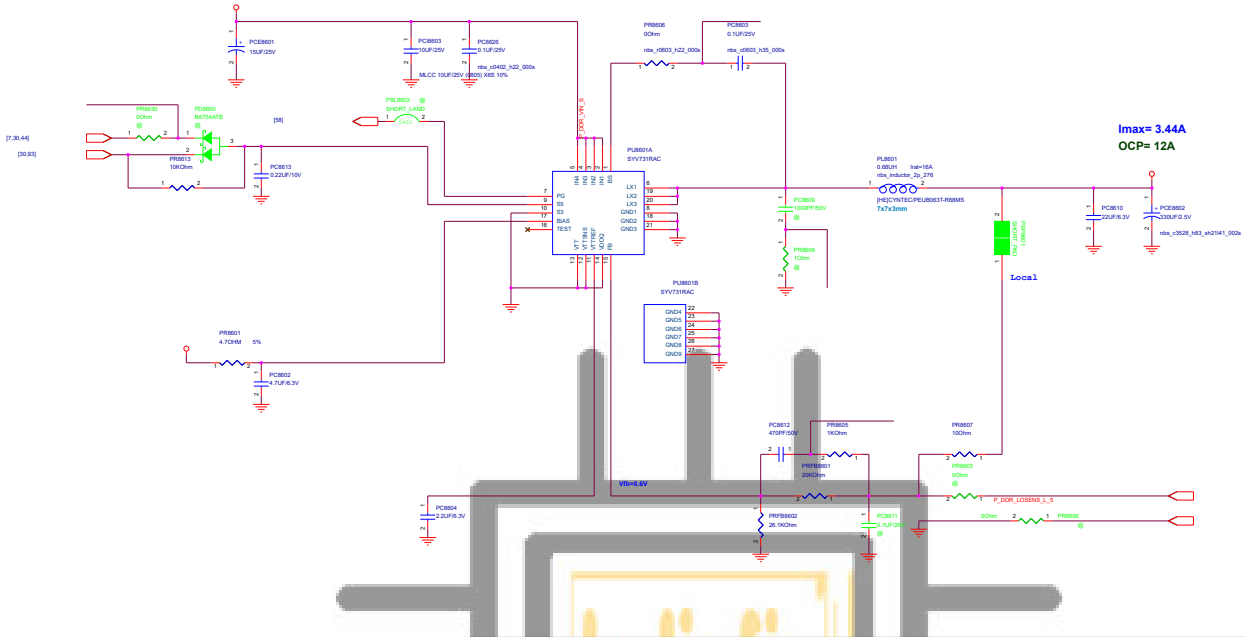
+VDD_MEM [For APU]



<Variant Name>

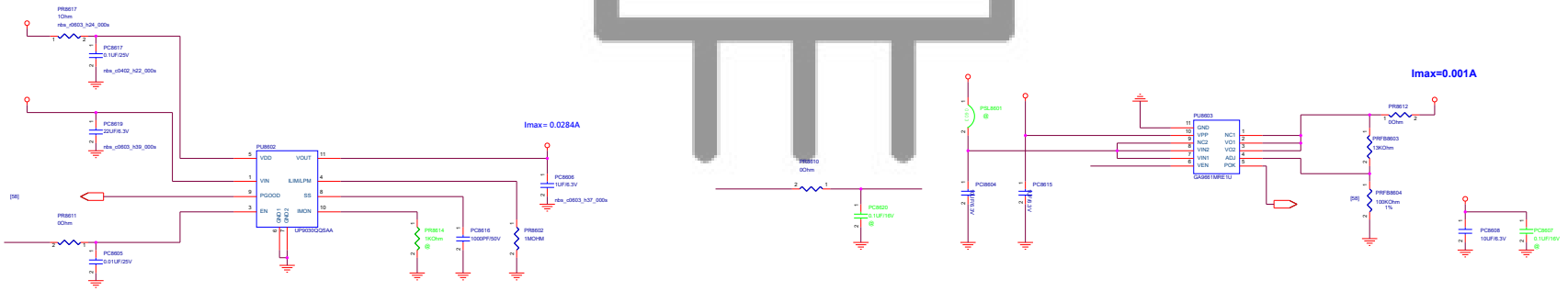
ASUS	Project Name: GU603E	Rev: R1.0
Title : PW_+1.8VSUS		
Size: A3	Dept.: NB Power team	Engineer: Hon
Date: Wednesday, January 18, 2023	Sheet 85 of 102	

+1.05V/+VTT[For Memory]



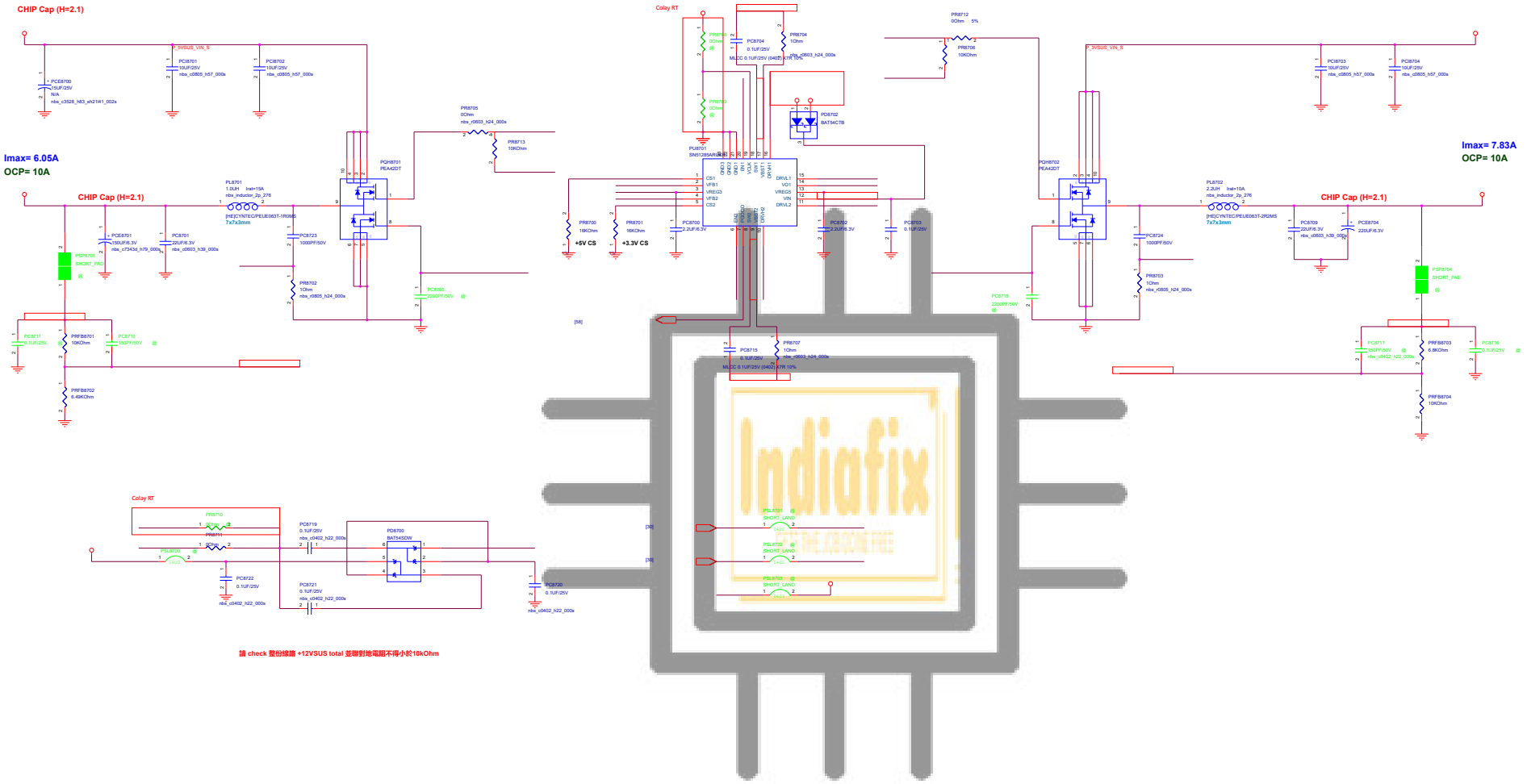
DDR Type	Vout	PU8601	PRFB8601	PRFB8602	PC8604	PR8604	PIN14 NET Name
DDR4	+1.2V	Silergy / SYV731RAC 06018-02990000	20K ohm 10G2122002140 30	19.6K ohm 10G2121962140 10	2.2UF/6.3V 11G2322225253 60	@	+1.2V
LPDDR4x	+1.1V	Silergy / SYV731DRAC 06018-02990100	20K ohm 10G2122002140 30	23.2K ohm 10G2122322140 30	@	0ohm 10G2120000040 30	+1.1V
LPDDR5	+1.05V	Silergy / SYV731RAC 06018-02990000	20K ohm 10G2122002140 30	26.1K ohm 10G2122612140 10	2.2UF/6.3V 11G2322225253 60	@	+1.05V

+1.8V/+0.9V[For Memory]



<Core Design>	Project Name: Project Name		Rev: R1.0
Title : PW_+1.05V / +VTT / +1.8V			
Size: A2	Dept.: NB Power team	Engineer: Power_RD	
Date: Wednesday, January 18, 2023	Sheet 86 of 104		

+3VSUS / +5VSUS [System Power]



Adaptor Mode (IMVP8)

	S0	CS	S3	DS3	S4	S5	S5 with USB Charger+
PS_ON	1	-	1	-	1	-	1
3VADSW_ON	1	-	1	-	1	-	1
3VSUS_ON	1	-	1	-	1	-	1
5VSUS_ON	1	-	1	-	1	-	1
1.35V_ON	1	-	1	-	0	-	0
SUSC_EC#	1	-	1	-	0	-	0
SUSB_EC#	1	-	0	-	0	-	0

Battery Mode (IMVP8)

	S0	CS	S3	DS3	S4	S5	S5 with USB Charger+
PS_ON	1	-	-	1	0	0	1
3VADSW_ON	1	-	-	1	0	0	0
3VSUS_ON	1	-	-	0	0	0	0
5VSUS_ON	1	-	-	1	0	0	1
1.35V_ON	1	-	-	1	0	0	0
SUSC_EC#	1	-	-	0	0	0	0
SUSB_EC#	1	-	-	0	0	0	0

PT870* 請放置 PU8700旁;並請放置Trace 上!

<Variant Name>

ASUS GA502IV	Project Name	Rev R1.0
Title : PW_+3VA_DSW/+5VSUS		
Size: A2	Dept.: NB Power team	Engineer: NR EE RD3
Date: Wednesday, January 18, 2023	Sheet 87 of 104	

uP9030 ILIM/LPM Setting 對應表		
ILIM PIN	LPM (C10)	LIMIT Current
GND	Off	3A
1M to GND	Off	5A
Float/VDD	On	8A

PR8838 & PC8838 close VDD PIN

Imax = 0.0201A

EE CHECK

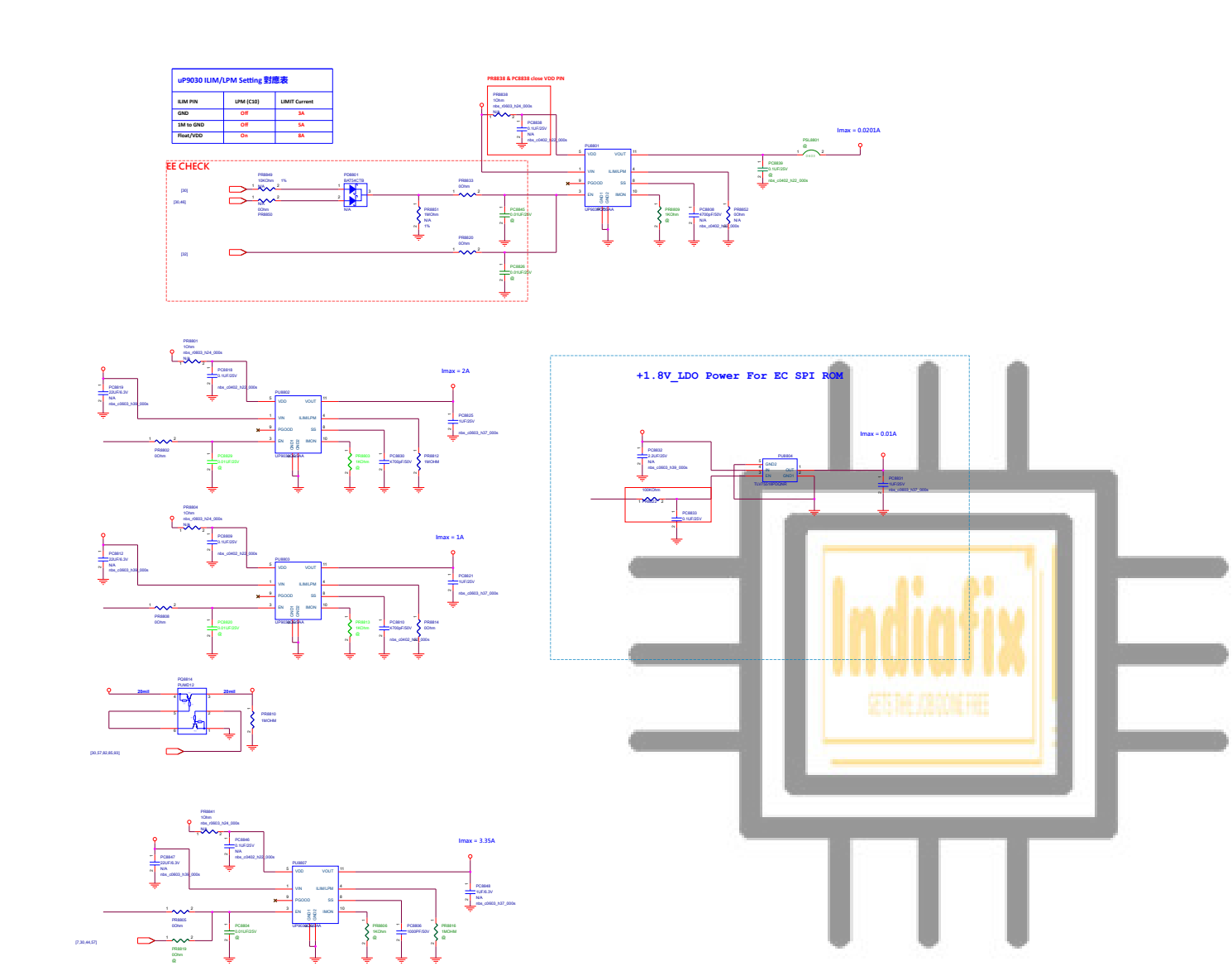
Imax = 2A

Imax = 1A

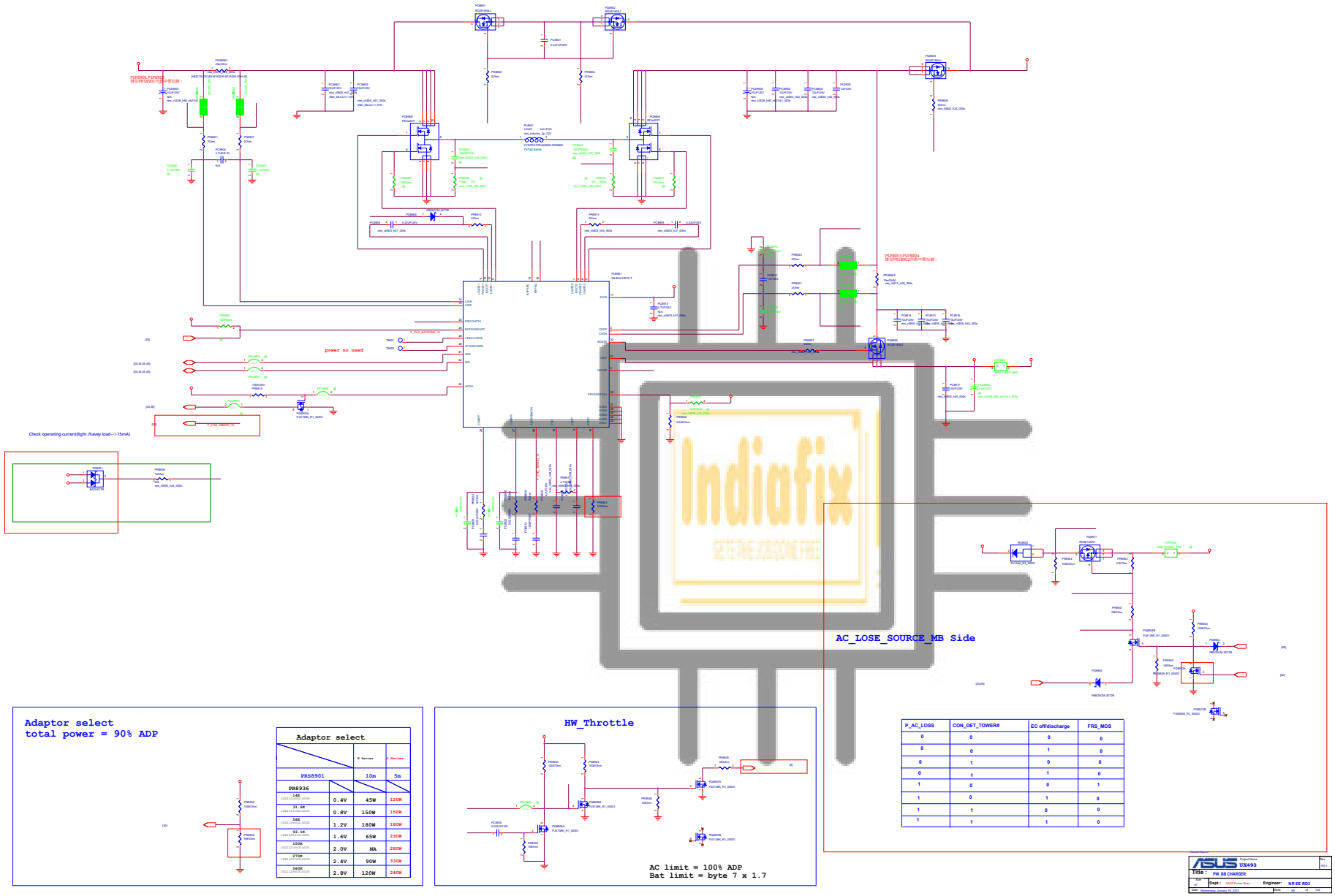
Imax = 3.35A

+1.8V_LDO Power For EC SPI ROM

Imax = 0.01A



ASUS	Project Name: GA502IV	Rev:
Title : PW_LOAD SWITCH		
Size: Custom	Dept.: NB Power team	Engineer: NR EE RD3
Date: Wednesday, January 19, 2022	Sheet 88 of 104	



Adaptor select
total power = 90% ADP

Adaptor select

	H Series	D Series
PR8901	10m	5m
PR8936		
14K	0.4V	45W	120W
31.6K	0.8V	150W	150W
56K	1.2V	180W	180W
93.1K	1.6V	65W	230W
150K	2.0V	NA	280W
270K	2.4V	90W	330W
560K	2.8V	120W	240W

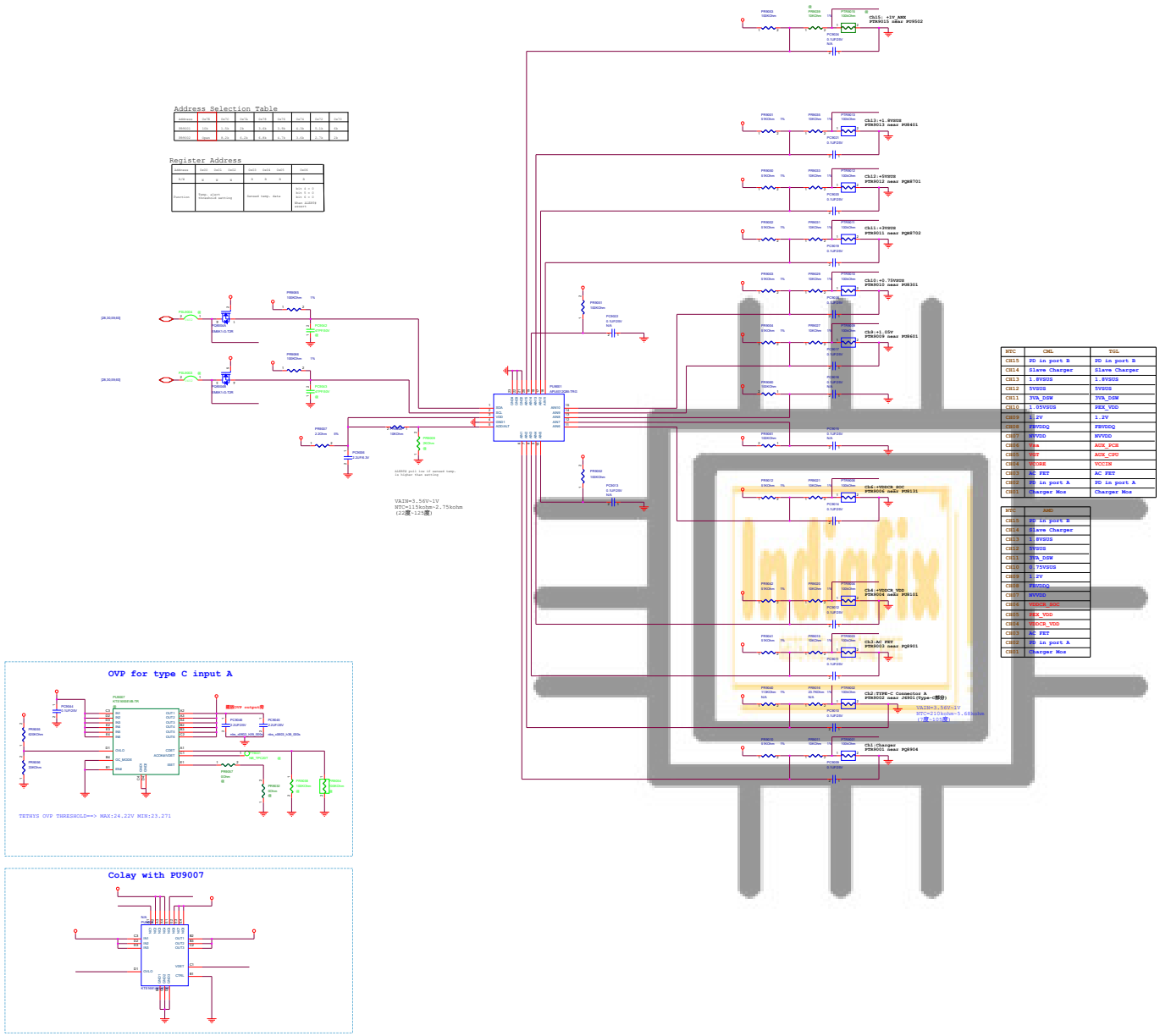
HW_Throttle

AC limit = 100% ADP
Bat limit = byte 7 x 1.7

P_AC_LOSS	CON_DET_TOWER#	EC offdischarge	FRS_MOS
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	0

ASUS Project name: UX493
Title : PW_BB CHARGER
Dept.: MSAS Power Team
Engineer: NR EE RD3
Date: Wednesday, January 08, 2020
Sheet 89 of 106

P90_PROTECTION

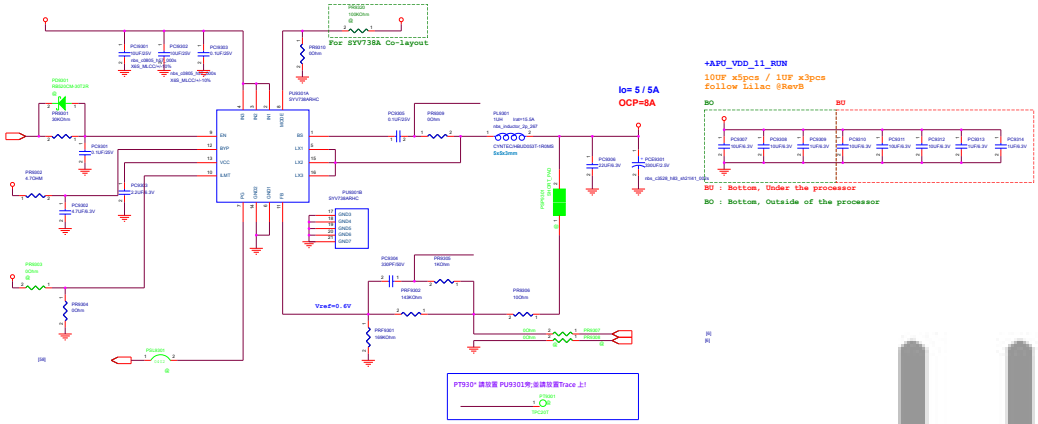


NTC	CML
CH15	PD in port B
CH14	Slave Charger
CH13	1.8VSUS
CH12	5VSUS
CH11	3VA_DSW
CH10	1.05VSUS
CH09	1.2V
CH08	PSVDDQ
CH07	MVVDD
CH06	VDDA
CH05	VNN
CH04	VDDCR
CH03	AC FET
CH02	PD in port A
CH01	Charger Mos

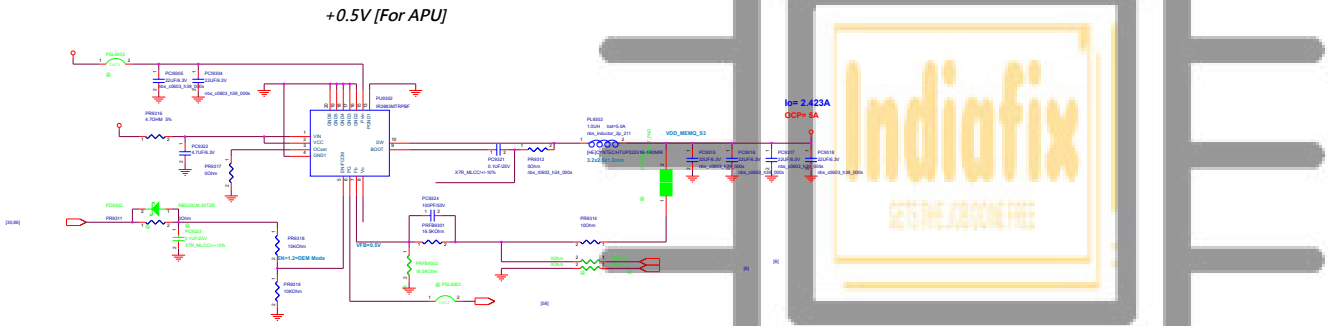
NTC	AMD
CH15	PD in port B
CH14	Slave Charger
CH13	1.8VSUS
CH12	5VSUS
CH11	3VA_DSW
CH10	1.75VSUS
CH09	1.2V
CH08	PSVDDQ
CH07	MVVDD
CH06	VDDCR_SOC
CH05	SEA_VDD
CH04	VDDCR_VDD
CH03	AC FET
CH02	PD in port A
CH01	Charger Mos

Project Name	GA502V
Title :	PW_PROTECTION
Dept.	MB Power Team
Engineer:	MB EE RD1
Date	Wednesday, January 19, 2022

+1.1VS [For APU]

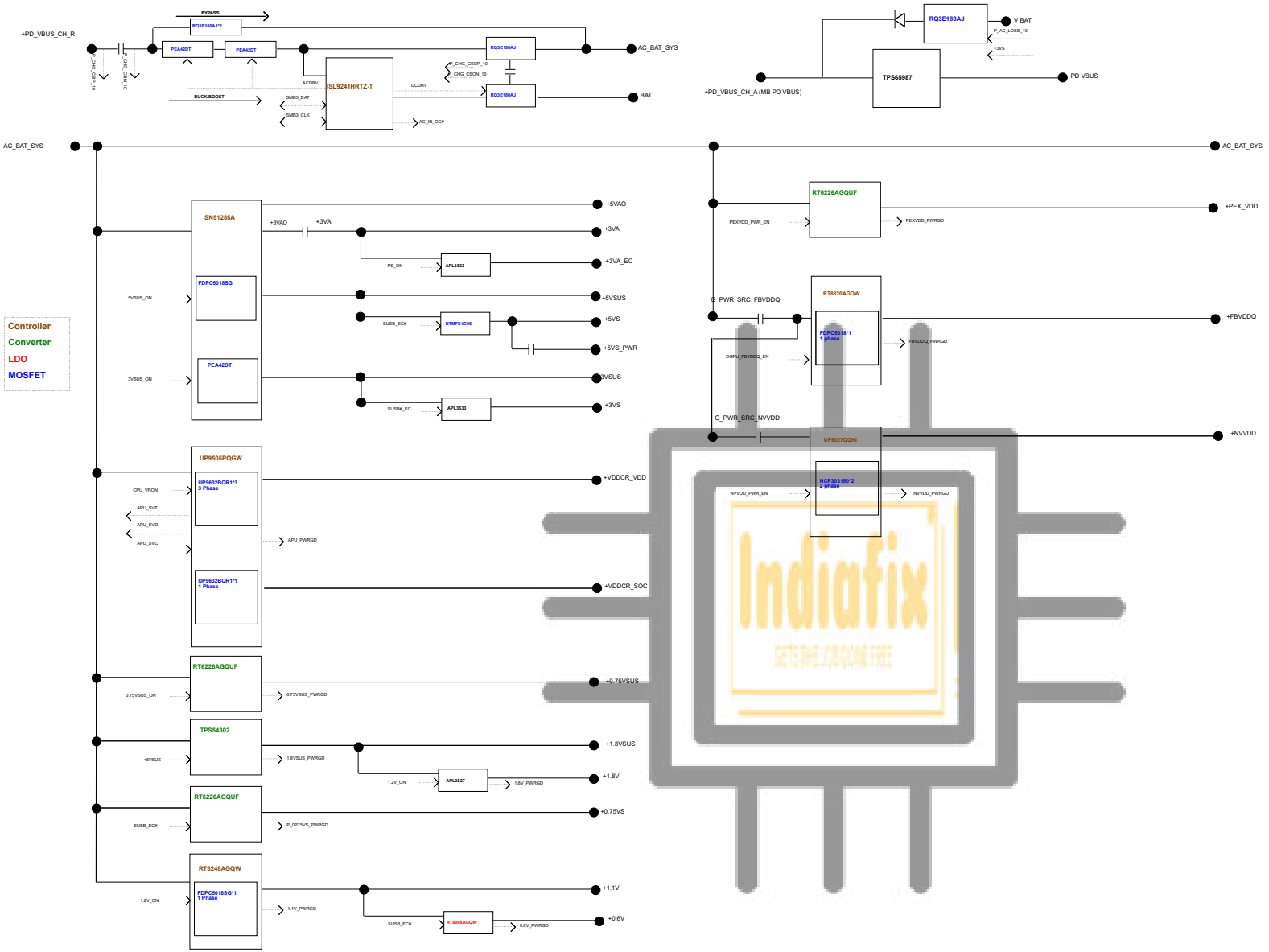


+0.5V [For APU]



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ASUS	Project Name: GU603E		Rev: R1.0
Title :	PW_+1.8VSUS		
Size: Custom	Dept.: NB Power team	Engineer: Hon	
Date: Wednesday, January 18, 2023	Sheet: 93 of 102		

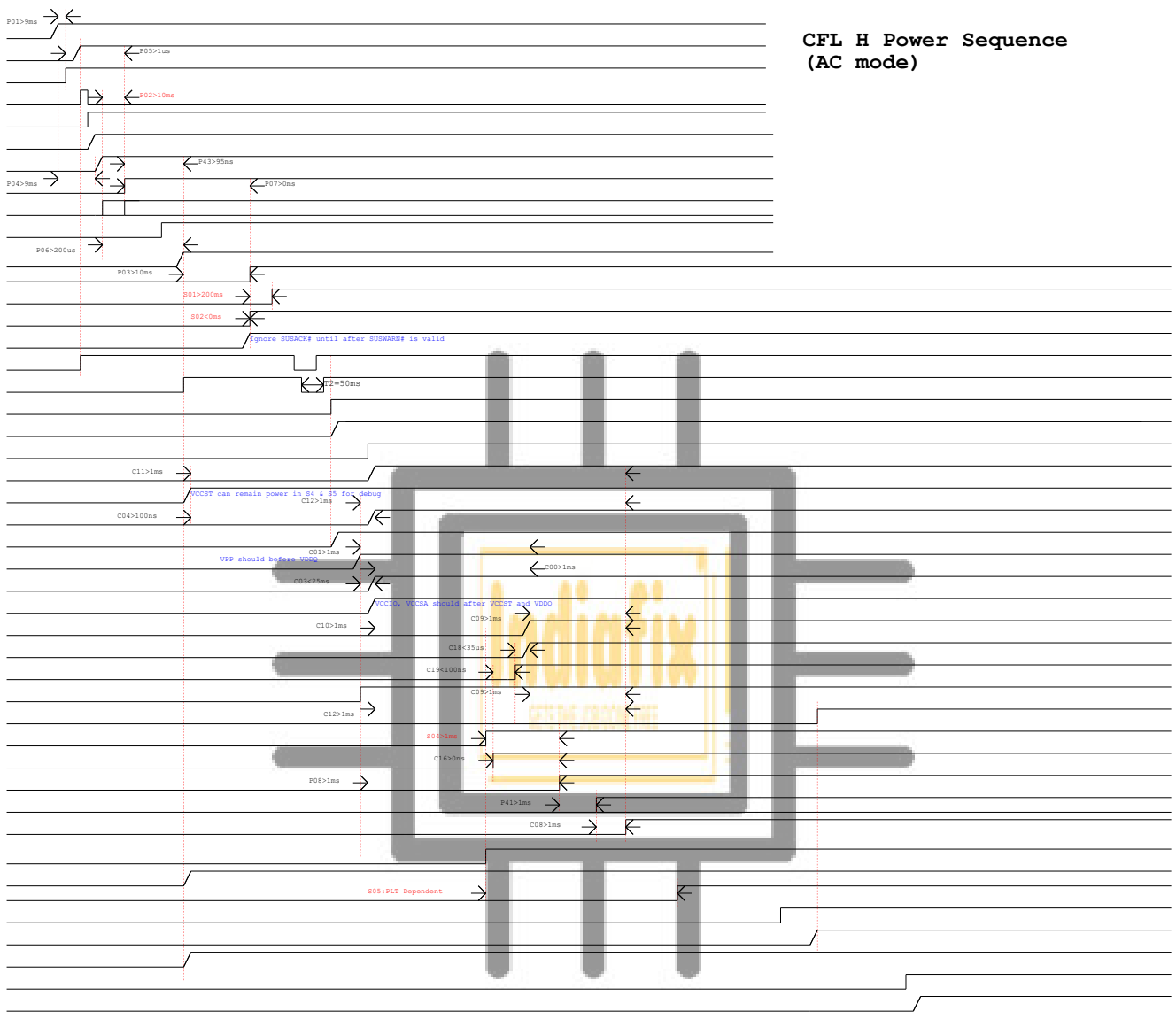

+PD_VBUS_CH_R
BYPASS
RQ3E180AJ*2
PEA42DT
PEA42DT
ACDRV
ISL9241HRTZ-T
BUCK/BOOST
SMB3_DAT
SMB3_CLK
DCDRV
P_CHG_CSOP_10
P_CHG_CSON_10
RQ3E180AJ
RQ3E180AJ
AC_IN_OC#
AC_BAT_SYS
BAT
RQ3E180AJ
V BAT
P_AC_LOSS_10
+3VS
TPS65987
PD VBUS
+PD_VBUS_CH_A (MB PD VBUS)
AC_BAT_SYS
Controller
Converter
LDO
MOSFET
SN51285A
+3VAO
+3VA
+5VAO
+3VA
+3VA_EC
PS_ON
APL3533
FDPC5018SG
5VSUS_ON
+5VSUS
SUSB_EC#
NTMFS4C06
+5VS
+5VS_PWR
PEA42DT
3VSUS_ON
3VSUS
SUSB#_EC
APL3533
+3VS
UP9505PQGW
UP9632BQR1*3
3 Phase
CPU_VRON
APU_SVT
APU_SVD
APU_SVC
APU_PWRGD
+VDDCR_VDD
UP9632BQR1*1
1 Phase
+VDDCR_SOC
RT6226AGQUF
0.75VSUS_ON
0.75VSUS_PWRGD
+0.75VSUS
TPS54302
+5VSUS
1.8VSUS_PWRGD
+1.8VSUS
1.2V_ON
APL3527
1.8V_PWRGD
+1.8V
RT6226AGQUF
SUSB_EC#
P_0P75VS_PWRGD
+0.75VS
RT8248AGQW
FDPC5018SG*1
1 Phase
1.2V_ON
1.1V_PWRGD
+1.1V
SUSB_EC#
RT9058AGQW
0.6V_PWRGD
+0.6V
RT6226AGQUF
PEXVDD_PWR_EN
PEXVDD_PWRGD
+PEX_VDD
G_PWR_SRC_FBVDDQ
RT8820AGQW
FDPC5018*1
1 phase
DGPU_FBVDDQ_EN
FBVDDQ_PWRGD
+FBVDDQ
G_PWR_SRC_NVVDD
UP9027QQKI
NCP303156*2
2 phase
NVVDD_PWR_EN
NVVDD_PWRGD
+NVVDD

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AC-IN Mode

C:CPU
P:PCH
S:PLT
Power
Signal

(+RTCBAT) +3VA_RTC
(AC_BAT_SYS) +3VA/+5VA
(+3VA_RTC) RTCRST# (PCH)
(Power) AC_IN_OC# (EC)
(EC) PS_ON (+3VA_EC)
(PS_ON) +3VA_EC (EC)
(3VADSW_ON) +3VA_DSW (3VA_DSW_PWRGD)
(EC) DPWROK_EC (PCH)
(+3VA_DSW) PM_BATLOW# (PCH)
(PCH) PM_SLP_SUS# (EC)
(VSUS_ON) +1.0VSUS_VCCPRIM (1.0VSUS_PWRGD)
(EC) PM_RSMRST# _PCH (PCH)
(PCH) SUSWARN# (EC)
(EC) ME_AC_PRESENT_PCH (PCH)
(EC) PCH_SUSACK# (PCH)
(PWR Switch) PWR_SW# (EC)
(EC) PM_PWRBTN# (PCH)
(EC) SUSC_EC# (Power)
(SUSC_EC#) +12V/+5V/+3V
(EC) SUSB_EC# (Power)
(SUSB_EC#) +12VS/+5VS/+3VS
(SUSB_EC#) +1.0V_VCCST, VCCPLL
(SUSB_EC#) +VCCIO, (+12VS) +VCCSTG
(1.2V_ON) +2.5V (2.5V_PWRGD)
(1.2V_ON) +VDDQ_CPU (1.2V_PWRGD)
(+12VS) +VCCPLL_OC
(SUSB_EC#) +VCCIO (VCCIO_PWRGD)
(ALL_SYSTEM_PWRGD) +VCCSA (IMVP8_PWRGD)
(DDR_VTT_CTRL) +0.6V
(CPU) DDR_VTT_CTRL (Power)
(Power) 1.2V_PWRGD (AND)
(Power) IMVP8_PWRGD
(AND) ALL_SYSTEM_PWRGD (CPU/PCH/EC/Power)
(ALL_SYSTEM_PWRGD) VCCST_PWRGD_CPU (CPU)
(EC) PM_PWROK_PCH (PCH)
(PCH) CLK_PCH_BCLK (CPU)
(PCH) H_CPUPWRGD (CPU)
(CPU) P_SVID_DATA_X2 (Power)
(EC) PM_SYSPWROK_PCH (PCH)
(PCH) PLT_RST# (CPU/EC/Device)
(P_IMVP8_DRVON) +VCCCORE (IMVP8_PWRGD)
(CPU) H_THERMTRIP# (PCH)
(PCH) DDR4_DRAMRST# (Memory)
+VCCGT



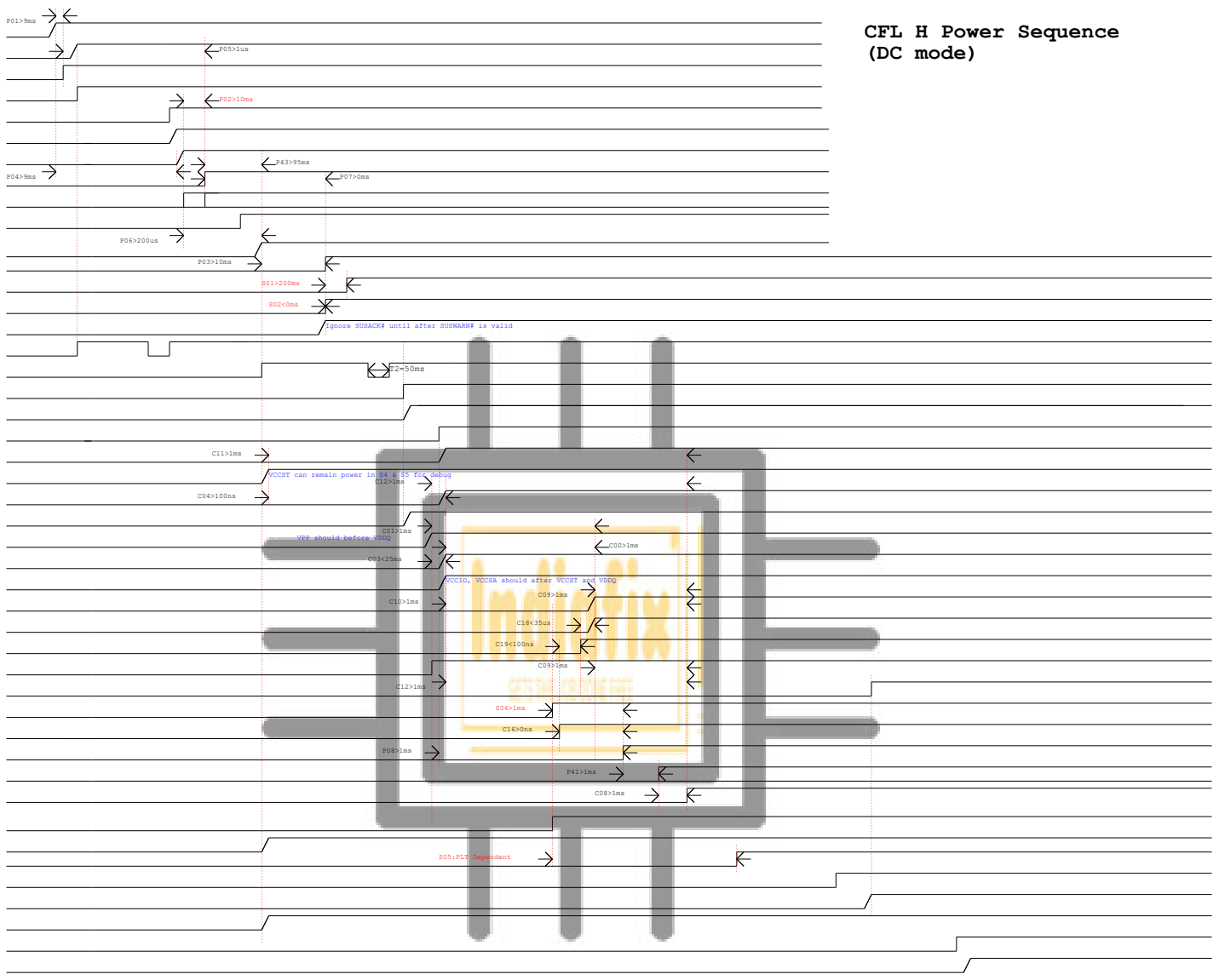
CFL H Power Sequence (AC mode)

ASUS	Project Name: GX502GX		Rev: 1.0
Title : Power On Timing--AC mode			
Size: Custom	Dept.: ASUSTeK COMPUTER INC.	Engineer: NR EE RD3	
Date: Wednesday, January 18, 2023	Sheet 100 of 104		

DC-IN Mode

C:CPU
P:PCH
S:PLT
Power
Signal

(+RTCBAT) +3VA_RTC
(AC_BAT_SYS) +3VA/+5VA
(+3VA_RTC) RTCRST# (PCH)
(Power) AC_IN_OC# (EC)
(EC) PS_ON (+3VA_EC)
(PS_ON) +3VA_EC (EC)
(3VADSW_ON) +3VA_DSW (3VA_DSW_PWRGD)
(EC) DPWROK_EC (PCH)
(+3VA_DSW) PM_BATLOW# (PCH)
(PCH) PM_SLP_SUS# (EC)
(VSUS_ON) +1.0VSUS_VCCPRIM (1.0VSUS_PWRGD)
(EC) PM_RSMRST#_PCH (PCH)
(PCH) SUSWARN# (EC)
(EC) ME_AC_PRESENT_PCH (PCH)
(EC) PCH_SUSACK# (PCH)
(PWR Switch) PWR_SW# (EC)
(EC) PM_PWRBTN# (PCH)
(EC) SUSC_EC# (Power)
(SUSC_EC#) +12V/+5V/+3V
(EC) SUSB_EC# (Power)
(SUSB_EC#) +12VS/+5VS/+3VS
(VSUS_ON) +1.0V_VCCST, VCCPLL (VCCST_PWRGD)
(+VCCIO) +VCCSTG
(1.2V_ON) +2.5V (2.5V_PWRGD)
(1.2V_ON) +VDDQ_CPU (1.2V_PWRGD)
(+12VS) +VCCPLL_OC
(SUSB_EC#) +VCCIO (VCCIO_PWRGD)
(ALL_SYSTEM_PWRGD) +VCCSA (IMVP8_PWRGD)
(DDR_VTT_CTRL) +0.6V
(CPU) DDR_VTT_CTRL (Power)
(Power) 1.2V_PWRGD (AND)
(Power) IMVP8_PWRGD
(AND) ALL_SYSTEM_PWRGD (CPU/PCH/EC/Power)
(ALL_SYSTEM_PWRGD) VCCST_PWRGD_CPU (CPU)
(EC) PM_PWROK_PCH (PCH)
(PCH) CLK_PCH_BCLK (CPU)
(PCH) H_CPUPWRGD (CPU)
(ALL_SYSTEM_PWRGD) P_IMVP8_EN_10 (Power)
(CPU) P_SVID_DATA_X2 (Power)
(EC) PM_SYSPWROK_PCH (PCH)
(PCH) PLT_RST# (CPU/EC/Device)
(P_IMVP8_DRVON) +VCCCORE (IMVP8_PWRGD)
(CPU) H_THERMTRIP# (PCH)
(PCH) DDR4_DRAMRST# (Memory)
+VCCGT



CFL H Power Sequence (DC mode)

ASUS	Project Name: GX502GX		Rev: 1.0
Title :	Power On Timing--DC mode		
Size: Custom	Dept.: ASUSTeK COMPUTER INC.	Engineer: NR EE RD3	
Date: Wednesday, January 18, 2023	Sheet 101 of 104		

Rev	Date	Description
1.0	2017.03.03	P21 Remove CR
	2017.03.04	Q5602, Q5601 change for DXF Limitation
	2017.03.10	+1V8_MAIN, +3VS_NV Follow G753 Power sequence setting
	2017.03.11	R7821 Follow GL702 GPU Power seq
	2017.03.11	+PEX_VDD GPU Sequence update
	2017.03.14	PCE8104, PCE8107, PCE8106 update Cap for DXF Limit
	2017.03.14	GX501VIK Remove useless nut in P67
	2017.03.15	J5301 Connector list update
	2017.03.15	R6716, R6706, R6724, R6723 Screw Hole update
	2017.03.16	J5402 change LID SW for Angle Simulation
	2017.03.17	EMI Reserve C5606, C5607 Cap
	2017.03.17	RN3002, L4512, L4513, L4504, L4515, L4505 SWAP for Layout
1.1	2017.03.21	P30, P37, P39 AMP Change SNL04M25555
	2017.03.21	EMI Reserve P30 C77360 Cap and P69 Caps and Spring
	2017.03.23	AMP Change Remove P37 MUTE CONTROL schematic
	2017.03.23	AMP Change Reserve Q3701
	2017.03.27	P71 +FBVDDQ Discharge setting Follow G753 Power sequence setting
	2017.03.29	P12 Remove TBT Power Low Switch for Layout
1.2	2017.03.30	Change Cap C7105, C7101 for thermal request
	2017.03.30	Thermal Request for R6728
	2017.04.05	EMI Reserve Cap and Spring in P69
	2017.04.06	Reserve C2601 for +VCCPGPP_EF Ripple
	2017.04.07	25.00MHz Cap C1126, C1124 change for Vendor Suggest
	2017.04.07	27MHz Cap C7811, C7801 change for Vendor Suggest
2.0	2017.04.11	P48 Reserve 0 ohm for SDA and SCL
	2017.04.17	P71 Add Caps for NVVDD Ripple
	2017.04.18	Q5704 Chnage MOS for DXF Limitation
	2017.04.20	P71 Add Caps for NVVDD Ripple
	2017.04.21	P67 Screw Hole R6705 update
	2017.04.21	P71 Add Caps for NVVDD Ripple C71119~C71122
	2017.04.25	P71 C7122, C7123, C7127, C7135, C7143, C7144, C7156, C7164, C7166,, C7186, C7187 change to 10UF
	2017.04.25	P70 C7007, C7002 change to 4.7UF
	2017.04.25	P70 C7020, C7024 change to 22UF
	2017.05.02	EMI Reserve C6948, C6949, C6950, C6951, C6952, C6953, C6954 and remove D6913
	2017.05.02	R6711, R6712 change screw hole type fot EMI Solution
	2017.05.03	Remove J4502 Pin 36 GND fot Factory Issue
	2017.05.05	P88, P58 Remove VCCST_PWRGD signal
	2017.05.05	P79, P45 Remove eDP Lan2 and Lan3 Signal
	2017.08.10	P31 R3121,R3122 change to unmount

Rev	Date	Description

Rev	Date	Description

Rev	Date	Description

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ASUSTeK COMPUTER

Title : Revision History

Engineer: NR EE RD3

Size	Project Name	Rev
A	GX502GX	1.0

Date: Wednesday, January 18, 2023 Sheet 1 of 1

EE PART

GV301QH SKU Table

Part Number	PCB	SKU	CPU	LPDDR4X	VRAM	PD ID	
60NR06C0-MB621A	R2.0	GV301QH MB._32G/R9-5980HS//R2.0(V4G)	R9-5980HS	32GB	V4G	WW	
60NR06C0-MB6110	R2.0	GV301QH MB._16G/R9-5980HS//R2.0(V4G)	R9-5980HS	16GB	V4G	WW	
60NR06C0-MB6010	R2.0	GV301QH MB._8G/R9-5980HS//R2.0(V4G)	R9-5980HS	32GB	V4G	WW	
60NR06C0-MB521A	R2.0	GV301QH MB._32G/R9-5900HS//R2.0(V4G)	R9-5980HS	32GB	V4G	WW	
60NR06C0-MB511A	R2.0	GV301QH MB._16G/R9-5900HS//R2.0(V4G)	R9-5980HS	32GB	V4G	WW	
60NR06C0-MB5010	R2.0	GV301QH MB._8G/R9-5900HS//R2.0(V4G)	R9-5980HS	32GB	V4G	WW	
60NR06C0-MB423A	R2.0	GV301QH MB._32G/R7-5800HS//R2.0(V4G)	R9-5980HS	32GB	V4G	WW	
60NR06C0-MB4110	R2.0	GV301QH MB._16G/R7-5800HS//R2.0(V4G)	R9-5980HS	32GB	V4G	WW	
60NR06C0-MB401A	R2.0	GV301QH MB._8G/R7-5800HS//R2.0(V4G)	R9-5980HS	32GB	V4G	WW	
60NR06C0-MBC210	R2.0	GV301QH MB._32G/R9-5980HS//R2.0(V4G)(GE)	R9-5980HS	32GB	V4G	WW	
60NR06C0-MBC110	R2.0	GV301QH MB._16G/R9-5980HS//R2.0(V4G)(GE)	R9-5980HS	32GB	V4G	WW	
60NR06C0-MBC010	R2.0	GV301QH MB._8G/R9-5980HS//R2.0(V4G)(GE)	R9-5980HS	32GB	V4G	WW	
60NR06C0-MBB210	R2.0	GV301QH MB._32G/R9-5900HS//R2.0(V4G)(GE)	R9-5980HS	32GB	V4G	WW	
60NR06C0-MBB110	R2.0	GV301QH MB._16G/R9-5900HS//R2.0(V4G)(GE)	R9-5980HS	32GB	V4G	WW	
60NR06C0-MBB010	R2.0	GV301QH MB._8G/R9-5900HS//R2.0(V4G)(GE)	R9-5980HS	32GB	V4G	WW	
60NR06C0-MBD210	R2.0	GV301QH MB._32G/R7-5800HS//R2.0(V4G)(GE)	R9-5980HS	32GB	V4G	WW	
60NR06C0-MBD110	R2.0	GV301QH MB._16G/R7-5800HS//R2.0(V4G)(GE)	R9-5980HS	32GB	V4G	WW	
60NR06C0-MBD010	R2.0	GV301QH MB._8G/R7-5800HS//R2.0(V4G)(GE)	R9-5980HS	32GB	V4G	WW	

N-KEY

	White 1SREG	1SRGB
R3104	0	10G212000004010
R3106	10G212000004010	0
R3107	0	10G212000004010
R3108	10G212000004010	0
R3564 R3574	0	10G212000004010
R3509 R3511	0	10G212100314010
Q3502 Q3505	0	07005-01920000

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ASUSTeK COMPUTER	Title : GM501GX_SKU Table_R3.0		
	Engineer: NR EE RD3		
Size: D	Project Name: GX502GX		Rev: 1.0
Date: Wednesday, January 19, 2022		Sheet 103 of 104	

13. Audio Codec : 02043-00130000 (663-VA4)